

Measuring Performance in Small and Medium
Enterprises in the Information & Communication
Technology Industries

A thesis submitted in fulfillment of the requirements for the degree of
Doctorate of Philosophy

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DECLARATION

I certify that except where due acknowledgement has been made, the work is that of the author alone; the work has not been submitted previously, in whole or in part, to qualify for any other academic award; the content of the thesis is the result of work which has been carried out since the official commencement date of the approved research program; and, any editorial work, paid or unpaid, carried out by a third party is acknowledged. Ethics procedures and guidelines approved by university for my project have been followed.

Date:

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LIST OF ABBREVIATIONS

Acronym	Meaning
ABS	Australian Bureau of Statistics
ABE	Australian Business Excellence (framework)
ANZSIC	Australian and New Zealand Standard Industrial Classification
BSC	The Balanced Scorecard
BRW	Business Review Weekly (leading business magazine in Australia)
CEO	Chief Executive Officer
CSFs	Critical Success Factors
CSIRO	Commonwealth Scientific Industrial Research Organization
EFQM	European Foundation for Quality Management
ICT	Information and Communication Technology
IT	Information Technology
ITAA	Information Technology Association of America
KPIs	Key Performance Indicators
MBNQA	Malcolm Baldrige National Quality Award
MD	Managing Director
NBSC	National Bureau of Statistics of China
NPD	New Products Development
OECD	Organization for Economic Co-operation and Development
PM	Performance Measurement
PM&M	Performance Measurement and Management
PMM	Performance Measurement Matrix
R&D	Research and Development
RBV	Resource-based View
SMART	Specific, Measurable, Achievable, Relevant and Time-bound
SMEs	Small and Medium Sized Enterprises
TQM	Total Quality Management
UN-ECE	United Nations Economic Commission for Europe
VCs	Venture Capital Companies

ABSTRACT

Small and medium-sized enterprises (SMEs) in the information and communication technology (ICT) industries play a crucial role in the new economy. Most SMEs operate in a fiercely competitive environment; as such, it is important to optimize business practices. It is believed that performance measurement (PM) tools can help to identify weaknesses, clarify objectives and strategies, and improve management processes. While many theories on performance measurement and performance management have been developed mainly for large organizations over the past two decades, few have been tailored for SMEs. In addition, research highlights that these tools are difficult to adapt for SMEs.

This research aims to identify and investigate the critical factors influencing the performance of SMEs in the ICT industries and, in doing so, develop a new PM framework that is able to effectively measure SME performance.

To improve the accuracy of the judgments in this research, methodological triangulation strategy, which mixes both qualitative and quantitative approaches, was employed. At the same time, data-source triangulation was applied to reduce the impact of potential biases. In qualitative research, twenty interviews were conducted, sixteen cases were analysed through cross-case analysis. In quantitative research, sixty-six valid responses to questionnaire surveys were collected.

This thesis has completed three major tasks:

- 1) Investigated critical success factors that affect the performance of SMEs.
- 2) Analysed existing SME performance measurement techniques in order to identify the best framework for SMEs in ICT industries.
- 3) Constructed a practical PM framework and implementation strategy for best-practice PM in ICT-related SMEs.

Research results show that an effective PM system for ICT SMEs should help the companies to formulate right strategies that can especially manage the uncertainty of the external environment in their development. Importantly, a PM process should be constructed which aligns with the SMEs' daily processes such as ordinary planning, budgeting and reporting processes. Such a PM system should involve a flexible PM framework that can be adapted dynamically to suit the changes in the PM process. It is also required that the PM framework help SMEs identify customer needs, manage and measure customer satisfaction.

A PM framework meets the above requirements can be built based on the structure of business excellence models and the theory of system management. At the same time, the KPIs should be incorporated in the PM system to help design the PM framework. The constructed PM framework involves measuring both performance determinants and performance results. The performance determinants, including both internal determinants and external determinants, are represented in following dimensions: capability building; resource developing and utilizing; environment adapting; strategies formulating; internal process managing and PM on innovation and learning. The performance results' dimension should include the financial results to satisfy the investors, the customer satisfaction indicators to meet customers' needs, the competitive indicators to reflect the competitive advantages, and the collaborative measures to measure the partnership.

CHAPTER 1 INTRODUCTION

1.1 BACKGROUND

In the new economy, small and medium sized enterprises (SMEs) play a critical role. For example, in Australia, SMEs represent 97 percent of all private sector businesses and provide 49 percent of all private sector employment (ABS 2002). Among SMEs, the fastest growth in the last several decades has occurred in the information and communication technology (ICT) industries.

Because of the important role of SMEs in the new economy, a lot of research focuses on their performance and critical success factors. In Australia, CSIRO (Commonwealth Scientific Industrial Research Organization) has studied the performance measurement (PM) system for SMEs (Barnes, Coulton et al. 1998). Studies have traditionally used management accounting to create performance benchmarks. By the late 1980s, studies had shown that historic financial data is not enough to satisfy the performance measurement in the new economy because of the increasing complexity of organizations and the markets in which companies compete (Kennerley and Neely 2002). This is because financial reports are now less indicative of shareholder value. As pointed out by Cumby & Conrod (2001), sustainable shareholder value is instead driven by non-financial factors, such as customer loyalty, employee satisfaction, internal processes, and an organisation's innovation. For the Standard and Poor 500, only 10 to 15 percent of market value is captured by traditional accounting measures (Webber 2000). Hence, a series of PM frameworks have been introduced during last two decades, such as the Balanced Scorecard (Kaplan and Norton 1996), the Performance Prism (Neely, Adams et al. 2001), the EFQM Business Excellence Model (Shergold and Reed 1996; EFQM 2003), and Integrated Performance Measurement (Nanni, Dixon et al. 1992), among others. Other than these holistic performance models, many performance tools, such as the TQM, KPIs, Benchmarking, etc., were also used in practice. However, most of these were designed for large organizations.

There is no consensus as to whether these models are apt for SMEs. Some scholars believe they are applicable (Jungman, Okkonen et al. 2004), while contradictory research (Hvolby and Thorstenson 2000; Hudson, Smart et al. 2001; Abouzeedan and Busler 2005) finds that the performance measurement in SMEs has its special characteristics and cannot be regarded as a small version of that in big enterprises. Though it is not clear, current research suggests that established bi-business PMs are not applicable – at least, as much as tailored techniques would be.

1.2 AIMS OF THE STUDY

This research aims to identify and investigate the critical success factors that influence the performance of SMEs in the ICT industries; furthermore, seeks to develop a new PM framework that is able to measure these companies' performance. Finally, this research will add to the PM framework by also developing comprehensive performance planning, diagnoses, and analyses tools.

1.3 SCOPE OF THE STUDY

The research focuses on the performance measurement in SMEs in the ICT industries. The emphasis of this study is placed on addressing the issue of how to measure ICT-related SME performance, from a practical perspective.

For that purpose the study concerns:

- Performance measurement (PM): Based on reviewing many scholars' definitions about PM, this study defines PM as a structured system and a process of gathering, monitoring, and assessing information related to an organization's activities, in order to achieve goals and objectives.
- Small and medium sized enterprises (SMEs): the definition of SMEs varies from country to country. In this study, most sample companies come from Australia. Hence, this study uses the definition of ABS (Australian Bureau of Statistics); therefore, SMEs are companies with between 1 and 199 employees.
- ICT industries: In this study, the definition of ICT industries uses the definition from OECD, which is followed by ABS. The ICT industries include the

industries of computer hardware, software, telecommunications, Internet, computer services, and e-business.

The sample companies for this study were selected from both Australia and South China. These were selected because the researcher has been working in ICT-related SMEs in these two countries for many years. The network helps the researcher obtain considerable research data. In addition, many ICT manufacturers are located in South China, which compensates for the lack of manufacturers among the sample companies in Australia.

1.4 RESEARCH QUESTIONS

To achieve the aims specified above, two primary research questions are to be investigated:

- What is an effective performance measurement system for SMEs in the ICT industries?
- How can SMEs implement the performance measurement system successfully?

To address the above questions, the following questions are investigated:

1. What are the knowledge gaps in the current literature in regard to performance and performance measurement systems for SMEs in the ICT industries?
2. What are the weaknesses and problems of the current PM models, when applied in SMEs in the ICT industries?
3. How can SMEs design an effective PM framework?
4. What are the critical success factors that drive SME performance?
5. What are the measures and indicators that SMEs in the ICT industries should measure?
6. How can SMEs successfully measure and manage their performance?

1.5 RESEARCH DESIGN

This study employs a research design of a mixed methodology, i.e. the researcher combines aspects of the qualitative and quantitative paradigms at many methodological steps in the research design (Creswell 1994; Creswell 2003). This methodology takes advantage of both the qualitative and the quantitative paradigms and reduces the limitations that are likely to be derived from a single methodological design (Bryman 1996). In this research, a concurrent triangulation strategy (see methodology chapter)

was used. This strategy helps this researcher “use two different methods in an attempt to confirm, cross-validate, or corroborate findings within a single study” (Creswell 2003). The methods of semi-structured interviews, structured-questionnaires surveys, and case studies were employed in order to collect qualitative and quantitative data.

Qualitative Research:

At the qualitative stage, a comprehensive and extensive literature review was conducted, analysing existing PM theories, models, and methods in order to identify the strengths and weaknesses of each PM model and method and summarize research regarding PMs in SMEs in the ICT industries. Twenty interviews have been conducted. Sixteen cases were analysed through cross-case analysis, to identify the critical success factors that drive ICT SME-performance and to investigate the implementation of PM in current SMEs in the ICT industries. Furthermore, the PM implementation in four SMEs was studied via multiple case studies.

Quantitative Research:

Based on a conceptual performance measurement framework, a structured questionnaire survey was developed to collect quantitative data. The objectives of the research at this stage are: (1) to test and revise the conceptual performance measurement framework through a quantitative approach, and (2) to assess the level of importance of factors and indicators in the PM framework.

1.6 CONTRIBUTIONS OF THE STUDY

The contributions of this study are both theoretical and practical:

1. During last two decades, many PM frameworks have been developed. However, most of them are for large organizations and can not be adopted by SMEs. This study identifies the requirements of PMs for effective use in SMEs, through the analysis of case studies.
2. There is no agreed viewpoint for an ideal PM model, though many holistic PM models have been developed. As stated by Meyer (2002), the issues of ‘what should we measure and what can we measure’ can not be well solved without practical, quantitative analysis. This study answered the question of ‘what is an effective PM framework for SMEs in the ICT industries’ by utilizing case studies in order to apply theory.

3. Most of the existing PM research on SMEs has examined only one or a few critical success factors that contribute to performance. Indeed, SMEs performance is an integrated result from multi-variables. The related issues should be addressed from a dynamic, balanced, comprehensive and integrated perspective. This research provides a comprehensive view on how to measure SMEs' performance effectively.

1.7 THESIS STRUCTURE

This thesis is comprised of five parts, presented through ten chapters.

Part I involves two literature review chapters (Chapter 2 and Chapter 3), wherein Chapter 2 clarifies the concepts of 'performance', 'performance measurement', and 'performance measurement framework'. To accomplish this, popular PM frameworks were studied and their strengths and weaknesses were compared. Chapter 3 studies the characteristics of PM in SMEs and the theoretical approaches to designing PM for SMEs in the ICT industries. Furthermore, a conceptual PM framework has been constructed based on an analysis of PM features and requirements in SMEs in the ICT industries.

Part II (Chapter 4) presents the research methodology.

Part III presents and discusses the findings from the present study. It includes Chapters 5, 6 and 7. Chapter 5 investigates the determinants that drive ICT SMEs' performance. Chapter 6 investigates the implementation of PM in the ICT industries, based on interview data and questionnaire survey results. The performance measures and indicators that SMEs employed were illustrated in this chapter. Chapter 7 presents multi-case studies. The implementation of PM in four SMEs in the ICT industries is studied. The lessons and experience from the implementation of PM in the case companies are discussed.

Part IV includes Chapter 8 and Chapter 9. A new PM framework for SMEs in the ICT industries is developed in Chapter 8, based on the empirical study result in Part III. A seven-step PM implementation procedure is suggested in Chapter 9.

Part V (Chapter 10) is the summary and conclusion chapter. It summarizes the findings, contributions and limitations of this study. The chapter concludes with suggestions on future research in the study field.

PART I LITERATURE REVIEW

INTRODUCTION

This Part clarifies related definitions regarding PM and its associated theories. The strengths and weaknesses of the typical PM frameworks are reviewed in Chapter 2. Chapter 3 studies the features of SMEs and ICT industries. The challenges, features of PM in SMEs in the ICT industries are also discussed. The knowledge gaps in regard to PM are presented. In addition, a conceptual framework for the present study is developed in Chapter 3.

The objectives of this part are:

- To identify the knowledge gaps in the current literature in regard to PM for SMEs in the ICT industries.
- To find the most effective theoretical approaches to designing PM framework for ICT SMEs through examining the strengths and weaknesses of each of the theoretical approaches based on PM features in SMEs in the ICT industries.
- To identify the features and requirements of PM in SMEs in the ICT industries.
- To construct a conceptual framework for the present study.

CHAPTER 2 PERFORMANCE MEASUREMENT

2.1 INTRODUCTION

To construct a practical PM framework for SMEs, two fundamental questions should be addressed: what is the concept of performance of SMEs? What is the performance measurement in SMEs? In this chapter, the definitions of performance and performance measurement are reviewed. The specific features of these concepts in SMEs are discussed. Furthermore, the typical existing performance measurement models are reviewed and compared, their strengths and weaknesses identified, and the useful components (that can be combined in the PM framework) analysed. The chapter is organized as follows: basic conception of performance is discussed in section 2.2; in section 2.3, PM and PM frameworks are analysed and their strengths and weaknesses are compared; finally, a summary is presented in section 2.4.

2.2 DEFINITION OF PERFORMANCE

Performance is a widely used concept in many areas. Usually, performance is a measure of how well a mechanism/process achieves its purpose. In enterprise management, Moullin (2003) defines an organization's performance as "how well the organization is managed" and "the value the organization delivers for customers and other stakeholders." For the purposes of this research, 'performance' is related to achieving stockholder/investor interests.

Measuring performance is a multi-dimensional concept. Effectiveness and efficiency are the two fundamental dimensions of performance; this is emphasized by Neely, Adams et al. (2002): "Effectiveness refers to the extent to which stakeholder requirements are met, while efficiency is a measure of how economically the firm's resources are utilized when providing a given level of stakeholder satisfaction". To attain superior relative-performance, an organization must achieve its expected objective with greater efficiency and effectiveness than its competitors (Neely 1998). To illustrate efficiency, effectiveness, and the value delivered, multi-measures should be used. Though their forms vary widely, financial indicators are traditionally used; Neely (1998) further expounded upon manufacturing performance measures, suggesting that

five key-dimensions should be assessed: quality, delivery speed, delivery reliability, price (cost), and flexibility. By measuring all of these factors, performance is thus balanced and multi-dimensional, better reflecting stockholder interest.

In this study, performance is defined as the extent to achieving proposed objectives using resource economically in the face of internal/external environment (stockholders, competitors, society).

2.3 PERFORMANCE MEASUREMENT

Before 1980, financial data was used in enterprises as the main performance measure. After late 1980s, scholars were aware that financial data, alone, does not capture comprehensive performance information and, hence, does not completely capture or predict future performance. Thus, a balanced and multi-dimension concept was developed. This section briefly reviews this concept.

2.3.1 Definition of Performance Measurement

Although much research has been conducted on the issues of performance measurement—the definition of performance measurement is still debated. Neely (1998) defines Performance Measurement as “the process of quantifying the efficiency and effectiveness of past actions through acquisition, collation, sorting, analysis, interpretation and dissemination of appropriate data”. Moullin (2003) thought that while Neely’s definition describes the process, “it does not give much guidance to organisations about what it is essentially all about.” He suggests that another definition may be more apt: “performance measurement is evaluating how well organisations are managed and the value they deliver for customers and other stakeholders” (Moullin 2003). Moullin (2003) argued that his definition clearly shows the purpose of performance measurement and emphasizes the assessment both of the value an organisation gives to its various stakeholders and the way the organisation is managed. Nanni et al. (1990) defined performance measurement as “a means of monitoring and maintaining organisational control which is the process of ensuring that an organization pursues strategies that lead to the achievement of overall goals and objectives.” Amaratunga and Baldry (2002) provided a more specific definition of performance measurement: “Measurement provides the basis for an organisation to assess how well

it is progressing towards its predetermined objectives, helps to identify areas of strengths and weakness, and decides on future initiatives, with the goal of improving organisational performance.” This definition illustrates the role and the process of performance measurement clearly from different aspects.

As identified from the above definitions, performance measurement is a structured system and a process of gathering, monitoring, and assessing the information about an organization’s activities, in order to achieve the proposed goals and objectives. In this study, the goals and objectives concern an organization’s strategic objectives, a business unit’s business goals and objectives, and personal business commission.

2.3.2 Functions of Performance Measurement

Generally, the function of performance measurement can be categorized into the following four aspects (Neely 1998):

- (1) *Checking position.* Establishment of current status and monitoring of progress over time and against benchmarks.
- (2) *Communicating position.* Communicate with shareholders, customers, or employees by releasing annual reports, etc.
- (3) *Confirm priorities.* Performance data provide insights into what is important to a business, thus exposing shortfalls that allow organisations to identify priorities.
- (4) *Compel progress.* The measures can help organisations focus on specific issues and encourage people to search for ways to improve performance. The measures communicate the priorities and can form the basis for reward.

The above summary actually suggests a clear flow map for performance measurement, i.e. (1)→(2)→(3)→(4). Indeed, many SMEs apply PM for quality management. It is observed that the above four points cover the roles of PM from the perspective of quality management (Oakland 2004):

- Tracking progress against organizational goals (represented in the above point (1));
- Identifying opportunities for improvement (represented in the above points (3), (4));
- Comparing performance against internal standards (represented in the above points (1), (2));

- Comparing performance against external standards (represented in the above points (1), (2)).

Therefore, the role of performance measures is to control processes and to enforce continuous performance improvement by quality improvement teams. That is, measures should supply information about how well processes and people perform, the goal of which is to motivate better future performance.

A in-depth illustration of the above functions (3) and (4) of performance measurement is made in (Godener & Soderquist 2004) , which summarized the use of performance into four groups, based on Kerssens-Van & Bilderbeek's (1999) study of 19 uses of performance on four different organizational levels.:

- Use of performance measurement results for personnel evaluation, promotion and incentives (promotion prospects, salary, project participation, bonuses)
- Use of performance measurement results for resource allocation (project participation, forming/dissolving teams, assignment of new projects and of resources)
- Use of performance measurement results for control/correction (control, correction, reorganisation)
- Use of performance measurement results for learning/continuous improvement

Note the usage of performance measurement results for resource allocation is highlighted in Godener & Soderquist (2004)'s summary. This is important for SMEs, whose development is usually limited by resource.

2.3.3 The Evolution of Performance Measurement

Before the 1980s, performance measurement was largely evolved within the large industrial firms, focusing on the achievement of a limited number of key financial measures (Johnson & Kaplan, 1987). By the early 1980s, as the increasing complexity of organizations and the markets in which companies compete, it was no longer appropriate to use financial measures as the sole criteria for assessing success (Kennerley and Neely 2002). According to Ghalayini & Noble (1996), the literature concerning performance measurement evolved in two phases, the first of which began in the late 1880s and concluded in the 1980s. In this phase, the emphasis was on financial measures such as profit, return on investment, and productivity. The second phase started in the late 1980s as a result of changes in the world market, specifically in

corporate environments; these organizations discovered that performance measurement, as traditionally practiced, is limited. Yenyurt (2003) and Gomes, Yasin & Lisboa (2004) summarised the major inadequacies of traditional metrics in their literature review; these weaknesses include:

- *Traditional accounting measures of performance are inadequate for strategic decisions (e.g. Kaplan and Norton, 1992);*
- *They are too historical and backward looking (e.g. Ittner and Larcker, 2003);*
- *They are lack of predictive ability to explain future performance (e.g. Ittner and Larcker, 2003);*
- *They provide information on root causes (e.g. Ittner and Larcker, 2003);*
- *They do not link the non-financial metrics to financial numbers (e.g. Kaplan and Norton, 1992)*
- *They report functional not cross- functional processes (e.g. Ittner and Larcker,2003)*
- *They do not consider intangible assets(e.g. lehn and Makhija,1996)*
- *There are too many measures, new measures are needed that have broader content, being able to describe more with less numbers(e.g. Kaplan and Norton, 1992; Frigao and Krumwiede,2000);*
- *Traditional metrics do not aggregate from an operational level to a strategic level (e.g. Kaplan and Norton, 1992; Frigo and Krumwiedw,2000)*

In response to this change in theory, a series of performance measurement systems were introduced, the most widely cited of which is Kaplan & Norton's (1992) Balanced Scorecard, which is based on the principle that a performance measurement system should help managers at all levels monitor results in their key areas. The system forces managers to look at the business from four important perspectives (see details in subsection 2.4.2). Experience has revealed that innovating CEOs used the Balanced Scorecard not only to clarify and communicate strategy, but also to manage it.

In contrast to the Balanced Scorecard, competing techniques were introduced, such as: the ABPA Activity-based Profitability Analysis (Meyer 2002)(); the Performance Prism (Adams and Neely 2002); Performance Pyramid (Lynch and Cross 1991); Integrated Performance Measurement (Nanni, Dixon et al. 1992); and Performance Measurement in Service Businesses (Brignall, Fitzgerald et al. 1991).

Different from constructing the general framework of PM for all kinds of organizations, many researchers focus on designing and adopting the performance measurement system based on definite organization's characteristics (Beamon 1999). To date, researchers have not adopted a universally accepted best-practice due to the following requirements on PM (Gomes et al., 2004):

- § *must reflect relevant non-financial information based on key success factors of each business (Clarke, 1995);*
- § *should be implemented as a means of articulating strategy and monitoring business results (Grady, 1991);*
- § *should be based on organizational objectives, critical success factors, and customer needs and should monitor both financial and non- financial aspects (Manoochehri, 1999);*
- § *must accordingly change dynamically with the strategy (Bhimani, 1993);*
- § *must meet the needs of specific situations in manufacturing operations and should be longterm oriented as well as simple to understand and implement (Santori and Anderson, 1987);*
- § *must make a link to reward systems (Tsang et al., 1999); and*
- § *financial and non-financial measures must be aligned and fit within a strategic framework (Drucker, 1990; McNair and Mosconi, 1987).*

Although it is suggested in Sedecon Consulting (1999) that “measures should be developed from strategy and tailored to match the specific organizational context and structure”, the gap between what are wanted to be measured and what can be measured is the main reason for performance measurement being so challenging (Meyer 2002).

Since a dominant theory has not been developed, most companies simply continue with what they've used in the past, rarely deviating from their established practices.

2.4 PERFORMANCE MEASUREMENT FRAMEWORKS

2.4.1 Overview of Performance Measurement Frameworks

To overcome the shortcomings of traditional measurement systems, various holistic performance measurement frameworks were developed after the late 1980s. The common, changed approach in each of the theories echoes a multi-dimensional approach, which seeks to balance financial and non-financial measures. Kennerley &

Neely (2000) had summarized the characteristics of performance measurement frameworks:

1. The measures used by an organization have to provide a ‘balanced’ picture of the business.
2. The framework of measures should provide a succinct overview of an organization’s performance.
3. The performance measures should be multi-dimensional.
4. The performance measurement matrix (PMM) provides comprehensive mapping. It is possible to map all possible measures of an organization’s performance onto the framework and identify where there is omission or a need for greater focus.
5. The performance measures should be integrated across the organization’s functions and through its hierarchy.
6. The performance measurement system can provide data for monitoring past performance and planning future performance. It implies the measures should measure both results and the drivers of them.

Although quality management models – such as the European Foundation for Quality management (EFQM) excellence model, the Malcolm Baldrige National Quality Award (MBNQA) and Australian Business Excellence (ABE) framework – are not designed for performance measurement, they act as tool of managing performance and are often used to help an organization to improve performance practice. These quality management models are also self-assessment frameworks. In performance measurement aspect, they provide:

1. Measuring both performance results and drivers of them. For example, the EFQM Excellence model divides the excellence criteria into two groups: the enablers and results.
2. Focusing-measures on strategy and process
3. Involve all employees in the continuous improvement process to achieve better performance.
4. Comprehensive measures including society indicators
5. Improved performance, based on innovation, learning, and information analysis.

Besides these models, many other PM theories and methods are applied in practice, such as key performance indicators system (KPIs), bench-marking, and so on. These

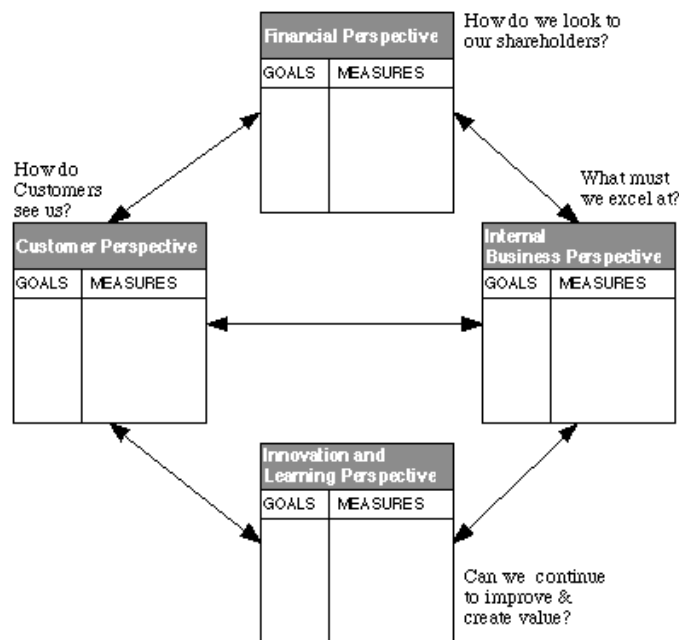
theories and models discuss the issue of PM from different perspectives, the relative strengths and weaknesses of which are presented in the following paragraphs.

2.4.2 The Balanced Scorecard

The Balanced Scorecard (BSC) (Kaplan and Norton 1996) is perhaps the most well-known performance measurement framework. Many people even believe it is the most important and widely used management theory of the 20th century. BSC suggests managers to view organization's performance from four perspectives (see Figure 2-1):

- (a) How do customers see us? - Customer perspective
- (b) What must we excel at? - Internal perspective
- (c) Can we continue to improve and create value? - Innovation & learning perspective
- (d) How do we look to shareholders? - Financial perspective

Figure 2-1 The Balanced Scorecard



Source: Kaplan and Norton (1992)

BSC incorporates financial and non-financial measures in one measurement system. The objectives and measures of BSC are derived from an organization's vision and strategy. The Balanced Scorecard provides executives with a comprehensive framework that translates a company's vision and strategy into a coherent set of performance measures.

According to Kaplan & Norton (1996) “the balanced scorecard not only allows the monitoring of present performance, but also tries to capture information about how well the organization is positioned to perform in the future”. Furthermore, the Balanced Scorecard has evolved to become a core management tool, in that it helps CEOs “not only to clarify and communicate strategy, but also to manage strategy.” In practice, companies use the BSC approach to accomplish four critical management processes:

1. Clarify and translate vision and strategy
2. Communicate and link strategic objectives and measures
3. Plan, set targets, and align strategic initiatives
4. Enhance strategic feedback and learning.

The four perspectives in the BSC model are regarded as a chain of cause-and-effect. For example, financial performance depends on a customer’s loyalty, which is influenced by an enterprise’s internal/business processes. Similarly, internal business processes are dependent on employee’s skills (learning and growth). A good Balanced Scorecard should have an appropriate mix of outcomes (lagging indicators) and performance drivers (leading indicators) of the business unit’s strategy (Kaplan & Norton 1996). Kaplan & Norton (2001) pushed the BSC to a new level: the ‘strategy map’. ‘The strategy map describes the process for transforming intangible assets into tangible customer and financial outcomes. It provides executives with a framework for describing and managing strategy in a knowledge economy.

Though the Balanced Scorecard is widely lauded, numerous authors have identified shortcomings:

- The Balanced Scorecard neglects the most fundamental question – the competitor perspective (Neely, Gregory et al. 1995)
- The approach used by the Balanced Scorecard is not consistent with a complete performance measurement system; rather, it merely provides senior managers with a tool to monitor performance against strategic and operational objectives (Brignall 1991; Brignall, Fitzgerald et al. 1991; Gomes, Yasin et al. 2004)
- The BSC is more like a strategic management tool than a true, complete PM system (Kaplan and Norton 1996)

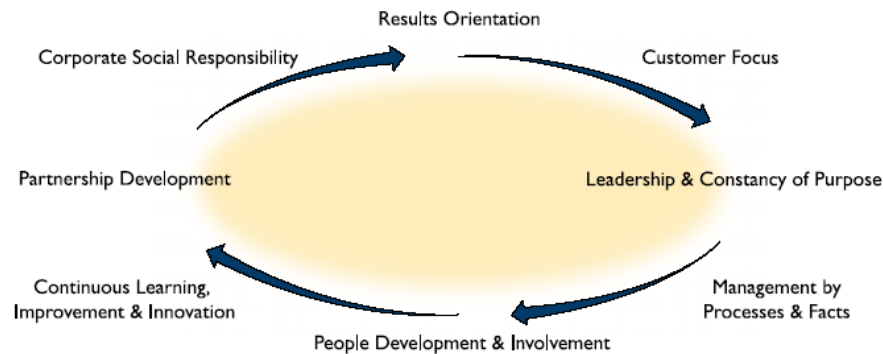
- The BSC does not address the stockholders (end user, employee, suppliers, regulators, pressure groups and local communities). The BSC does not take a broad enough view of the stakeholders who interact with an organization (Neely and Adams 2001).
- “The scorecard has floundered as a device for measuring and rewarding performance.” (Meyer 2002)
- The BSC is just taken as a boilerplate. It is hard to work in many enterprises (Ittner and Larcker 2003).
- It does not reflect different dimensions of performance addressed by the SMART pyramid or Results and Determinants Model. Neither the customer nor internal perspective are defined in terms of performance-dimensions that regard success, such as the generic strategic-objectives of quality, cost, delivery (speed and reliability), and flexibility (Neely 2002).
- Kaplan and Norton promote the application of strategy maps (as noted above); they only weakly develop this, however, since they fail to break them down into their vital components – the potential for success and the potential for failure. Organizations have many opportunities, but they also face several threats; their measurement systems need to be able to capture both so that executives can manage the business with a clear view of both scenarios (Neely 2002)

2.4.3 The EFQM Excellence Model

The EFQM Excellence Model was introduced by EFQM in 1991. The EFQM is the organization which was formally established by 14 European companies in 1988 to guide organizations to improve performance. The EFQM Excellence Model is a non-prescriptive framework to help guide an organisation and improve its performance. The model is based on eight Fundamental Concepts of Excellence: Results Orientation, Customer Focus, Leadership and Constancy of Purpose, Management by Processes and Facts, People Development and Involvement, Continuous Learning, Innovation and Improvement, Partnership Development, and Corporate Social Responsibility (see Figure 2-2). EFQM believes that ‘Truly excellent organisations are those that strive to satisfy their stakeholders by what they achieve, how they achieve it and what they are likely to achieve’. According to the definition of EFQM, the stakeholders include those individuals or groups that impact upon, or have an impact on, the organisation, such as

customers, employees, partners, suppliers, the society in which the organisation operates, and those with a financial stake in the organisation.

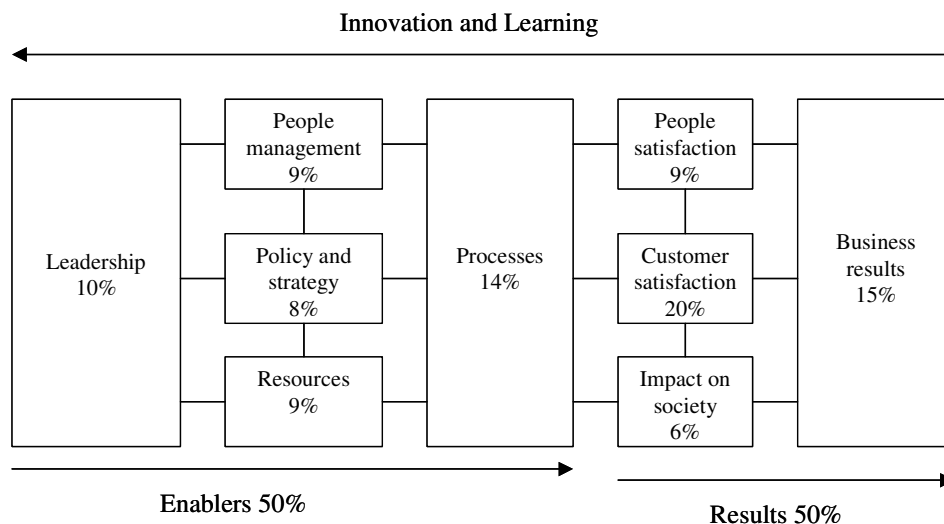
Figure 2-2 The EFQM's Fundamental Concepts



Source: the brochure of EFQM (2003)

The EFQM Excellence model consists of nine criteria (Figure 2-3), which can be divided into two groups: Enablers and Results. The five 'Enabler' criteria cover what an organisation does. The four 'Results' cover what an organization achieves. The logic interrelation between the 'Results' and 'Enablers' is as follows: 'Results' are caused by 'Enablers', and 'Enablers' are improved using feedback from 'Results'.

Figure 2-3 EFQM Excellence Award Criteria



Source: EFQM. The EFQM business excellence Model is a registered trademark
Cited from: Wongrassamee, Gardiner & Simmons (2003)

The EFQM Excellence Model is a practical tool that can be used in a number of different ways:

- § *As a tool for Self-Assessment*
- § *As a way to Benchmark with other organisations*
- § *As a guide to identify areas for Improvement*
- § *As the basis for a common Vocabulary and a way of thinking*
- § *As a Structure for the organisation's management system*

To satisfy the requirement of the management in specific subject area, a series of frameworks have been developed from the EFQM Excellence Model (since 2003): Corporate Social Responsibility, Risk Management, Innovation, Knowledge Management, and HR Management. The Frameworks are non-prescriptive, holistic tools that help approaches that focus on the specific subject area. These frameworks follow the EFQM Model, consisting of the same 9 criteria and the RADAR logic (determines the **R**esults required→ plan & develops **A**pproaches→ **D**eploy→ **A**ssess & **R**eview).

Summarily, EFQM includes 8 fundamental concepts, 9 criteria measurement model, some management framework, and may be implemented by following these ten steps (Oakland 2004):

1. Set direction through leadership
2. Establish the results it wants to achieve
3. Establish and drive policy and strategy
4. Set up and manage appropriately its approach to process, people, partnerships and resources
5. Deploy the approaches to ensure achievement of the policies, strategies and thereby the results
6. Assess the 'business' performance, in term of customers, its own people and society results
7. Assess the achievements of key performance results
8. Review performance and areas for improvement
9. Innovate to deliver performance improvements
10. Learn more about the effects of the enablers on the results

The significant feature of EFQM excellence model is that the model distinguished the result area (Results the organisation has achieved (WHAT)) and organisation areas

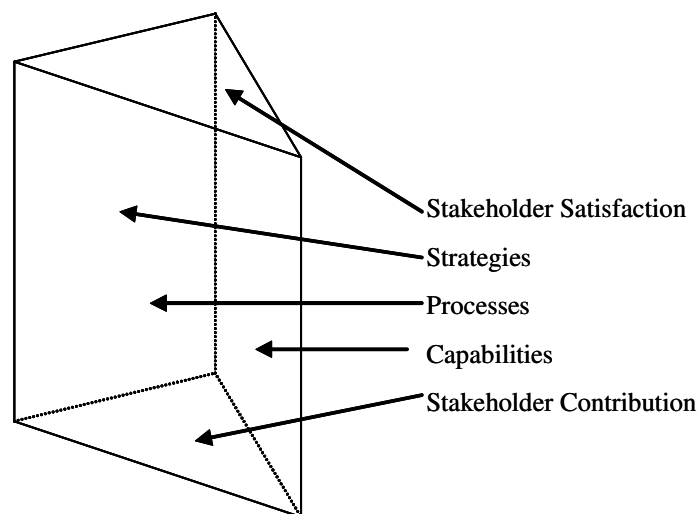
(Management of the organisation (HOW)) (Westerveld 2003; Wongrassamee, Gardiner et al. 2003).

In fact, the business excellence models (EFQM and Baldrige Award) take a broader view of performance and include references to a wider set of stakeholders than does the BSC. However, they also contain a host of dimensions that are effectively unmeasurable (Neely, Adams et al. 2001).

2.4.4 The Performance Prism

The performance prism was introduced by Neely, Adams et al. (2001) based on three fundamental premises. First, the organizations should think about the wants and needs of all of their important stakeholders and endeavour to deliver value to each of them if the organization wants to survive and prosper in the long-term. It is no longer acceptable for organizations to focus on one or two of their stakeholders. Secondly, organizations have to align and integrate strategies, processes, and capabilities in order to deliver real value to its stakeholders. Thirdly, the relationship between organizations and their stakeholders is reciprocal – stakeholders have to contribute to organizations as well as to expect something from them. The performance prism, as shown in Figure 2-4 , is considered as a second-generation performance measurement framework. It builds on and strengthens existing measurement framework on shareholder value (the BSC, the EFQM excellence model, the Malcolm Baldrige Award criteria).

Figure 2-4 The Performance Prism



Source: Neely, Adams et al. (2001)

The Performance Prism consists of five interrelated perspectives of performance that pose specific vital questions about:

- Stakeholder Satisfaction – who are our key stakeholders and what do they want and need?
- Stakeholder Contribution – what do we want and need from our stakeholders on a reciprocal basis?
- Strategies – what strategies do we need to put in place to satisfy the wants and needs of our stakeholders while satisfying our own requirements too?
- Processes – what processes do we need to put in place to enable us to execute our strategies?
- Capabilities – what capabilities do we need to put in place to allow us to operate our processes?

The significant feature of the Performance Prism is that the performance measurement should be derived from the stakeholder satisfaction. It changes the usual opinion that is adopted by most performance measurement framework or methodologies, i.e. the performance measure should be derived from the strategy technique. Neely, Adams et al. (2001) believe that the purpose of measurement and the role of strategy is, in this way, fundamentally misunderstood. Instead, they believe that performance measures “are designed to help people track whether they are moving in the direction they want to. They help managers establish whether they are going to reach the destination they set out to reach. Strategy, however, is not about destination. Instead, it is about the route you choose to take- how to reach the desired destination”.

In the Performance Prism framework, an organization’s key stakeholders usually include

- Investors (principal shareholders, but other capital providers as well);
- Customers and intermediaries;
- Employees and labour unions;
- Suppliers and alliance partners;
- Regulators, pressure groups and communities.

To some extent, the performance prism was instrumental in developing the Balanced Scorecard. For example, the concept of stakeholders is broadened in the latter, which

not only considers shareholders and customers, but also include employees, suppliers, alliance partners or intermediaries. However, the two theories also differ: Neely, Adams et al. (2001) disagree that the performance measurement should be derived from the strategies; instead, “strategies should be put in place to ensure the wants and needs of the stakeholders are satisfied”. In short, the Performance Prism is not a prescriptive-measurement framework; instead, it is a tool (framework) that helps management teams to think about key questions and strategies to address them. The very same benefits that make the Performance Prism a strong, comprehensive model, however, also make it difficult to easily utilize.

2.4.5 The Malcolm Baldrige National Quality Award

The Malcolm Baldrige National Quality Award (MBNQA) is a widely-used performance self-assessment framework. It was developed in the late 1980s in the United States. The Baldrige criteria for performance excellence are designed to help organizations manage performance through an integrated approach. The goal of the Baldrige criteria includes:

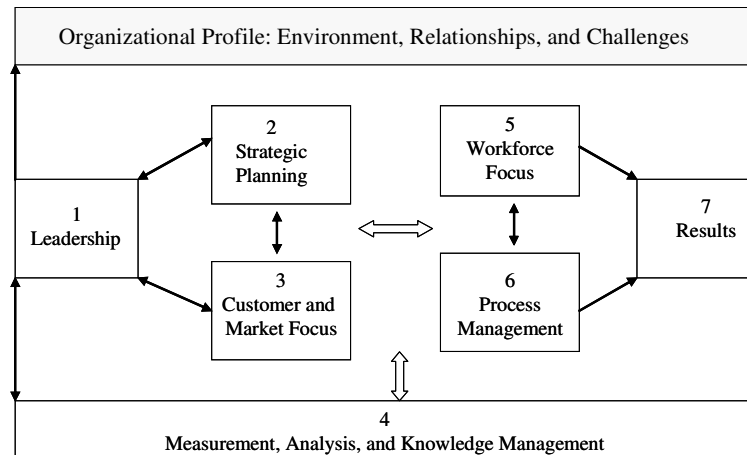
- Delivery of ever-improving value to customers and stakeholders, contributing to organizational sustainability
- Improvement of overall organizational effectiveness and capabilities
- Organizational and personal learning

The Baldrige criteria are built upon a set of interrelated core values and concepts:

- Visionary leadership
- Customer-driven excellence
- Organizational and personal learning
- Valuing employees and partners
- Agility
- Focus on the future
- Managing for innovation
- Management by fact
- Social responsibility
- Focus on results and creating value
- Systems perspective

As shown in Figure 2-5, the Baldrige performance criteria consist of seven categories: 1 Leadership, 2 Strategic Planning, 3 Customer and Market Focus, 4 Measurement, analysis and Knowledge Management, 5 Human Resource Focus, 6 Process management, 7 Results.

Figure 2-5 Baldrige Criteria for Performance Excellence Framework



Source: Malcolm Baldrige National Quality Award, (2009) '2009-2010 Criteria for performance excellence', US National Institute of Standards and Technology, Gaithersburg, Maryland, USA

These seven categories were divided into two triads: Leadership (Category 1), Strategic Planning (Category 2), and Customer and Market Focus (Category 3) represent the Leadership triad which emphasizes on the importance of leadership focusing on strategy and customers, wherein senior leaders set organizational direction and seek future opportunities for organization; and Workforce Focus (Category 5), Process Management (Category 6), and Results (Category 7) represent the Results triad.

In the Baldrige criteria, all activities are toward the results:

- (1) Product and service outcomes
- (2) Customer-focused outcomes
- (3) Financial and market outcomes
- (4) Workforce-focus outcomes
- (5) Process effectiveness outcomes, including key internal operational performance measures
- (6) Leadership and social responsibility outcomes

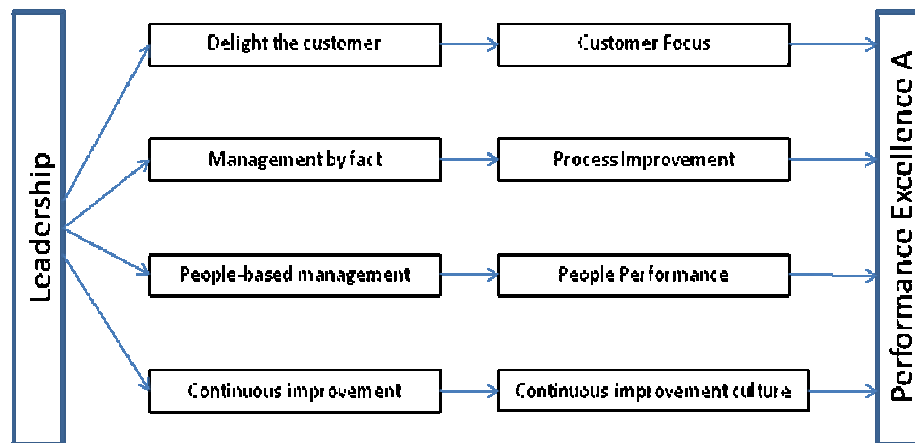
The Baldrige criteria are system-perspective, in that they are non-prescriptive and

adaptable, thus supporting goal-based diagnoses.

2.4.6 Kanji's Business Excellence Measurement System

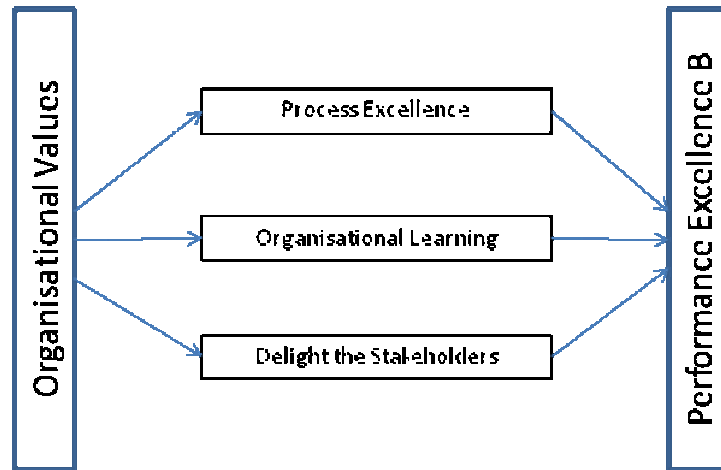
Kanji's Business Excellence Measurement system (KBEMS) is grounded on Critical Success Factors (CSFs), which correspond to the drivers of performance. By adopting conceptual model of Business Excellence, the CSFs are embedded in Kanji's Pyramid Model, in which two structural models were developed: Kanji's Business Excellence Model (KBEM: Kanji, 1998) and Kanji's Business Scorecard (KBS: Kanji & Sá, 2002). KBEM (see Figure 1) is dedicated to the measurement of performance from the internal stakeholders' perspective, whereas the KBS (see Figure 2) assesses the performance from the external stakeholder' point of view. Internal and external scores are finally combined to calculate the final organizational performance excellence index (OPI), which gives an aggregate measure of the excellence of the organization in managing all the CSFs. The KBEM and KBS should be applied simultaneously, since they form a single and complementary view of organizational performance.

Figure 2-6 Kanji's Business Excellence Model (KBEM)



Source: Kanji, G. K. (1998)

Figure 2-7 Kanji's Business Scorecard (KBS)



Source: Kanji & Sá (2002)

KBEMS has the following properties:

- The whole system is driven by leadership and organizational values.
- Internal and external stakeholders are actively involved in the assessment process.
- It supports a balanced view of performance measurement.
- It highlights communication problems and calls attention to the need for information sharing and cooperation among the various stakeholders.
- It is value-based, and has a long-term orientation and is dynamic in nature.

In KBEMS, the methodology suggested is essentially quantitative and questionnaires are the key pieces in measuring the organization's performance. The mathematical and statistical reasoning provided by KBEMS allows identifying the strengths and areas for improvement. Actually, KBEMS has shown its success in measuring performance in public and service sectors (Kanji, 2008). All these depend on a high feedback rate in information collection from questionnaires, which is not often the case when ICT SMEs tries to construct an effective performance measurement system.

2.4.7 Comparing PM frameworks: Strengths & Weaknesses

The following table (Table 2-1) compares the characteristics, premises, relative strategy, logical relationships, proposed functions, historical background, basic structure, and evaluation process of each discussed PM framework.

Chapter 2 – Performance measurement

Table 2-1 Comparing the Performance Measurement Framework

	The Balanced Scorecard	EFQM Excellence Model	The Performance Prism	MBNQA
Premises	A performance measurement system should monitor results in key areas. The managers should look at the business from four important perspectives: customer, internal, and learning & growth, financial perspective.	Excellent results with respect to Performance, Customers, People and Society are achieved through Leadership driving policy and Strategy, which is delivered through People, Partnerships and Resources, and Processes.	The performance measurement should consider both the needs of stakeholders (investors, customers, employees, suppliers, regulators and communities) and what the organisation needs from them.	Customers drive the excellence. An organization's performance measurements need to focus on key results. Results should be used to create and balance value for key stakeholders—customers, employees, stockholders, suppliers, partners, the public, and the community.
The position of strategy in the model	Provide a mechanism for translating strategy into specific objectives, measures and targets, and monitoring the implementation of the strategy. It is a mechanism for strategy implementation, not for strategy formulation.	Leadership drives Policy and Strategy, which are delivered through People, Partnership and Resources, and Processes.	The measures should not be derived from strategy. The strategy is just to ensure that the wants and needs of stakeholders are satisfied.	Leaders should ensure the creation of strategies, systems, and methods for achieving performance excellence. The values and strategies should help guide all of an organization's activities and decisions.
Logical relationship between the criteria	Cause-and-Effect: Learning and growth → internal/business process → customer → financial results	'Results' are caused by 'Enablers' and 'Enablers' are improved using feedback from 'Results'.	Strategies, process and capabilities should be in alignment with each other to deliver real value to its stakeholders. Stakeholders should contribute to organizations as well as to expect something from them.	Made up of results-oriented requirements.
Proposed functions	Provides a framework, a language, to communicate mission and strategy	The model was used in five ways: A Self-Assessment tool, A way to Benchmark with other organization, a guide to identify the improvement area, A basic way of communicating and thinking, a structure for the organization's management system.	The performance Prism is not a prescriptive measurement framework. Instead, it is a tool to help management teams to influence their thinking about what the key questions are that they want to address when seeking to manage their businesses.	To help improve organizational performance practices, capabilities, and results; To serve as a working tool for understanding and managing performance and for guiding organizational planning and opportunities for learning
Background when it	To produce a performance measurement model to overcome the	As a framework to judge the applicants for European Quality	To overcome the shortcomings in the balanced scorecard approach, the	A framework for quality and organisational self-assessments. The

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was founded	weakness of traditional financial measure.	Award when the model was founded in 1991.	Performance prism extended the view of stakeholders.	criteria for American National Quality Award.
Basic Structure (features)	The balanced scorecard measures organizational performance across four balanced perspectives: financial, customers, internal business processes, and learning and growth. The Balanced Scorecard retains an emphasis on achieving financial objectives, but also includes the performance drivers of these financial objectives.	Eight fundamental concepts, the Business Excellence model which includes 9 criteria and many EFQM frameworks for specific subject management area.	It consists of five interrelated facets: stakeholder satisfaction, strategies, processes, capabilities and stakeholder contribution.	Seven categories were divided into two triads: leadership triad and results triad.
Evolution process	In 1992, the Balanced Scorecard was introduced by Kaplan & Norton, which combined the financial and non-financial measures in a holistic framework. From 1993 to 1996, the Balanced Scorecard evolved to a strategic management system. In 2001, the 'strategy map' was introduced, which is a logical and comprehensive architecture for describing strategy.	A model was constructed based on the 9 fundamental concepts in 1991. The model had evolved to be applied in Public Sector and SMEs in 1995. To satisfy the requirement of the management in specific subject area, the model is developed to a series of frameworks since 2003.	The model is constructed based on five questions. The success and failure/risk maps were introduced in 2002.	

Source: (Kaplan and Norton 1992; Kaplan and Norton 1993; Kaplan and Norton 1996; Kaplan and Norton 2001)

(Neely and Adams 2001; Neely, Adams et al. 2001; Neely 2002; Neely, Marr et al. 2003)

The brochure of EFQM (2003); Wongrassamee, Gardiner & Simmons (2003); Malcolm Baldrige National Quality Award (2009)

Based on different background and premises, each framework has its strengths and weaknesses in the aspect of performance measurement, see Table 2-2.

Table 2-2 the Strength and Weaknesses of Performance Measurement Framework

	Strengths	Weaknesses
BSC	- o Overcomes the shortcoming of traditional performance measure - o Strong strategies management tool - o The measurement linkages of Cause-and-effect relationship help managers and employees to consider balanced key perspectives of performance.	- o Only focuses on the internal performance measurement and management. - o Does not measure external environment, cannot reflect the competitive performance - o Does not think about all of the stakeholders
EFQM	- o Measure both 'result' and 'Enabler'. - o Strong benchmarking function - o Non-prescriptive framework, adaptable. - o Can be a diagnostic tool to identify the areas that need to be improved. - o Feedback from 'Result' helps to improve 'Enabler'	- o Need lots of resource and time to conduct measurement. - o A host of dimensions that are effectively unmeasurable - o Did not provide idea about how to construct and conduct effective performance measurement
Performance Prism	- o Tests each stakeholder's wants and needs - o Consider the stakeholders' contribution to performance	- o Many indicators are not effective in practice. - o Short of logic among the measures - o There is no necessary feedback chain between the results and drives
MBNQA	- o Non-prescriptive framework, adaptable. - o Focus on result, customers and market drive performance-improvement - o Support goal-based diagnosis - o A system perspective - o Consider external environment, challenges and own profile.	- o Need a lot of resource and time to conduct measurement. - o A host of dimensions that are effectively unmeasurable - o Did not provide idea about how to construct and conduct effective performance measurement and management system.

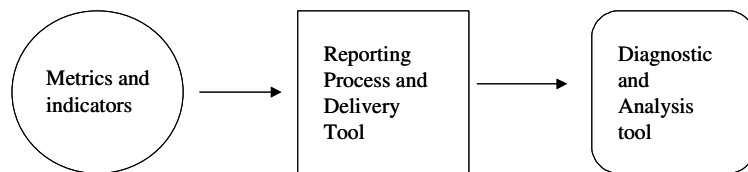
2.5 PM STURCTURE AND PROCESSES

There is an old saying: "you get what you measure" or "what gets measured gets done." It shows that performance measurement not only helps an organization gather

performance information and identify performance situations, but also serves as a powerful performance management tool, used to help executive strategy-objectives and motivate employee activities.

Hence it is necessary to discuss PM components, structure and processes. According to Tesoro and Tootson (2000), a practical performance measurement system is composed of three simple elements: (1) a set of metrics and indicators; (2) a reporting process and delivery tool; and (3) a diagnostic and analysis tool (as shown in Figure 2-8).

Figure 2-8 Three Elements of a Practical PM System



Source: Tesoro and Jack (2000)

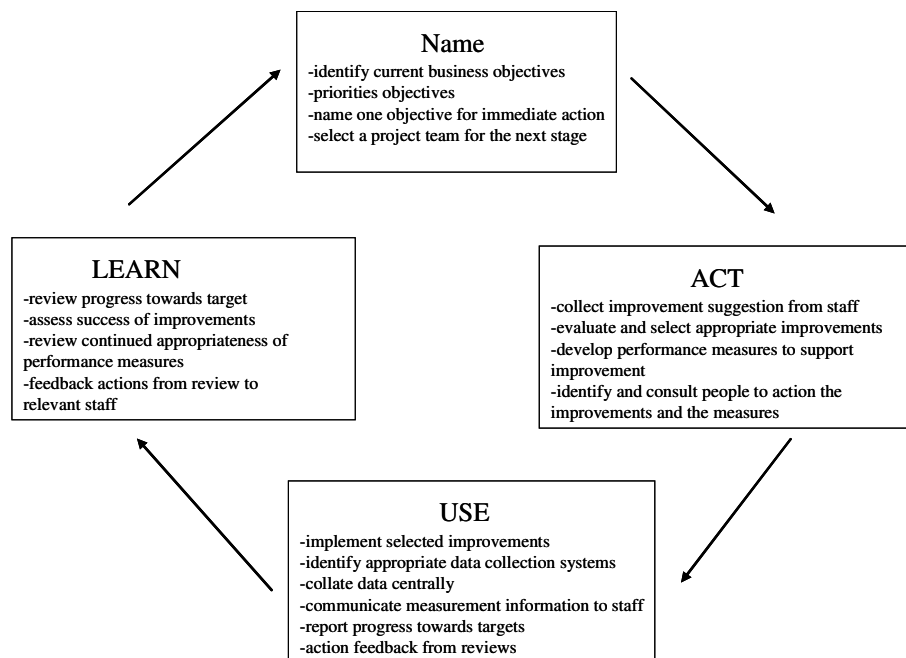
Cokins (2004) suggested a recipe for implementation of PM. The steps were shown as the following:

1. Agree on the vision, mission, and strategic internet of the enterprises, and define the strategies.
2. Define the strategic objectives that support step 1.
3. Map the interrelated strategic objectives with their cause-and-effect linkages.
4. Define initiatives to decrease the performance gap for each strategic objective, and scale back non-supportive projects.
5. Select appropriate strategic measures and cascade them to relevant parts of the organization.
6. Select the target levels for each KPI for relevant time period and identify the performance deficiency gap.
7. Collect the actual KPIs, display the scores, and compare to the targets.
8. Manage performance gaps in order to steer the organization, by interpreting and reacting to the score, then revising the established action plans.

The study of PM process for SMEs is conducted by Hudson (2001). The research reviewed a serial of PM systems that were developed during last two decades and found

that “no current approach to the development of SME PM Systems is wholly suitable for use in this sector. This is due to the level of resources required, the lack of flexibility and the orientation of these approaches.” Based on case studies, the authors comprised and recommended an original performance measurement process for SMEs (Figure 2-9). The goal of this method is to make each incremental performance quicker and more efficient by focusing resources on one strategic objective at a time; this approach enables performance measures to be updated regularly, in order to reflect strategic changes. However, this strategy is not complete – it lacks a framework that helps SMEs identify and execute strategic objectives and measurements. It does not provide the analysis tools either that help SMEs to identify what should be measured and what is needed to be improved.

Figure 2-9 Continuous Strategic Improvement Processes for SMEs



Source: Hudson (2001)

From above illustration of performance measurement structure and process, it can be identified following criteria need to be met for an organization to successfully measure its performance. First, an organization should construct an appropriate performance management infrastructure, encapsulating: performance management policy; performance management procedures; and guidelines, templates, and necessary organizational structures. Secondly, the organization should institute a performance

measurement framework that identifies performance metrics and indicators. Third, it should institute a performance-intelligence system, which includes: data collection; conveying and storage systems; data analysis; and, a results-communication system. Finally, the organization should establish a performance-results implementation system, through which actions (such as decision-making) continue to improve team-member and corporate efficiency, reward positive contributions with appropriate compensation, and properly measure performance.

In short, a PM framework:

- Formulates and plans Strategy;
- Maps interrelated strategic objectives with cause-and-effect linkages;
- Chooses correct performance metric and indicators which reflect the performance objectives;
- Collects performance information and analysis; and,
- Takes action based on performance measurement results.

2.6 SUMMARY

The concepts of performance and performance-measurement were reviewed and discussed in this chapter. The literature provides a multi-dimension approach, reflecting two main categories of performance measurement: effectiveness and efficiency.

This chapter also reviewed typical PM frameworks. The BSC approach is a strong strategy-implementation tool. Organizations can use it to clarify goals, communicate strategies, and set-up performance objectives. In contrast, non-prescriptive self-assessment frameworks (EFQM, MBNQA) are more appropriate for benchmarking an organization's performance.

CHAPTER 3 PERFORMANCE MEASUREMENT FOR SMEs IN THE ICT INDUSTRIES

3.1 INTRODUCTION

During the past two decades, SME performance has been studied by a number of scholars. Most research focuses on the analysis of performance determinants, in which critical success-factors were identified by researching relationships between input-factors and performance. This chapter builds on previous research about SME performance measurement, in which the features and requirements for PM in SMEs are generalised and discussed. Further, a conceptual PM framework for SMEs is developed.

This chapter is structured as follows: definitions of SMEs and ICT industries are introduced in section 3.2 and 3.3; the features and challenges of PM in SMEs are discussed in section 3.4; knowledge gaps are identified in section 3.5; and in section 3.6, theoretical-design approaches are discussed.

3.2 SMALL AND MEDIUM SIZED ENTERPRISES AND THEIR FEATURES

There is no strict definition of small and medium sized firms (SMEs). Different countries have different definition of SMEs (as shown in Table 3-1). The typical definition is based on categorization by the maximum number of staff and annual turnover. In Australia, ABS defines SMEs as companies with staffing between 1-199, wherein companies with 1-4 staff are termed ‘micro-company’, companies with 5-19 staff are termed ‘small company’, while those with 20-199 employees are termed ‘medium company’. In this study, the Australian definition is used (i.e. SMEs are those with 1-199 employees), as most sample cases hail from the region.

Table 3-1 Definition of SMEs

	ABS Definition	Micro	Small	Medium	SME	Large
Australia	No. of Staff	1-4	5-19	20-199	1-199	≥200
China	NBSC Definition	Varies with industry, for manufacturing:				
	No. of Staff		<300	300- 2000	1-2000	≥ 2000

Chapter 3 –PM in SMEs in the ICT industries

EU	EU Definition	Micro	Small	Medium	SME	Large
	No. of Staff	< 10	< 50	< 250	1-250	≥ 250
	Turnover	<€2 mil	<€10 mil	<€ 50 mil		
OECD	OECD	Micro	Small	Medium	SME	Large
	Definition	1-9	10-49	50-499	1-499	≥500
	No. of Staff					

Source: Temperley et al., (Temperley, Galloway et al. 2004), , UN-ECE (1996) , NBSC (2003)

Around the world, SMEs play a critical role in employment and economy. For example, in Australia, SMEs represent 97 percent of all private sector businesses and provide 49 percent of all private sector employment (ABS 2002). The typical features of SMEs are:

- SMEs' primary competitive advantage is its flexibility; they are often better able to quickly meet customer requests and needs (Temperley, Galloway et al. 2004; Antony, Kumar et al. 2005; Edwards, Delbridge et al. 2005; Murphy and Ledwith 2007).
- Innovation through a research and development program is vital to the existence of a typical SME. However, SMEs tend to focus more on incremental innovation, as opposed to radical innovation (Temperley, Galloway et al. 2004; Oke, Burke et al. 2007).
- Rapid decision-execution in order to mitigate external threats and capitalize on opportunities (Antony, Kumar et al. 2005; Murphy and Ledwith 2007).
- SMEs are less vertically-integrated than their counterparts, as there are fewer layers of management and bureaucracy. This helps SMEs simplify their management, but also brings the disadvantages that most SMEs focus on operational matters, rather than planning (Antony, Kumar et al. 2005; Deros, Yusof et al. 2006; McAdam, Keogh et al. 2007).
- The personality of an SME's chief executive officer or managing director is often a key element in the direction, growth, and success of the company; in this way, SMEs are often people-oriented (Temperley, Galloway et al. 2004).
- In SMEs, the working relationship is often loose and informal; the process is often absent of standardization (Antony, Kumar et al. 2005).
- Policy-making procedures and resource utilization that are appropriate for large companies are not necessarily appropriate for SMEs (Welsh and White 1981; Deros, Yusof et al. 2006).

3.3 INFORMATION AND COMMUNICATION TECHNOLOGY INDUSTRIES AND THEIR FEATURES

The Definition of ICT Industries

According to the Information Technology Association of America (ITAA), the information and communication technology industry (ICT) describes companies in the fields of computer hardware, software, telecommunications, Internet, e-business, e-education, computer services, and more. However, the Australian Bureau of Statistics (ABS) uses the definition promoted by the Organisation for Economic Co-operation and Development (OECD); accordingly, this study adopts the very same definition, which encompasses the following classes of ANZSIC:

- ☐ ☐ Class 2841, Computer and business machine manufacturing;
- ☐ ☐ Class 2842, Telecommunication, broadcasting and transceiving equipment manufacturing;
- ☐ ☐ Class 2849, Electronic equipment manufacturing n.e.c.;
- ☐ ☐ Class 2852, Electric cable and wire manufacturing;
- ☐ ☐ Class 4613, Computer wholesaling;
- ☐ ☐ Class 4614, Business machine wholesaling n.e.c.;
- ☐ ☐ Class 4615, Electrical and electronic equipment wholesaling n.e.c.;
- ☐ ☐ Class 7120, Telecommunication services;
- ☐ ☐ Class 7831, Data processing services;
- ☐ ☐ Class 7832, Information storage and retrieval services;
- ☐ ☐ Class 7833, Computer maintenance services; and
- ☐ ☐ Class 7834, Computer consultancy services.

Note: Cited from (ABS 2004); n.e.c. not elsewhere classified

Attributes of the ICT Industry and Typical Corporate Structures

ICT industries' rapid-growth plays an important role in many countries' economics. In fact, 8 percent of the US economic output in the year 2000 was due to ICT-related industries, while one-third of the real growth between 1995 and 1999 came from ICTs. In the Netherlands, ICTs account for about 4 percent of economic output, but 17 percent of the growth over 1996 - 98 was attributable to ICTs (Anderson 2001). Australian data

(2002 – 03) shows that ICT accounted for 4.6 percent of Australia's GDP contributing more to the economy than agriculture, forestry and fishing, and slightly less than mining. As of 30 June 2007, there were approximately 300,000 persons employed in Australia's ICT industries (ABS 2008).

Innovation is a typical feature of ICT companies, as roughly 3 to 5 percent of total expenditures are spent on R&D and innovation. Central to this cause is the development of new products and services; this is especially true within manufacturing companies and telecommunications providers. (MultimediaVictoria 2007)

Because of the features of ICT products and the industries chain, collaboration with large ICT companies, education providers, government and non-ICT companies is a typical feature. In Victoria, Australia, seventy-seven percent of surveyed ICT companies report that they are engaged in some form of collaborative activities (MultimediaVictoria 2007). Indeed, in high-technology industries, it is common for SMEs to become a part of a knowledge network to learn, adapt to technological change, and innovate, in order to overcome problems of resource and information limitations (Mohannak 2007). However, ICT firms currently tend to use more own skilled staffs rather than rely on other forms of collaboration. This lack of outsourcing and linkages is understandable, given that many entrepreneurs in the ICT industries are technological specialists without business training or experience. This certainly influences ICT SME performance, as organizational success or failure seriously affected by the managerial competencies of owner-managers (Pansiri and Temtime 2008).

Note quality analysis of information technology service level management (ITSML) is discussed in (Bigio, Edgeman et. al, 2004). The measurement of ITSML was intended to identify strengths, weaknesses, opportunities, and threats in and to ITIL Service Delivery approaches and subsequently, to enhance the performance.

3.4 SME PERFORMANCE MEASUREMENT

3.4.1 Challenges and Features of Performance Measurement in SMEs

Measuring SME performance is complex and challenging work (Brush and Vanderwerf 1992; Murphy, Trailer et al. 1996; Sapienza and Grimm 1997; Amason, Shrader et al.

2006). The challenges are usually distinct from those of large organizations and, because most existing performance measurement systems were designed for the latter, few tools are available for SMEs. The main challenges to measuring performance in SMEs are as follows:

First, collecting performance information from privately held SMEs is often difficult due to a lack of historical information and accessibility. The information is often imperfect and the accuracy is hard to be checked even if the information can be obtained. For example, traditional financial measures of performance are often unavailable (Brush and Vanderwerf 1992, Chandler and Hanks 1993; Wang and Ang 2004)

Second, financial data is difficult to interpret (Barnes, Coulton et al. 1998). This is because SMEs usually have small starting base, enormous and erratic growth rate and uneven record-keeping (Sapienza and Grimm 1997)

Third, many measures, such as future profits and survival, require a longitudinal sample-design. It is inappropriate to use such measures on an SME, however, due to the group's typically short operation-history (Brush and Vanderwerf 1992, Chandler and Hanks 1993; Wang and Ang 2004).

Fourth, financial data is often influenced by industry-related factors (Wang and Ang 2004). The performance measures for ICT SMEs present a different connotation from that for traditional industries.

Fifth, there exists possible source bias, e.g. owner/founder might manipulate the related information in propaganda (Brush and Vanderwerf 1992).

Sixth, SMEs' future and potential performance is more important than lagged-performance. This requires that performance measurement systems not only measure lagged performance, but also capture future performance (Kaplan and Norton 1992).

Seventh, Most SMEs focus on day-to-day operations. There may not be enough resources to execute comprehensive PM measurement (Stephens 2000).

Finally, the decision-making processes in SMEs are always not formalized and their strategies are often poorly planned, which influences the standard PM system employed in SMEs (Garengo, Biazzo et al. 2005).

Because of the above challenges, ICT-related PM in SMEs is often inhibited by the following:

1. Very few SMEs adopt a systematic performance measurement. Research shows that only a few SMEs with some track record in quality management had structured their measurement system (Barnes, Coulton et al. 1998). The reasons may be that the managerial structures are often simple and, as such, most SMEs do not need a complex process to support its operations.
2. Small firm leaders tend to spend more time dealing with day-to-day operational concerns (Stephens 2000). They emphasise the operational and financial performance measures rather than a balanced measurement. There are three reasons for this: (1) the leaders in SMEs are not aware of the existing integrated PM model, (2) the balance concept in SMEs might be different to that in large organizations because SMEs cannot achieve the balanced objective based on their limited resources; thus, the balanced PM in SMEs might be on a dynamic balance, and (3) there is no applicable PM model for SMEs (Wu 2006).
3. The performance measurement in SMEs is often casual. Performance indicators tend to be discrete events rather than continuous measures (Barnes, Coulton et al. 1998). They also often lack formal planning. Consequently, SMEs do not take advantage PMs to introduce strategic planning (Garengo, Biazzo et al. 2005). Instead, PMs are often used to address specific problems, short-term problems or goals (Barnes, Coulton et al. 1998).
4. An SME's future and potential performance is more important than lagged performance. This requires that the PM system not only measure lagged performance, but also capture future performance (Kaplan and Norton 1992).

Briefly, the PM characteristics in SMEs can be summarized as follows:

Non-institutional

A reliance on key performance indicators

Objective-orientated

Data usually derived from non-financial and intangible indicators

3.4.2 The Requirements of PM in SMEs in the ICT Industries

Many scholars (Storey 1994; Garengo, Biazzo et al. 2005) believe that there are three fundamental differences between small and large firms: environmental uncertainty, innovation in products and service, and sustained evolution. Hence, PM is often used in SMEs to manage uncertainty, to innovate products and services, improve their processes, and benchmark against competitors (Garengo, Biazzo et al. 2005).

For SMEs in the ICT industries, one critical success factor is related to tracking technological trends in order to formulate informed strategies. In this way, PM in SMEs not only helps to deploy and execute strategies, but also to correct current strategies. Many scholars (Bititci, Carrie et al. 1997; Bourne, Wilcox et al. 2000) agree with the opinion that a PM system should also support the definition, development, and evolution of strategy. This is a minority view, however, that contrasts with the widely accepted opinion that a PM system should derive from the company strategy; in their counter, these scholars argue that a PM system and strategy should be separated but interrelated. In SMEs in the ICT industries, developing a right business strategy is more difficult than deploying a proposed strategy. Hence, it is critical for PM to help SMEs develop its strategies.

SMEs are not typically well-endowed with resources and, as such, often cannot conduct institutional PM. In any formulation of SME PM, its engineers should align the PM's resource-demand with a business' existing, daily work-processes. It should also be balanced, tailored, flexible and dynamic; all stakeholders should be considered. Of course, this recommendation should be tempered against over-analysis and its associated costs.

In summary, the requirements of ICT-related PM in SMEs include the following key elements:

Managing uncertainty (by measuring internal and external environmental factors)
Help the innovation of products and services
Sustain evolution and change processes
Competitive measures
Developing strategy
Align with ordinary process
Balanced measures
Tailored PM system
Flexible adaptability
Dynamic adaptability

In fact, the above requirements on PM target the properties of ICT companies listed in Section 3.3. We have the following observation:

- 1) ICT industries' rapid-growth plays an important role in many countries' economics. Actually, the establishing and expansion of a large amount of small and medium sized ICT companies contribute to this growth, which also means a very competitive environment for ICT SMEs. This requires the PM to manage uncertainty in the operation of a company. The PM system should involve competitive measures for the company to assess its position in the market and show the direction to put the resource in for development of the company, i.e. helping to develop suitable strategy.
- 2) Innovation is a typical feature of ICT companies. Research and Development is the core department for many ICT SMEs that stand in market because of new products. For those SMEs supplying IT service, innovation often means improving service based on new technology. It is necessary to notice that innovation is always accompanied by adventure. Accordingly, performance measurement system should assess the innovation property, e.g. the quality of innovation, which is very important to predict the success or not of a new product or service.
- 3) To overcome problems of resource and information limitations, it is common for ICT SMEs to collaborate with large ICT companies, education providers,

government and non-ICT companies. The PM system could attributed these actions as external environmental factors and assess all the influence in a dynamic way. Actually, the limited resource also determines that the PM in ICT SMEs should be tailored and flexible, aligning with ordinary process.

- 4) Many entrepreneurs in the ICT industries are technological specialists without business training or experience. This actually could be a barrier for ICT SMEs to establish a PM system to support the operation of the company, since the owners might not realize the function of the PM. On the other hand, once a company had such a system built up (which should always be the case if the company wants to survive in the highly competitive market), the managerial competencies of owner-managers should be counted for performance assessment.

3.4.3 Performance Dimension, Measures and Indicators in SMEs

The performance measures of SMEs vary widely. Murphy, Trailer & Hill reviewed the performance dimensions and measures used in literature then examined the relationship between performance variables. Following the 1996 research, this study reviewed thirty-five published papers (from year 1997 to 2006) focusing on empirical study of SME performance. Most of these papers were published in the Journal of Business Venturing. The criteria for selecting the papers was three-fold: (1) the study should be empirical research, (2) the study should include firm performance as a dependent variable (with the word of 'performance' in the document title), and (3) the sample should be composed of new ventures or SMEs. Such a selection is based on the considerations that: (1) the authors of the paper reviewed had checked much document on how to measure a SMEs' performance when they studied the correlativity between SMEs' performance and the independent variables, (2) the indicators used in these studies come from empirical data, and (3) comprehensive performance measurement dimensions can be captured through reviewing pervious empirical studies.

Growth and profitability were found to be the two performance dimensions most frequently used in the empirical research. Table 3-2 shows the measures of performance being used.

By examining fifty-one published articles from the years 1987-1993, Murphy, Trailer et al. (1996) found little consistency in performance measurement recommendations. Their results (the frequency of measurement-criteria cited in these studies) is shown in Table 3-3.

Table 3-2 Performance Measures and Frequencies of Measures in the Articles Reviewed (years 1997 -2006)

Dimension used by researcher	Measures	Frequency	Measure	Frequency
Efficiency	Return on assets (ROA)	10	The Internal rate of return (IRR)	1
	Return on investment (ROI)	4	Earning before interest and taxes (EBIT)	1
	Return on equity (ROE)	3	The ratio of gross profits to sales	1
	Revenue per incremental change in cost (dR / dC)	1		
Growth	Growth in sales	18	Acquiring capital on a timely basis	1
	Change in employees	5	Assets growth	1
	Growth in market share (GMS)	5	the change in return on sales	1
	New product/process development	3	Profitability growth	1
	Market development	2	Increase available capital	1
	The growth of margins	1	Revenues growth	1
Profit	Net profit	12	Stock market returns	1
	Return on sales	4	Profitability relative to competitors	1
	Net profit margin	1		
Size /liquidity	Number of employees	6	Net cash flow (NCF)	2
	Gross revenues	5	Cash flow relative to competitors	1
	Sales share	3		
Success/Failure	Survival	4		
Other	The number of patents applied	4	Customer satisfaction	3
	Operating efficiency	3	Financial stability	1
	The actual return as compared to industry return (RINDU)	1	The actual return compared to business plan return (RPLAN)	1
	Initiate strategic alliances	1	1-year Sharpe's measure(e.g. the 1 year risk-adjusted stock price returns controlling for market effects)	1

Source: 35 published papers from year 1997 to 2006 on Journal Business Venturing

Table 3-3 Performance Measures and Frequencies of Measures in the Literature (years 1987- 1993)

Dimension used by researcher	Measure	Frequency	Measure	Frequency
Efficiency	Return on investment	14	Average return on assets	2
	Return on equity	9	Net sales to total capital	1
	Return on assets	9	Return on average equity	1
	Return on net worth	6	Internal rate of return	1
	Gross revenues per employee	3	Relative product costs	1
Growth	Change in sales	23	Job generation	1
	Change in employees	5	Company births	1
	Market share growth	2	Change in present value	1
	Change in net income margin	2	Number of acquisitions	1
	Change in CEO/owner compensation	2	Change in pre-tax profit	1
	Change in labour expense to revenue	1	Loan growth	1
Profit	Return on sales	11	Stock price appreciation	1
	Net profit margin	8	Price to earnings	1
	Gross profit margin	7	Respondent assessment	1
	Net profit level	5	Earnings per share	1
	Net profit from operations	5	Average return on sales	1
	Pre-tax profit	3	Average net profit margin	1
	Clients estimate of incremental profits	1	Market to book	1
Size liquidity	Sales level	13	Number of employees	5
	Cash flow level	6	Cash flow to sales	1
	Ability to fund growth	5	Inventory turnover	1
	Current ratio	2	Accounts receivable turnover	1
	Quick ratio	2	Cash flow to total debt	1
	Total asset turnover	1	Working capital to sales	1
	Cash flow to investment	1		
Success/Failure	Discontinued business	4	Operating under court order	1
	Research subjective assessment	1	No new telephone number	1
	Return on net worth	1	Salary of owner	1
	Respondent subjective assessment	1	Change in gross earnings	1
Market share	Respondent assessment	3	PIMS value	1
	Firm product sales to industry product sales	1		
Leverage	Debt to equity	2	Long-term debt to equity	1
	Times interest earned	1	Stockholders capital to total capital	1
Other	Change in employee turnover	1	Relative quality	1
	Dependence on corporate sponsor	1		

Source: Murphy, Trailer et al. (1996)

Comparing these two summaries, the performance indicators that were used in recent research are:

- (1) Growth indicators, such as sales growth, were adopted by two thirds of the research.
- (2) Efficiency indicators, such as return on assets (ROA), were used frequently. The former, however, was used more frequently in recent studies, while return on investment (ROI) was more frequently used ten years ago.
- (3) More and more intangible and subjective indicators (for example, customer satisfaction and managerial satisfaction) were used, as compared to ten years ago.

There are many arguments on whether some indicators suit small and medium companies. For example, many scholars use the indicators of ROA, ROI or ROE to measure micro and small companies' performance. But studies had shown that self-reported return measures are not entirely reliable, and it is suggested that these indicators were not appropriate for micro- and small firms (Welsh and White 1981; Chandler and Jansen 1992; Chandler and Hanks 1994).

Most SMEs in the ICT industries are service companies. In this capacity (service industries) Fitzgerald, Johnston et al. (1991) provided a results and determinants framework in their research, as shown in Table 3-4. They argue that six dimensions of performance should be measured in service-industry businesses. These six dimensions can be categorized into either the 'performance results' or 'determinants' category.

Table 3-4 Performance Measures across Six Dimensions

	Dimensions of performance	Types of measures
Results	Competitiveness	Relative market share and position Sales growth Measures of the customer base
	Financial performance	Profitability Liquidity Capital structure Market ratios
Determinants	Quality of service	Reliability Responsiveness Aesthetics/tidiness Cleanliness/tidiness Comfort Friendliness Communication Courtesy Competence Access Availability Security
	Flexibility	Volume flexibility Delivery speed flexibility Specification flexibility
	Resource utilization	Productivity Efficiency
	Innovation	Performance of the innovation process Performance of individual innovations

source: Fitzgerald, Johnston et al. (2003)

3.4.4 Theoretical Approaches to Designing PM Framework for SMEs

The issue of PM in SMEs has been addressed by different approaches. For instance, Hvolby & Thorstenson (2000) analysed the use of Balanced Score Card (BSC) and suggested that PM indicators should be prioritized because of SMEs' constrained resources. They believed that use of a very limited number of performance indicators might have further advantages. Hudson, Smart et al. (2001) compared different PM models. They found that, because SMEs always have limited resources, the administered PM should be resource-effective and dynamic.

The studies of theoretical PM approaches for SMEs

Many important studies into implementation of a variety of PMs in SMEs have been conducted. These studies address issues of PM in SMEs from different aspects.

1. The approach of Balanced Scorecard (BSC)

As discussed in Section 2.4.2, the objectives and measures employed in the BSC are derived from an organization's vision and strategy. The BSC incorporates financial and non-financial measures in one measurement system and provides executives with a

comprehensive framework that translates a company's vision and strategy into a coherent set of performance measures.

Research shows that there exist a number of problems when the BSC is applied to SMEs (Kaplan and Norton 1992; 1996; 2001; McAdam 2000). These problems can not be alleviated by simply reducing the rigor in the BSC. First, the BSC's mechanization and inflexibility does not fit the flexible environment of SMEs. The BSC focuses on a long-term measure; in contrast, SMEs are usually changing their operations in accordance with market conditions. Second, SMEs often keep a closer relationship with their customers, as compared with large organizations. A large number of employees in SMEs have direct customer contacts (McAdam 2000). Third, the processes within SMEs are much more temporal and less defined than those in large organizations. The formalized process of large organizations might restrict the rapid and spontaneous cross functional process in SMEs.

2. The approach of Business Excellence Models and Total Quality Management (TQM)

The typical business excellence models are the EFQM model, Baldrige criteria, and the Australian Business Excellence (framework). They are non-prescriptive frameworks to help guide an organization to improve its performance.

Stephens (2000) evaluated the implementation of Baldrige criteria in SMEs. His study identifies the importance of the Baldrige criteria to small firms and to what extent these criteria were used by small businesses. It surveyed management practices of business, asking leaders to rate them from least to highly important; the results showed that strategy development processes and leadership were ranked higher than other items. Usually, the operational items are ranked highly, which suggests that managers in small firms spend more time on day-to-day operations.

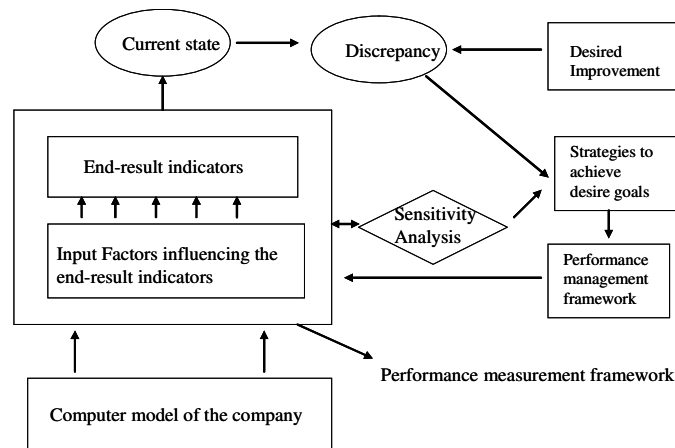
Most previous studies support the notion that management quality significantly affects a firms' performance (Lai 2003; Prajogo and Sohal 2003; Cho and Pucik 2005). Research also shows that TQM implementation leads to better product quality, regardless of the firms' size (Ahire and Golhar 1996; Chandler and McEvoy 2000). Particularly, many schools have studied the implementation of TQM in SMEs. The conclusion is that SMEs have benefited from achieving the ISO 9000 and TQM (Bayati and Taghavi 2007; Pinho 2008; Prajogo and Brown 2006). Rahman (2001) studied TQM implementation

and organizational performance of SMEs in Western Australia, with and without ISO 9000 certification. His research shows that the TQM approach based on the Australia Business Excellence (ABE) framework is a valid and reliable instrument to assess organizational performance for SMEs. However, there is no significant difference between the impacts of TQM practices on organizational performance for firms with and without ISO 9000 certification.

3. The Approach of System Theory

Jackson (2000) believes an organization can be regarded as a system, made-up of interrelated parts. Each part contributes to the system and ensures its survival and continuity. To achieve this objective, managers have to understand the various parts of their organization and the relationship between the parts and with the external environment. Based on this assumption, Ali (2003) constructed a PM model for SMEs. In his model, the performance measure is divided into two categories: end-result indicators and input factors that influence end-results indicators. A company's final PM framework is determined by sensitivity analysis. His proposed model of performance measurement based on the system dynamics approach is shown in Figure 3-1. As SMEs are more dependent on the external environments than large organizations do, such a dynamic PM model helps SMEs to measure both internal and external influence.

Figure 3-1 A PM Framework Based on System Dynamic Approach



Source: Ali (2003)

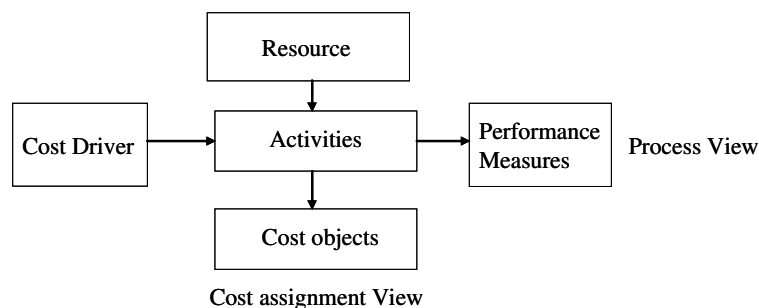
Similarly, the Australian CSIRO (Barnes, Coulton et al. 1998) introduced the Organizational Performance Measurement (OPM) system for SMEs, which is based on

the Open System Theory and Zones of Management. The OPM divides the PM into three levels: strategic, tactical, and operational. The Open Systems theory considers enterprises as systems located within a larger system-environment. It states that a dynamic relationship exists between an enterprise and its environment. Indeed, such a relationship should be considered building PM for ICT SMEs.

4. *The Approach of Activities Based on Costing (ABC)*

Activity-based costing (ABC) was introduced in the 1980s as a method to assign costs within an organization (Turney 1992). ABC assigns an organization's resource costs through activities to the products and services provided to its customers. It is generally used as a tool for understanding product and customer cost and profitability. According to Turney (1992), an ABC model has two facets (see Figure 3-2), the first of which is Cost Assignment; this assigns costs to activities and cost-objects, in order to analyse critical decisions. The second facet is the Process View, which measures and analyses information regarding workplace and employee motivation as well as work-quality. ABC helps organizations to create a PM system that supports decisions about budgeting, performance goals, employee mix, customer service, and process reengineering. Organizations can then analyse costs at multiple levels, including program-by-program, region, or department. Organizations also can improve operational control by targeting the activities, departments, or processes that are most inefficient (Gearhart 1999).

Figure 3-2 ABC Model: Cost Assignment View and Process View



Source: Turney (1992)

Laitinen (2002) introduced a dynamic integrated performance system (IPMS) for SMEs, based on the theory of ABC. His proposed IPMS consists of seven main-factors and a

causal chain connecting them. The factors are further classified into two categories: external factors (financial performance and competitiveness) and internal factors (costs, production factors, activities, products, and revenues). In the causal chain, a factor is regarded as a determinant of the factor that succeeds it. However, it is difficult to employ ABC to measure external factors. Note the external factors (in terms of financial performance and competitiveness through the profitability of the products and growth in the revenues of the products) considered in IPMS are actually lagged indicators, which, as discussed previously, are not enough for performance measurement in SMEs.

Comparing the Theoretical PM Approaches Used for SMEs

Hudson, Smart et al. (2001) developed a typology of strategic PM system-development SME-based processes. The typology has identified three related areas of PM framework design: development process, characteristics of performance measures, and dimensions of performance. Based on analysis of PM challenges, characteristics, and requirements in SMEs (see Section 3.4.1 and Section 3.4.2), they also introduce an SME PM design (see Table 3-5)

Table 3-5 Typology for PM Design for SMEs in the ICT Industries

PM requirement	PM characteristics	Dimensions of performance
• Manage uncertainty • Help innovation on product and service • Sustain evolution and change process • Competitive measures • Developing strategy • Align with ordinary process • Balanced • Tailored • Flexible adaptability • Dynamic adaptability	• Non-institutional PM system • Key performance indicators are employed • Non-financial and intangible indicators are mainly employed • Objective-orientated	• Performance results • Internal determinants • External determinants

In Section 3.4.5, issues of PM in SMEs from different approaches are addressed. To identify the strengths and weaknesses of each approach, the typology (in Table 3-5) is used as a basis for analysis. Unlike the study of Hudson, Smart et al. (Creswell 2003) who compared and analysed the different definite PM frameworks for SMEs, this study only focuses on the theoretical approaches in designing PM frameworks for SMEs. Not surprisingly, each approach has its strength and weaknesses (see Table 3-6).

Table 3-6 Comparison of Current PM Approaches against Typology of PM Framework for SMEs

<i>Typology</i> \ <i>Theoretical Approaches</i>	<i>BSC</i>	<i>EFQM & Baldrige</i>	<i>System Theory</i>	<i>ABC</i>
<i>A PM framework for SMEs should:</i>				
Manage uncertainty			√	√
Help innovation on product and service	√	√		√
Sustain evolution and change process	√	√		√
Competitive measures			√	
Develop strategy			√	
Align with ordinary process				√
Be balanced	√	√	√	
Be tailored				
Have flexible adaptability		√	√	
Have dynamic adaptability		√	√	
<i>The measures in a PM framework for SMEs should include:</i>				
Non-institutional PM system		√	√	
Key performance indicators	√	√		√
Non-financial and intangible indicators	√	√		√
Objective-orientated measures				√
<i>A PM framework for SMEs should measure:</i>				
Performance results	√	√	√	√
Internal determinants	√	√	√	√
External determinants			√	

BSC is a good strategy-management tool; it reviews the entire organization from four balanced perspectives. However, BSC is not sufficient to help SMEs because it does not examine many competitive and external factors. Furthermore, previous research has shown that BSC does not fit the flexible environment of SMEs because of BSC's inherent mechanization and inflexibility (McAdam 2000).

The EFQM Excellence Model and Baldrige criteria are non-prescriptive frameworks to help an organization to manage performance using an integrated approach. In these two frameworks, all measured criteria are divided into two groups: Enablers and Results. The logic interrelation between the 'Results' and 'Enablers' helps an organization not only analyse performance results, but also focus on the performance determinants. This structure is quite suitable for SME performance-measurement because: (1) for SMEs, performance 'enablers' are often more important than performance 'results'. Focusing on only a SME's performance 'results' often leads to incorrect conclusions, and (2) SMEs usually have a short operating history, so performance 'enablers' is more available to be measured than performance 'results'. As illustrated in Table 3-6, the

Excellence Model meets the SME PM-requirements, in that it is a dynamic, flexible, non-institutional PM system. However, falls short because it does not fully measure the competition and other external factors.

System Theory regards an organization as a holistic system. Each part of the organization contributes to the system and ensures its survival and continuity. To achieve this objective, managers should not only understand the various parts of their organization and interconnection, but also the relationship of the system to its external environment (Jackson, 2000). Table 3-6 shows that System Theory can help SMEs to set-up a dynamic and flexible PM system which can measure both internal and external information, including competitive performance. On the other hand, System Theory does not provide a feasible PM framework with which to measure performance. It needs to combine with other PM frameworks in order to construct a dynamic, flexible, and comprehensive PM framework for SMEs.

Finally, the ABC approach measures the cost of a resource used to perform organizational activities and then links the activity to the costs of the outputs. As previously discussed, ABC can hardly measure external factors for SMEs and does not provide a balanced, comprehensive performance measurement.

Comparison in Table 3-6 shows that Business Excellence models and System Theory satisfy more requirements on performance measurement for SMEs than other theoretical approaches. Further, these two theoretical approaches compensate each other in building a PM framework for SMEs. Therefore, they are employed in the later PM framework construction for ICT SMEs.

3.5 KNOWLEDGE GAPS

The previous theories leave knowledge gaps in the following ways:

1. There is no agreed viewpoint on an ideal PM model despite the development of many holistic PM models and theories over the last two decades. As stated by Meyer (2002), the issues of ‘what should we measure and what can we measure’ cannot be solved. Indeed, different PM models are derived from different

approaches to address PM, wherein every model has its strengths as well as weaknesses.

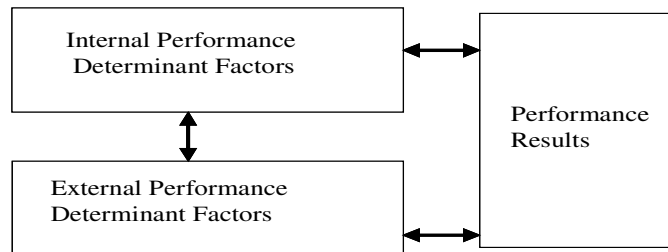
2. Most of the previous PM research on SMEs just examines one or several critical success factors that contribute to performance. There is not one unitary framework that comprehensively measures SME performance.
3. Many researchers have argued that a PM system designed for large organizations is not adaptable to SMEs. However, this issue is still very controversial. Some scholars have the opposite view. To date, no empirical studies address the issue.
4. While many empirical studies focus on PM implementation in SMEs, none answer the following questions: What is the effective performance measurement system for SMEs in the ICT industries? How can SMEs implement the performance measurement system successfully?

3.6 DEVELOPING A CONCEPTUAL FRAMEWORK FOR PRESENT STUDY

3.6.1 The Key Areas of PM Measurement

Performance results and drivers (enablers or determinants) should be part of any comprehensive PM framework, and are in most cases. For example: BSC measures three divisions from customer perspective, internal perspective, and innovation and learning perspective, in addition to financial data; the model by Brignall, Fitzgerald et al. (1991) – the Results and Determinants Framework – states that performance results and determinants should be both involved in PM framework; and, the Business Excellence models (EFQM, MBNQA and ABE) each divide performance criteria into two groups – enablers and results. Furthermore, both lagged and leading indicators should be measured (the latter of which is the most important). Finally, an SME's sustainable development and its long-term outcome should be considered in the measurement.

SMEs should measure both performance results and performance determinants, i.e. performance drivers. Therefore, three main interrelated components – (1) internal performance determinant factors, (2) external performance determinant factors, and (3) performance results (as shown Figure 3-3) – should be used.

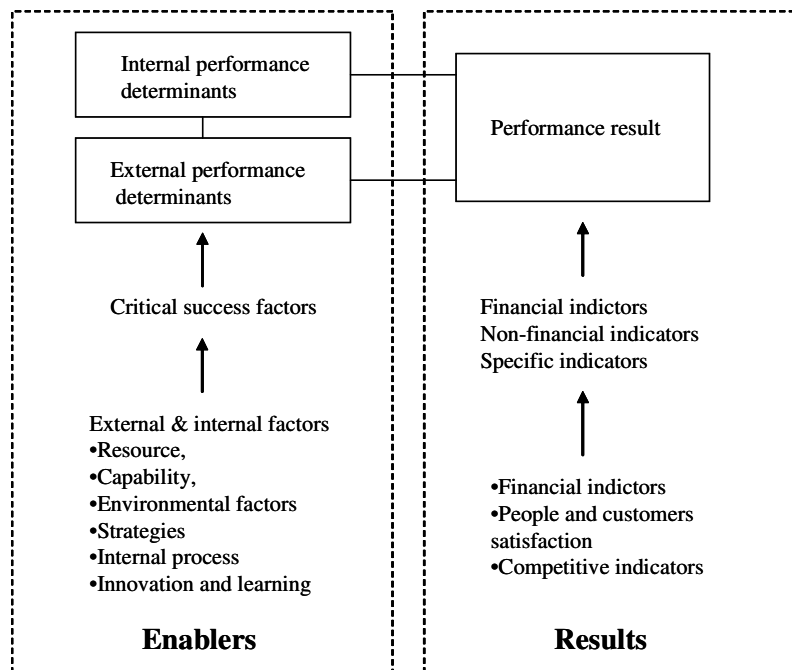
Figure 3-3 the Components of Performance in SMEs

Source: Wu and Zhao (2008)

3.6.2 The Conceptual Framework and Propositions for This Study

This study has found that Business Excellence models and System Theory are the most appropriate theoretical approaches. The typical feature of excellence models is dividing measuring criteria into two groups: performance enablers and performance result.

In System Theory (Jackson 2000), companies are simply a collection of smaller divisions/components. Furthermore, most ICT SMEs are components of a larger product-chain (for example, they are often service or design firms). Therefore, to best compete in the market; these companies must measure external as well as internal factors. This is demonstrated in Figure 3-4.

Figure 3-4 the Conceptual Framework for Present Study

The following propositions are developed in this study:

- Internal and external performance determinants, in addition to performance results, should be measured.
- Internal factors include SME resources; capability development; strategic objective formulation; internal process management; and, innovation and performance management. External factors include: environmental factors – including the influences from customers; strategic partners; competitors; and, regulation.
- SME-performance results should be measured in terms of both financial and non-financial indicators, customer satisfaction, and other competitor-oriented factors.
- SME performance depends on whether the company can adopt appropriate strategies in order to best align the internal/external resources with its objectives.

Among the internal and external factors, the following are considered in this paper's proposed PM methodology:

1. Capability

This refers to 'organizational capability', i.e. it not only considers the management team's static ability, but also measures whether the personnel ability can be transformed into the organizational capability.

2. Resources

Resources can constrain development and expansion. As such, an enterprise must efficiently manage human, financial, social (government or community support), technological, and physical resources. To do so, an SME can answer the following three questions: (1) What resource is needed to achieve the proposed performance results? (2) What resource can be obtained? and, (3) Was the resource utilized correctly?

3. Environment

Compared with large organizations, SME-survival depends more on business conditions. Measuring business environments is therefore necessary. To do so, SMEs should measure the competitive, technological, industrial, economic, and social environment.

4. Strategies and process

SME strategies are derived from an enterprise's vision and objectives. They are also decided by the enterprise's capability, resources, external environment, and these factors' interaction.

5. Performance results and objectives

Performance results are the outcomes of past activities. Performance objectives are the expected performance targets based on the current situation. In SMEs, the performance results and objectives cannot be mechanized; rather, they should be flexible. The key performance indicators vary according to specific SMEs. Generally, SMEs in the ICT industries prefer growth indicator and technological progress indicators for measuring performance.

Based on the conceptual framework, two research tasks were carried out in this study: (1) investigating the critical success factors in SMEs in the ICT industries to identify the internal and external performance determinants, and (2) studying the implementation of performance indicators in existing SMEs, to decide what indicators should be included in the PM framework. Furthermore, the implementation of PM was investigated in this study to suggest an appropriate PM solution for SMEs in the ICT industries.

3.7 SUMMARY

This chapter reviews the definitions of SMEs and the ICT industries. The definition of SMEs is adopted from the ABS guideline – i.e. companies with fewer than 199 employees. The definition of ICT is adopted from OECD (which describes information and communication technology related goods and services), which is also used by ABS.

The features of SMEs and ICT industries were discussed in this chapter. It is significant for an ICT-related SME to be innovative and collaborative with other companies. One should take these features into account while building PM framework for SMEs in the ICT industries.

SMEs' PM characteristics and requirements were discussed. PM systems help SMEs manage uncertainty, innovation, measure competition, formulate strategy, and guide their daily processes. An SME-specific PM system should be balanced, tailored, flexible, and dynamic. This chapter also reviewed the performance dimensions and indicators used by previous empirical studies. The results show that different scholars have advocated varying performance indicators when they conduct research.

The past and present research on PM in SMEs was also reviewed. A typology was developed, against which theoretical PM approaches were compared. This study concluded that Business Excellence frameworks and System Theory are best suited for SMEs in the ICT industry.

In the next chapter, this study's research methodology is presented.

PART II METHODOLOGY

CHAPTER 4 METHODOLOGY

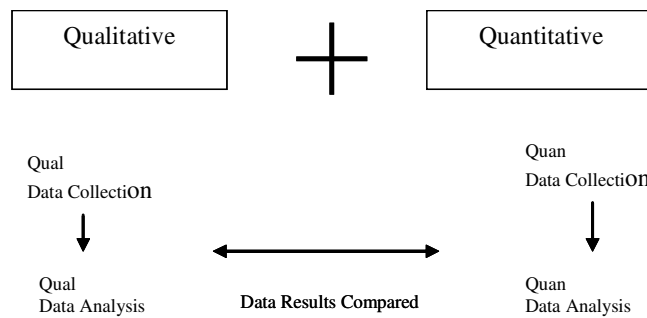
4.1 INTRODUCTION

According to Creswell, qualitative research is exploratory and useful when important variables are not examined (Creswell 1994; 2003). In contrast, quantitative approaches emphasize measurement and analysis of the causal relationship between variables, as opposed to processes (Denzin and Lincoln 2003). This research studies the issues of performance measurement and management in SMEs in the ICT industries; the main objective is to construct a practical PM framework. Literature in Chapter 2 and Chapter 3 shows that most of the related researchers focus their study on PM in large organizations. There is very little research; however, that addresses the issue of SME performance, especially from a holistic perspective. A combination of qualitative and quantitative methodologies can draw on the strengths and minimise the weaknesses of each research paradigm (Johnson and Onwuegbuzie 2004). Specifically, a qualitative approach is a feasible when one needs to identify performance measurements for SMEs. On the other hand, the qualitative approach alone has limited value. Hence, a mixed research method is employed in this research, which benefits from both qualitative and quantitative approaches.

The chapter is organized as follows. The study's research strategy is discussed in section 4.2; research methods, data collection, and analysis are presented in section 4.3-4.6; the study's reliability and validation are discussed in section 4.7; finally, ethics issues associated are discussed in the chapter's final section.

4.2 RESEARCH STRATEGY

For a mixed research approach, six research strategies can be applied (Creswell 2003) : sequential explanatory strategy; sequential exploratory strategy; sequential transformative strategy; concurrent triangulation strategy; concurrent nested strategy; and, concurrent transformative strategy. In this research, the concurrent triangulation strategy (see Figure 4-1) was used. This strategy helps researcher use different methods in an attempt to confirm, cross-validate, or corroborate findings within a study (Creswell 2003).

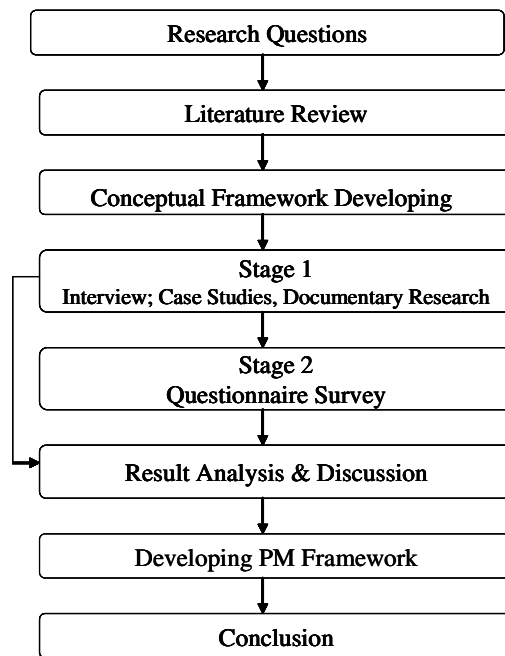
Figure 4-1 Concurrent Triangulation Strategy

Source: Creswell (2003)

Concurrent Triangulation Strategy minimizes the inherent weakness within one method by strengthening others; this is done by using separate quantitative and qualitative methods. According to Creswell (2003), this strategy is advantageous because researchers are familiar with it and its findings can be well-validated and substantiated. However, the limitations of this strategy are: (1) it requires great effort and expertise to adequately study a phenomenon; (2) it is difficult to compare the results of two analyses using data of different forms; and, (3) it is difficult to resolve discrepancies that arise in the results.

Ideally, the priority would be equal between the qualitative and quantitative methods, but in practical applications priority may be given to either the qualitative or quantitative approach (Creswell 2003). In this research, the qualitative approach was undertaken prior to the quantitative approach. The research design consists of the following procedure (see Figure 4-2):

Figure 4-2 Research Procedures



Research Questions

To construct a PM framework for SMEs in the ICT industries, the following fundamental research questions were answered:

- What is the effective performance measurement system for SMEs in the ICT industries?
- How can SMEs implement the performance measurement system successfully?

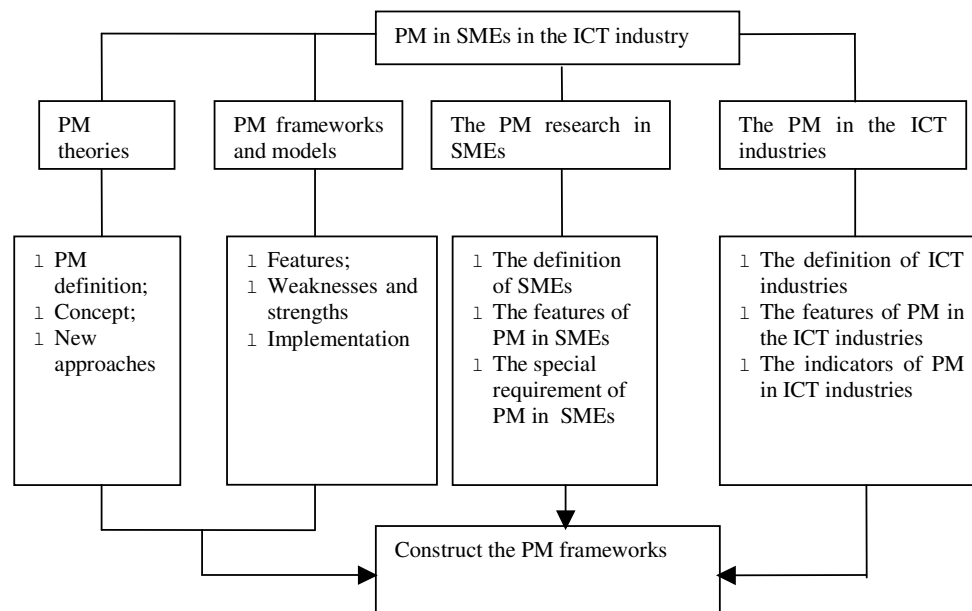
To address the two fundamental questions, the following were investigated:

1. What are the knowledge gaps in the current literature in regard to performance measurement for SMEs in the ICT industries?
2. What are the weaknesses and problems of the current PM models when applied in SMEs in the ICT industries?
3. How can SMEs design an effective PM framework?
4. What are the critical success factors that drive SME performance?
5. What measures and indicators should be measured?
6. How can SMEs successfully measure and manage their performance?

Literature Review

The literature review in this research concentrates on the following aspects – firstly, an overview of PM theories and frameworks; secondly, the practice of the PM theories, frameworks and models; thirdly, the research on PM in SMEs; fourth, the research about the characteristics and methods of the PM in the ICT industries. The literature review map is shown in Figure 4-3.

Figure 4-3 Literature Review Map for Developing a PM Framework



Conceptual Framework Developing

A conceptual framework is developed for this study and is used as the research framework (see Section 3.6.2). This is based on the review of the related PM theories and frameworks by comparing the features of current PM implementation and requirements.

Stage 1

Stage 1 focuses on studying: the performance determinants; the measures and PM frameworks that employed by SMEs; and, the implementation of PM in SMEs, through qualitative approaches. At this stage, a qualitative data collection method is used.

Semi-structured interviews:

20 semi-structured interviews were conducted with 16 SMEs in ICT and 3 venture capital companies whose main investment area is in ICT. The interviewees from ICT SMEs include 6 CEOs, 1 Founder-manager, 1 CTO, 1 General Manager, 2 R&D Project Managers, 2 Sales Managers, and 4 Managing Directors. Among these 20 interviews, 14 are face-to-face while the others were conducted over the telephone.

Documentary studies:

After interviews, relevant PM data and documents was compiled and studied. The documents include PM policies, procedures, working temperate, and measurement criteria.

Multiple Case studies:

Executives (CEO and/or Managing Director) were interviewed based on semi-structured questions. After the interviews, to get more detailed information on PM implementation, a questionnaire was completed by the executives. Further, the relevant PM documents in the case companies were studied.

At qualitative investigation stage, the following objectives were achieved:

- o Understanding the current implementation of PM in SMEs in the ICT industries;
- o Identifying the critical success factors that impact SME performance, i.e. performance determinants;
- o Studying PM metrics that SMEs employed;
- o Investigating the requirements, the problems, and barriers facing PM implementation.

Stage 2

At this stage, a questionnaire-survey was conducted. It was a two-step process: Questionnaire Survey I was given to the interviewed-company representatives – it was designed to collect comprehensive data on PM implementation and to compare the importance of the critical success factors that were drawn-out from interviews; and, Questionnaire Survey II was sent to 1,863 SMEs in the ICT industry, each of which were located in Victoria, Australia – this was done to identify and verify important PM variables in SMEs

Results Analysis and Discussion

The data obtained from Stage 1 and Stage 2 were compiled and analysed.

PM Framework Development

Based on above data analysis, the researcher built a practical PM framework for SMEs in the ICT industry.

Conclusion

The research results were summarized in conclusion, which also provides the suggestion for future research in this area.

4.3 TRIANGULATION

Triangulation refers to a method that answers research questions in more than one way (Collis and Hussey 2003); it was broadly defined by Denzin (1978) as “the combination of methodologies in the study of the same phenomenon.” Triangulation can help researchers improve the accuracy of their judgments by collecting different kinds of data bearing on the same phenomenon and reduce the impact of potential biases that can exist in a single method (Jick 1979). Triangulation can be used in both qualitative and quantitative research. Scholars suggest that accepted triangulation protocols include: data source triangulation; investigator triangulation; theory triangulation; and, methodological triangulation (Denzin 1989; Stake 1995).

There are two types of triangulation in this research. As mentioned in previous section, this study employed methodological triangulation – concurrent triangulation strategy, which mixes both qualitative and quantitative approaches. At the same time, data-source triangulation was applied in this study. In qualitative research, the data are derived from interviews and documentary studies. In quantitative research, questionnaire surveys were employed.

There are some challenging issues in designing a multi-triangulated research study (Creswell 1994). For example, researchers must locate a common subject of analysis to which multiple methods can be applied, and deal with the discrepancies that arise in the results and compare the results of two analyses using data of different forms. To address

these challenges, the topic was divided into two sub-topics: the drivers of performance (performance determinants) and the implementation of PM.

4.4 CASE STUDY

A case study is an appropriate strategy when the research tries to answer ‘how’ and ‘why’ questions (Coffey and Atkinson 1996).

The essence of a case study, the central tendency among all types of case study, is that it tries to illuminate a decision or set of decisions: why they were taken, how they were implemented, and with what result (Schramm 1971; Yin 2003).

This study addresses the following research topic: “How can SMEs in the ICT industries implement PM system successfully?” To do so, case studies were used to answer the following related questions: 1) how do SMEs implement PM? 2) Why do SMEs employ PM?; 3) What are the results?; and, 4) What are the experience and lessons learned from PM implementation in SMEs?

Yin (2003) has suggested that six sources be used to collect evidence in case studies: documentation, archival records, interviews, direct observation, participant observation, and physical artefacts. In this study, the researcher mainly collects data from interviews and documentation. Senior executives (Chief Executive Officer, Managing Director and so on) have been interviewed based on semi-structured questions. After the interviews, questionnaires were completed by the executives. Following, the researcher compiled this data in order to analyse it with collected document-sourced company data.

4.5 DATA COLLECTION

In this study, a triangulation method of data collection was employed.

4.5.1 Sampling Procedure

Different sampling techniques were applied to the qualitative and quantitative stages of this study. According to Patton (2002), quantitative methods typically depend on larger samples, selected randomly. In the qualitative research stage, purposeful sampling focuses on selecting information-rich cases whose study will illuminate the questions

under study (Patton 2002). It typically focuses in-depth on relatively small samples, even single cases, which have been purposefully selected. In this research, well-rounded cases were used, i.e. the samples are information-rich cases that manifest the phenomenon intensely.

The samples for the present study are selected from Victoria, Australia, and Southern China, which include both SMEs that have successfully employed a PM system and those that have faced problems. The main reason for selecting these companies stems from the researcher's familiarity and networked connections, i.e. he has worked with all of them in the past and has adequate access to data. The interviewed companies were selected for two additional reasons: (1) They were apart of the BRW Fast 100 from 2003 to 2006, which recognises some of Australia's most innovative, smart and fast growing small and medium businesses – among them, 63 are in the information technology sector, especially those located in Victoria, Australia; and, (2) The intensity of their PM implementation.

In quantitative research, the questionnaire survey sample was drawn from SMEs in the ICT industries (ANZSIC: 2841, 2842, 2849, 2852, 4613, 4614, 4615, 7120, 7831...34) in Victoria, Australia. These industries were selected based on the definition of ICT industries by ABS (Australian Bureau of Statistics). The present sample was from a database "Directory of IT companies in Victoria, Australia" developed by Victoria-online (State Government of Victoria).

4.5.2 Documentary Research

According to Yin (2003), documentary data has following advantages, in that it:

- Is stable, and can be repeatedly reviewed;
- Is unobtrusive, i.e. the results are not created for the purpose of the research;
- Is exact, i.e. data collected contains exact names, references and details of events;
- Has broad coverage, i.e. information spans a long period of time, which includes many events and many settings
- Corroborates and augments evidence from other sources

The weaknesses of documentation include: irretrievability; biased selectivity; reporting bias; and, access difficulties (Creswell 2003; Yin 2003). To address these weaknesses, this study collects documentary data from multiple sources. The publicly available information has been collected besides the documents from case companies.

The following documents were collected from the interviewed companies: (1) the performance policies; (2) performance record files; and, (3) working template. Besides the above documents, publicly-available information about the interviewed companies and ICT industries – such as ICT industries survey report by states government and reports about interviewed companies contained in newspapers, magazines, and internet websites – were also collected.

4.5.3 Interviews

Interviews can be conducted by face-to-face, by telephone, or in a group. Based on Creswell (2003), advantages of interviews include:

- Useful when participants cannot be observed directly
- Participants can provide historical information
- Allows researcher ‘control’ over the line of questioning

Furthermore, Oppenheim (1992) mentions that interviews can help research become rich, as spontaneous answers from participants and open-ended questions probe for more data. In addition, interviews often have higher response rate than mailed questionnaires and gives the researcher an opportunity to explain the purpose of the study.

There are also a number of disadvantages when researcher uses interview to collect data. It provides ‘indirect’ information to the participants, as filtered through the interview process and provides information in a designated method rather than the natural field setting. Participants may become bias because of the presence of researcher. Participates may have different ways of presenting their ideas (Creswell 2003). Furthermore, interviewees might face ethical concerns or societal mores, thus changing their answers in a face-to-face dialogue; for example, an interviewee might be upset by

some of the questions and the interview may have to be abandoned (Oppenheim 1992). To address the disadvantage of interviews, the interview themes and issues were mailed to interviewees before the interviews were started. All interviewees are senior managers in each company. To minimize bias, the research proposal was transcribed in plain language, sent to each participant before the interview.

This study employed semi-structured interview, which obtains more information than focused ‘structured’ interview does (Denzin and Lincoln 2005). A semi-structured interview does not restrict the answer to each question (to just ‘Yes’ or ‘No’). Instead, it allows the interviewees have more space to describe their thoughts; hence, it broadens the data that can be collected.

Interview Schedule

Based on literature review, open-response questions were constructed to obtain the data. The interview schedule contains 14 questions (see Appendix A). Following, themes and issues were covered within the one-hour discussion:

Table 4-1 Themes and Issues Studied Using Qualitative Approach

Theme	Issues
Business details	• The company’s size; business areas; employee number; business history.
Performance determinants	• The factors that impact the company’s performance; • The importance of strategies, capabilities, resources, environment, internal process, performance measurement and innovation.
The implementation of PM	• The most important performance indicators • The requirement and problems of PM implementation • The performance measurement model that is being used
Suggestions on implementation of PM	• Modifying the performance measurement • Critical success factors in implementing PM • The key components that need to be involved in PM system

In the interviews, the following were investigated: critical success factors, the current PM – which includes the PM structures, the key performance indicators employed, and the weakness of the performance measurement – and PM modification. Interview questions were grouped into three sections (see Appendix A). The first section was about the company’s background and the basic information in implementing performance measurement and management. The second section was about the performance determinants that impact the company’s performance. The third section was about the issues in implementing performance measurement and management.

In the first section, the following information has been collected: the company's employee number, main business area and history, and the PM framework the company employed. To study the performance determinants in SMEs, an open-question was first asked in the second section. In addition, based on the conceptual framework, the influence of eight factors on SME performance was investigated. Hence, the following questions have been addressed in the second section:

1. What are the key factors that impact your company's performance?
2. How does strategy influence your company's performance?
3. What are your company's core competences? In your opinion, how does one company's capability influence the company's performance?
4. What are the resources your company needs? Would the lack of resources influence the company's performance? How does one overcome this issue?
5. How does the business environment influence your firm's performance?
6. How does the internal process influence your company's performance?

In the last section, the implementation of PM in the case companies were investigated (themes are located in Table 4-1). In the third section, the following questions were asked:

1. What are the key financial indicators and non-financial indicators selected to measure the performance in your company?
2. In your opinion, what are the most important performance indicators to measure a company's performance?
3. What is the main performance measurement model being used in your company?
4. What are the weaknesses and problems of the performance measurement framework currently being used in your company?
5. In your opinion, what is an effective performance measurement framework?
6. How does one modify the performance measurement framework in your company?

Invitation Process & Participant Information

As mentioned above, senior executives from SMEs in the ICT industries were targeted for participation in this research. The research proposal and invitation letter were sent to sample companies by email. The targeted participants were Chief Executive Officers (CEOs), Managing Directors (MDs), and other senior managers (senior human resource managers, chief technological officer, senior sales managers). Fourteen face-to-face interviews were arranged at the participants' workplace for their convenience in Victoria, Australia and South China; six telephone interviews were conducted when face-to-face interview were difficult to conduct. The sample companies include different sized SMEs at different developing stages in the ICT industries. The participants' information is shown in Table 4-2. All interviewees are senior executives in SMEs in the ICT industry or senior managers from venture capital companies whose investments are mainly focused in the ICT industry. The total sample was comprised of 6 CEOs, 1 Founder-manager, 1 CTO, 1 General manager, 2 R&D Project Managers, 2 Sales Managers, 4 Managing Directors, and 3 Senior Managers from venture capital companies.

Table 4-2 Companies and People Interviewed

	Company	Main Business Activities	Interviewees
1	Company A	Internet Security Company	CEO and CTO
2	Company B	Education software design and consulting	CEO
3	Company C	Electronic design and consulting	CEO
4	Company D	System control and telecommunication solution	Founder Manager
5	Company E	Online job seeking company	General Manager
6	Company F	Telecommunication products research	Project Manager
7	Company G	Telecommunication products research	Sale Manager
8	Company H	Telecommunication design	CEO
9	Company I	IT technical consulting, planning, mentoring and training	Sales Manager
10	Company J	IT Technical consulting and implementation services	Managing Director
11	Company K	Electronics company	Managing Director
12	Company L	Online marketing service company	CEO
13	Company M	Electronics manufacturer specializing in telecommunication power	Managing Director
14	Company N	Telecommunication R & D company in 3G	Project Manager
15	Company O	Software designer and consultant company	Managing Director and founder
16	Company P	Online cards selling	CEO
17	Company Q	VC and Consulting	Director in research and consult department
18	Company R	Investment	Senior Investment manager
19	Company S	VC and finance consulting	Investment manager

The interviews took 45 minutes to 2 hours to complete, each of which was recorded by digital recorder (given the permission from the participants). Once an interview was

conducted, a transcript of the interview was written. The interviews were conducted between November 2006 and October 2007.

4.5.4 Questionnaire Survey

For quantitative data collection, the questionnaire was sent to proposed participants in the sample companies by post and email as well. The following process was observed:

Stage 1: the questionnaire survey was conducted in the interviewed companies. After interviews and documentary studies, a questionnaire was completed by the executives in the company.

Stage 2: The questionnaires were sent to SMEs in the ICT industry in Victoria, Australia by email and posted mail. Two-pronged strategy was adopted to ensure the maximum participation of the intended participants as well the convenience of providing choices/options (both on-line and hard copy questionnaire) to survey participants. It was also anticipated that it would help in avoiding or reducing the likelihood of a situation where the invitation for participation through email or post could be mistaken as a junk mail by the server. Therefore, a printed questionnaire was also posted to the senior managers' postal address.

The Questionnaire

The questionnaire developed for this study is composed of four sections (see Appendix B). The first section (four questions) is about the background of the responding company, which includes the information about the company's number of employees, the company attributes, and years in its business area, and the respondent's position in the company.

The second section consists in five questions about the implementation of PM in the company, which includes following aspects:

- the kind(s) of performance measurement that is (are) implemented in the company
- the performance measurement framework and tools that are used in the company
- the primary reasons for the company to implement the performance measurement system

- the barriers of implementation of performance measurement in the company
- the first three significant problems faced when the company implemented its performance measurement system

In the above five questions, the first four questions are multi-choice. Respondents can also provide other information. The last question is open-ended. Respondents are required to fill the content based on their own company's operation. Answering every question before submitting the feedback is not mandatory since some items are not applicable in some SMEs (if they did not employ a performance measurement system).

The third section refers to performance measures, which consist in 56 statements on the company's performance practices to identify the importance of performance determinants. The respondents were asked to rate the degree of importance of each factor according to five-point Likert scale from 1 (low importance) to 5 (high importance). The 56 performance determinants cover the aspects in Table 4-3.

Table 4-3 the Important Performance Determinants in SMEs

Capability	Response to customer Manage people and resource Flexible to external environment
Resource	Capital Human resource Management system Technological resource Enterprise culture
Strategy	The process of formula strategies Executing strategy
Internal process	Key process Support process
Performance measurement and management	Objective setting and deploying Implementation of PM tools and framework
Innovation and learning	Investment in NPD Employee training Management innovation

The fourth section consists of 20 statements about the company's performance results. Respondents were asked to rate the importance level, according to a five-point Likert scale from 1 (low importance) to 5 (high importance). These 20 indicators of performance results include financial indicators, customer satisfaction, and employee satisfaction.

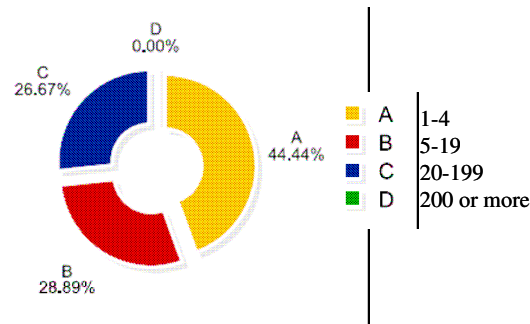
The Response

The questionnaire was initially pre-tested on a small pilot sample of 46 SMEs. Based on the result of pilot survey, the revised questionnaire was emailed to 1863 ICT-related

SMEs through online survey, resulting in 45 returns. Excepting 421 sending failures, the response rate is 3.12 percent. There are many reasons for the low response rate in this study. The three main reasons are: (1) To attain valuable data, the participants were restricted to be senior managers in the companies. Most senior managers are very busy. It is hard for them to spend time on filling a questionnaire for academic research; (2) The invitation was sent to participants by email, which is likely to be ignored by receivers by taking it as a junk mail; and, (3) Many SMEs do not employ performance measurement. Due to this, it is difficult for them to fill such a questionnaire.

Figure 4-4 shows the breakdown of respondent companies by number of employees. Among the returns, 44.44 percent companies were micro-companies, 28.89 percent belong to a small-company categorization, whilst 26.67 percent were medium companies, according to the definition of SMEs used in Australia (ABS 2002).

Figure 4-4 Number of Employee in the Sample Companies



The business-type of respondent companies is shown in Table 4-4. Of the 45 companies that participated in the survey, 41 companies (91.1 percent) are service companies. 2 companies (4.4 percent) are ICT retailers, one is a manufacturer, and one a research and development (R&D) company.

Table 4-4 the Business Type of Respondent Companies

Type of the business	Percentage	Responses
Manufacture	2.2	1
Service	91.1	41
R&D	2.2	1
Whole sales	0.0	0
Retailer	4.4	2

Figure 4-5 shows the operating length of each business, in years, of each respondent company. Figure 4-6 shows that all respondents are senior executives in their companies.

Figure 4-5 the Years in the Business in the Respondent Companies

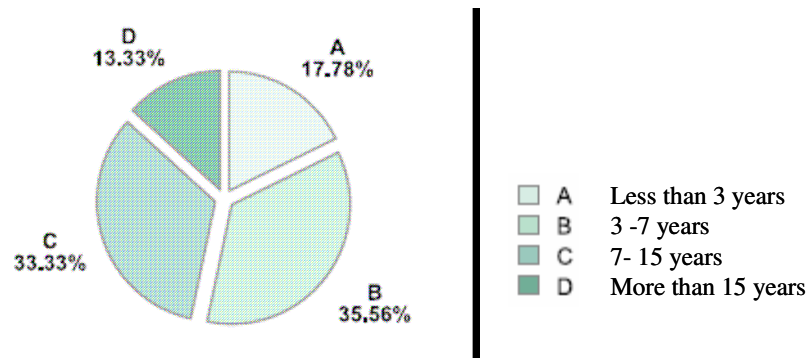
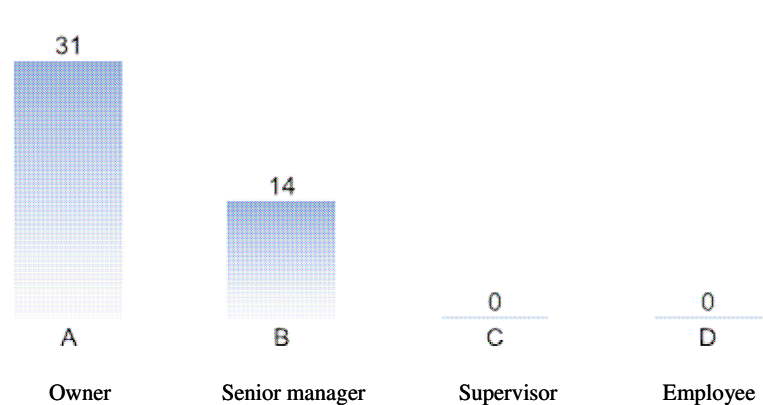


Figure 4-6 Job Position of Respondents



4.6 DATA ANALYSIS

The data analysis includes qualitative data analysis and quantitative data analysis. The interview transcripts were entered into the software tool NVivo. The quantitative data was analysed using SPSS package v13.0 (descriptive statistics).

The characteristic of qualitative analysis is that it deals with data presented in words. The aim of qualitative data analysis is to transform and interpret qualitative data in a rigorous and scholarly manner (Sarantakos 2005). In the research, the qualitative data analysis follows the strategy of *analytic induction*, e.g. constructing abstract concepts from a study of specific data (Joseph, Anderson et al. 1995). Analytic induction refers to

examining of similarities between various social phenomena in order to develop concepts or ideas through a systematic approach. Analytic induction allows for modification of social concepts and their relationships throughout the process of doing research (Ragin 1994). The analysis process includes:

1. Define and describe the topic of PM in the SMEs in the ICT industries.
2. Examine raw data produced through the study.
3. Formulate a working hypothesis from following seven categories: (1) the characteristics of PM, (2) the challenges of PM, (3) the requirement of PM model, (4) the performance determinants, (5) the importance of PM factors and indicators, and (6) the implementation of PM framework.
4. Examine the specific cases to confirm, redefine, and revise the hypothesis of issues of PM in SMEs in the ICT industries.
5. Continue comparing theory against cases until the analysis leads to a theoretic PM framework for SMEs in the ICT industries.

4.6.1 Content Analysis

Content analysis is a research technique for making replicable and valid inference from texts (or other meaningful matter) to the contexts of their use. Neuendorf (2002) provided a six-part definition of content analysis:

"Content analysis is a summarizing, quantitative analysis of messages that relies on the scientific method (including attention to objectivity-intersubjectivity, a priori design, reliability, validity, generalizability, replicability, and hypothesis testing) and is not limited as to the types of variables that may be measured or the context in which the messages are created or presented."

As a research technique, content analysis provides new insights, increases a researcher's understanding of particular phenomena, and informs practical actions (Krippendorff 2004). In this study, content analysis is used to answer the research questions four and five (see Section 4.2). To study the measurement dimensions and indicators of SME-performance, thirty-five published papers from year 1997 to 2006 were reviewed and analysed. In addition, data documents from each company (wherein at least one officer was interviewed) were analysed employing content analysis. Weber (1990) notes: "To make valid inferences from the text, it is important that the classification procedure be

reliable in the sense of being consistent: Different people should code the same text in the same way". To make the study most valid, the content analysis is based on the conceptual framework in Chapter 3.

4.6.2 Cross-case Analysis

The interview data was analysed based on the conceptual framework discussed in Figure 3-4; this was coupled with one cross-case analysis, which was employed to analyse the 16 case studies. The interview data across all of the cases was analysed, to identify the similarities and the differences at the aspects of PM determinants and PM implementation. According to Yin (2003), studying multiple cases makes it possible to build a logical chain of evidence. The cross-case analysis is applied to seek the evidence for the framework, to generalize and analyse the factors that impact SMEs' performance, and to identify how SMEs in the ICT industries implement PM based on the framework. The case companies are first classified into four groups, based on their performance. Then the critical success factors that impact SME-performance is analysed, based on the conceptual framework. The analysis results are located in Chapter 5, after which PM implementation in the case companies is discussed in Chapter 6.

As previously mentioned, the interviewed companies were divided into four groups (shown in Table 4-5). The classifying criterion is based on the executives' comments on their own firms and the public performance report on BRW.

- Excellent performance companies (Case code: EP01 to EP05): This includes the companies that have achieved good performance during last several years. They have been listed in Australia BRW fast 50 or have achieved excellent performance.
- Mediocre performance companies (Case code: MP01 to MP04): this includes the sample companies whose development is normal.
- Temporary frustration companies (Case code: TF01 to TF03): This includes those companies that have achieved good performance before but are now facing temporary frustration. These kinds of companies usually have constricted its business or reorganized their business or products.

- New venture (Case code: NV01 to NV04): This includes the companies that just start their business. They are on the stage of new products development without profit or with low profit during the first 5 years, which makes it hard to identify their performance good or bad from financial and tangible indicators.

Table 4-5 A Summary of Case Companies, Interviews and Data Collection Instruments

Case code	Field of business	Business size (employees)	Role of interviewees	Research instruments
EP01	System control and telecommunication solution	100	Director & Founder	I, Q
EP02	Online job seeking	50	General Manager	I, Q
EP03	IT consulting and training, software system	65	MD & Founder	I, Q
EP04	IT service and consulting	50	Sale manager	I
EP05	Telecommunication designer	47	CEO	I, Q, D
MP01	Electronics manufacturer	95	CEO, MD & HR manager	I, Q, D
MP02	IT power	220	MD	I, Q, D
MP03	Telecommunication R & D	100	Project manager	I
MP04	Software designer and consultant	80	MD& founder	I, Q, D
TF01	Electronic design and consulting	20	CEO	I, Q, D
TF02	Online marketing service company	50	CEO	I, Q
TF03	Telecommunication R&D	40	Project manager	I, Q
NV01	Internet Security	20	CEO and CTO	I, Q
NV02	Software design, consulting	16	CEO	I, Q
NV03	Telecommunication-VoIP	60	Sale manager	I, Q, D
NV04	Online cards selling	20	CEO	I, Q

Notes: Case code: NV= New ventures; EP = Excellent performance; MP= Mediocre performance; TF= Temporary frustration; Research instrument: I= Interview; Q=Questionnaire; D= Document study

4.6.3 Descriptive Statistics

SPSS package v13.0 was used to analyse quantitative data. Because there are only 45 valid surveys received, this study employed descriptive statistics, alone, in order to analyse the level of importance of performance measures in ICT SMEs.

4.7 RELIABILITY AND VALIDITY

Reliability and validity are the basic principles of social research. The former refers to the ability of an instrument to produce consistent results. It is also characterised by precision and objectivity. Validity means the ability to produce accurate result and to measure what is supposed to be measured. A valid measure produces true results that reflect the true situation and condition of the environment it is supposed to study

(Sarantakos 1998). There are different types and checking methods that address in reliability and validity in qualitative and quantitative research. This research combines both qualitative and quantitative approaches. A number of strategies were employed to minimize bias, to ensure the reliability, and to improve the validity.

4.7.1 Reliability

Reliability includes internal reliability as well as external reliability. Internal reliability refers to the consistency of results within a particular site, and the plausibility of data within that site. External reliability refers to the consistency and duplicative attributes of data across the sites (Neuman 1994). To ensure the internal reliability, low inference descriptors were used in the qualitative research stage in order to create a careful audit trail, by recording the data and interviews using an appropriate device (with permission). In quantitative research, reliability deals with an indicator's dependability, which means that the information provided by indicators does not vary as a result of the characteristics of the indicator, instrument, or measurement device itself (Sarantakos 1998). To improve reliability, the measurement variables in this research are not only those taken from other associated researches, but also include those proved to be important in qualitative research results. Furthermore, to maximise reliability of the instrument used, the survey was constructed as follows: 1) a pilot survey was conducted to ensure the reliability of the questionnaire; 2) each question was framed succinctly to reduce ambiguity and minimize bias, thereby ensuring the high statistical value of the data; and, 3) each participant in the survey was asked to state their job position to make sure participation was at senior-manager level.

4.7.2 Validity

According to Sarantakos (1998), validity includes face validity, content validity, construct validity, and internal and external validity. In this research, validation was checked based on above aspects.

Face Validity

Face validity is concerned with how a measure or procedure appears. In other words, does it seem like a reasonable way to gain the information the researchers are attempting to obtain? Does it seem well designed? Does it seem as though it will work reliably? As mentioned by a senior manager, SMEs differ across industries and, as such,

it is very hard to find the common features. Considering these differences, the samples are limited to the ICT industries. The sample companies' size follows the established criteria of employees (between 1 to 199 persons).

Content Validity

A measure is supposed to have content validity if it covers all possible aspects of the research topic (Sarantakos 1998). To ensure content validity, the researcher interviewed senior managers from different positions (CEO, MD, CTO, sales managers) and three venture capitalists whose invest area is mainly in ICT industry. In addition, the whole research was divided into two stages: qualitative research stage and quantitative research stage. This design enhances the reliability of the instrument that is used in quantitative stage. The materials were collected and pre-tested at the qualitative stage (multi-case study and interviews). Based on the results, the variables and the conceptual structure were adjusted to improve the instruments' content validity.

Construct Validity

"A measure can claim construct validity if its theoretical construct is valid" (Sarantakos 1998). The following techniques were used to minimise the risk of construct validity in this study: (1) The PM theories, evolution, and frameworks were studied in addition to analysing the research on PM in SMEs when the researcher developed the propositions and conceptual framework; (2) The theoretical construct is based on typical PM frameworks that employed widely, and (3) The features of SMEs and the ICT industry were analysed when the conceptual framework was developed.

Internal and External Validity

Internal validity concerns making inferences from data by considering alternative explanations and using converged data and related tactics. External validity reflects how accurately the results represent a phenomenon and whether results can be generalized (Yin 2003). "The former relates to the instrument's significance for the study situation; the later is associated with the generalisability of the findings gathered by means of the instrument in questions" (Sarantakos 1998). In case studies, lack of generalization has been the major issue. To ensure internal and external validity, the triangulation strategies were employed in methodologies and data resource. Multi-case studies and cross-case analysis were used to improve external validity.

4.8 COMPLIANCE WITH ETHICS

This research involves collecting data from SMEs in the ICT industry. The methods include semi-structured interviews of the senior managers in these companies and surveys using structured questionnaires. The ethics approval has been obtained from the RMIT Business ethical Sub-Committee. The project falls in the category □, i.e. MR (formerly Minimal Risk) in accordance with RMIT Ethical Principles. To meet the requirements of Ethical Principles and ensure confidentiality of interviewees and sample companies, the following measures have been taken:

- o The objectives, issues, risks and benefits of the project were well conveyed to the participants. In invitation letters, the research proposal which provides a clear picture of this study were attached and forwarded to each proposed company.
- o Formal consents have been obtained from the participants prior to the start of the interviews. The interviews were conducted at each participant's workplace.
- o The participants who agree to be interviewed are able to withdraw partially or completely at any time, or to refuse to answer any questions.
- o The privacy of the participants and the confidentiality of data that were obtained from the participants were strictly maintained in such a manner that the participants can not be identified in the report or any related publications.
- o The interview record and subsequent transcripts were stored in a secure area at RMIT University, where only the researcher had access to the raw data.

4.9 SUMMARY

This chapter has outlined the research methodology employed in this research. A mixed methodology was adopted using both qualitative and quantitative methods to draw the strengths and minimise the weakness of each approach.

Triangulation data collection was employed at qualitative research stage including interviews, survey, documentary studies and case studies. Questionnaire design was based on both literature review and qualitative research results.

Qualitative data analysis follows the strategy of *analytic induction*, e.g. constructing abstract concepts from study of specific data. All interview data were translated into a transcript and were initially coded into proposed categories. The transcript was then

analysed based on the hypothesized conceptual frameworks. In this study, quantitative data was analysed, employing the SPSS v13.0 package.

PART III FINDINGS AND DISCUSSION

INTRODUCTION

Part I reviews the literature of PM theories and PM implementation in SMEs. Based on that, a conceptual PM framework is developed. In Part II, the study's methodology is illustrated in detail. Part III presents the findings from the interviews, surveys, and case studies. An in-depth discussion of the findings is presented in Part III, based on the conceptual framework developed in Chapter 3. These findings are categorized into two themes: the critical success factors for SMEs in the ICT industries, and the implementation of PM in SMEs in the ICT industries. These two themes are discussed in the three proceeding chapters (chapters 5, 6, 7) in Part III. Critical success factors are examined in order to identify the determining factors of SME performance (i.e. the measures in 'enablers' group). Finally, PM implementation in SMEs is investigated and, following the results, a new PM framework for ICT SMEs is developed.

The following Chapters are as follows:

Chapter 5 discusses performance determinants that impact SMEs' performance, which helps SMEs identify important attributes for successful performance. The discussion is mainly based on the data from interviews in accordance with the conceptual framework in chapter 3.

Chapter 6 discusses the issues of PM implementation in ICT SMEs. The discussion is based on both interview data and survey results. The PM implementation issues discussed in this chapter include: the performance measurement areas targeted by SMEs; performance frameworks and tools employed by SMEs; the reasons to implement PM in SMEs; the barriers to implementation of PM in SMEs; the significant problems in implementing PM in SMEs; the key performance indicators employed; and, suggestions on PM implementation, as solicited from ICT industry executives. Analysing the above issues revealed important PM attributes that provided the basis for developing this study's proposed PM system.

Finally, Chapter 7 analyses four case studies in order to explore PM implementation issues.

CHAPTER 5 DETERMINANTS OF PERFORMANCE IN SMEs IN THE ICT INDUSTRIES

5.1 INTRODUCTION

In the previous chapters, the definition of performance measurement, the performance measurement framework, and performance measurement in SMEs, were discussed. While most existing PM theories are suitable for large organizations, they are not usually suitable for SMEs. This chapter focuses on studying the critical success factors that impact SME performance in order to identify attributes for PM.

This chapter employs both qualitative and quantitative methodologies to tackle two major research questions: 1) what are the factors (success factors or critical factors) that drive SMEs performance? and, 2) How important each factor is? The first question was investigated through interviews, the structure of which follows the framework in Chapter 3. Cross-case analysis was employed to analyse interview data. The interviewed companies were divided into four groups (see Chapter 4); the interview data was analysed to identify the similarities and differences regarding critical success factors that impact SME performance. As for the second question, questionnaires were used to rank the importance of each factor identified in Question 1.

The findings from interviews are summarised in Section 5.2, followed by a comparison of the practices against performance determinants in the case companies (Section 5.3). Section 5.4 presents the findings from the survey. An in-depth analysis of the key critical success factors and their relationship with PM is made in Section 5.5. The chapter ends with a summary of the most important findings.

5.2 FINDINGS: THE PERFORMANCE DETERMINANTS IN SMEs IN THE ICT INDUSTRIES

The framework developed in Chapter 3 identifies internal and external performance determinants for ICT SMEs. The internal performance determinants include company

resources; company capabilities; employed strategies; and, PM processes used by the company. The external performance determinants include: the level of competition; the degree of industry concentration/fragmentation; entry barriers in the industries; size and growth rate of the industries; and, the evolution stage of the industries and government regulation in the industries.

“All enterprise factors would influence a company’s performance.”

– Executive F

A company’s performance is determined by many factors. Every factor that is related to an enterprise would influence the company’s performance. According to system theory (Jackson 2000), a company is a system which operates in a larger systemic environment. The systems interact with each other and exchange the factors. Therefore, every internal and external factor influences a company’s performance.

Following specific factors are believed to most significantly influence a small and medium ICT company’s performance (as discovered via interviews).

5.2.1 A Highly Competent and Competitive Team

SMEs in the ICT industries face a very fierce competitive environment. All the executives interviewed believe that a high-quality team is a crucial factor that influences a company’s performance.

“I think the most important factor is the team, which includes senior managers and ordinary employees”.

– Executive F

According to Executive F, a high-quality team should involve all employees and leaders.

“Employees have the same expectation and value. The culture is upward and active. The senior managers understand the detailed business and can work out appropriate business models. At the same time, the team should be a learning organization, because IT companies compete with each other strongly. The team should have strong ability to adapt to the environment”.

To build a competitive team, *“First, the leader is very important, who can influence, motivate other team members. Second, emphasizing on team building and building a culture that can be aligned with works.”*

Executive H agreed with above opinion, stating that a *“team’s performance start from the leader... Leadership is more important than the reputation.”*

Executive D also agrees that building a constructive working environment is critical to performance. *“In a small company, the key factor of performance- we want to create the environment here where people are happy to work, and they gain satisfaction from their work.”*

In company C, Executive C believes that it is important to set upward culture and expectations. *“In small companies, the directors of the business were very much involved in the day-to day operation of the business. So it is very connected to kind of environment. We have total staff of 50; about half of them are here. We see interaction of all of the staff, so I think it’s an important aspect of setting culture and expectations, kind of lead to how we like things done, how we like to conduct and treat each other, and treat our customers.”*

Executive A agrees, saying: *“First, for a company to go, people have to like each other. They have to be compatible. And the management is different when you put a lot of very clever, creative people together. They actually manage themselves very well. If you put a team of ordinary people together, you’ll put management at a higher rank like a big company. Every team of a big company has a manager who reports to a higher manager. But that’s not work for small creative teams. They make much more freedom, creativity. And because of the sort of people who are very good at holding new things individually, they do not like other people to disturb them. And not as the traditional sort of management that applies to IBM, all the things you know are about product development. Small business is quite different.”* The Executive continues, saying that *“they all change. So this is why you need an atmosphere in your company that looks for change, it is always critical. Since we’re a small company, all the employees should be involved. This is a team, they should work together. And the company will work better if*

the communication works better. In fact, they are very smart people, you should respect them. That is the intelligent asset. You can't measure it by money, you measure by the people. It's not an accounting business."

Executive G stated that *"The company had employed nearly 40 NPD fellows. Among them, 15 employees are PhD. This kind of structure did not help the company accelerate its R&D; on the contrary, it slows down the speed of NPD seriously. The team manager has not enough power to manage the NPD team. The team manager has no right to recruit new team members and has no right to refuse team member. The management in NPD is poor. The structure of team members influences the collaboration in the NPD team. Each team member has an idea for the technological direction. They always quarrel with each other. It is short of a long-term design and long-term objectives. Management skill is quite poor. All employees came from the technological background. Poor management leads to poor performance. The significant weakness was presented on human resource management."*

Perhaps most concisely, Executive B stated that, for success, *"You have to have an environment where people are happy working, where they cooperate well, interact well. If you have problems in the way people work, it could terribly affect the performance."*

The above statements are samples from the interviews, each of which highlight the importance of creative people and a highly efficient team for SME success. In summary, a team with high performance for SMEs has the following characteristics:

- The team is made of smart and creative people.
- The team should have strong leadership and an appropriate management structure. Even though the team members' average educational levels in the ICT industries are higher than those in many other industries, optimal team structures require strong leaders.
- The team should be made-up of the people with different attributes. It needs not only leadership, but also people who can critically analyse problems and develop solutions.
- Team-leader requirements are different in ICT SMEs than those of a traditional team. The leader should be familiar with the business and have detailed technological knowledge.

- Building a compatible and upward culture in the team is very important. This means that every team member should be pushed to their maximum potential. Similarly, teams should be comprised of similarly-advanced team members that enjoy working with one another.
- Management is quite different in SMEs. The traditional sort of management that applies to big organizations does not meld with a small R&D team. However, as the company grows, standard and normative management becomes more and more important.
- The team should be a learning-organization. The members should be sensitive to the change of technology, track the right technology, and adapt to the changing external environment.

5.2.2 Right Strategy

A company's strategy is the second critical factor that influences performance. In the ICT industries, the changes in technology are so rapid that tracking its development-direction is very important. Therefore, formulating a good strategy, based on the company's unique competence and the external environment, is critical factor for the SME's future performance.

“In an R&D company, most resource is spent on research. So you really need to match your R&D outcome to product development. And product development needs to strategically in line with market. So you need to position the product in the market. If you begin a product not suitable to the market, which we've done a few times, then you are burning your resources that could be useful for other projects. So if you get in a wrong product, then you get stuffed at trouble.”

- Executive C

Based on the interview data, every ICT SME seems to have 3- to 5-year strategic objectives. Every interviewed executive agrees: strategy influences a company's survival. Among the companies that have been interviewed, well-performing companies commonly show following characteristics employing strategies:

- First, they have a deeply understanding of the industry and the technology's development-direction.
- Second, they know their own advantages and core competences. The company's

strategy should be aligned with the competences. *“I think if you have good R&D, you can control your intellectual base, and you can match that against your strategic plan. And if that strategy plan is in line with the market expectation, then that components will be well matched usually by having good key stuff, well trained and strategically well placed, which provides the company with its core-competence. And I guess the direction is in the right strategy. So you need to make sure that the competence is in line with the strategy.”* - Executive C

- Third, they know where the competitors are and are aware of competitor progress.
- Fourth, they understand what resources are needed for strategic objectives and how to get them.
- Finally, these companies typically position themselves in the proper sector of the industry/market. They usually cooperate with strategic partner in the industry. *“I think the best thing to do is to be in-line with the big companies. It’s like to have a big brother. But the problem is that the big brother will choke you as well. If you grow to threat their propagation, they will try to cut you. It is very hard to be independent. But you need strategic partners. And if you can broaden that base, then you are strategic to avoid other company controlling your company.”* - Executive C

Based on the discipline of market leadership (Treacy and Wiersema 1995), a company’s strategy can be categorized into: operational excellence, product leadership, and customer intimacy. However, most SMEs in the ICT industries usually choose two different strategies: to partake in a partnership with bigger companies and later find its own position in the industry; and, to develop original technology and lead the market, particularly where there are few competitors.

This study found that most SMEs in the ICT industry use the first strategy, due to limited resources; additionally, the second strategy not only requires more resources but is also more risky. Only a few SMEs, usually those with advanced technological resources, choose the later. These two strategies are not mutually exclusive; however, in both situations, a company usually needs original technology in order to form a partnership or comparative advantage.

Over-ambitious strategies should be cautioned. It is too often that these strategies end in failure.

5.2.3 Developing Core Competence Aligned with Strategy

Every executive interviewed agrees that the company's competence is very important to the company's performance.

“The company's competence influences the performance very much. Well, I think our directors have strong competence in computing technology, communication technology, and I think in the business and reform process. I think in the business sector, we are very strong in these areas, our external directors; his core competence is in strategy planning business, finance and overall company governance. So we've got some well-potential external directors that stand on us. Now the gentleman on our board is the former chairman of the major director of an investment bank. So in terms of competency at the board level, we are fairly sound in that area, and the board are very much focusing on growth strategy, on the managed basis, whereas we have lots of opportunities.”

- Executive D

A company's competence is multi-dimensional and, even in the ICT industry, requires a command of more than technology. According to interviewed executives, a company's core competences include:

- 1) Research development capabilities.
- 2) Managerial skills. Successful SMEs pay attention to building a network to improve their resource network, strategic planning, and managerial capability. They look for smart persons from funding organization and managerial experts to help them build the network. A familiar method is to invite external directors and independent board members to join their company.
- 3) Exclusive-resource capabilities. Companies with exclusive resources are less influenced by external environmental factors. For example, Company D has historical data of open-water channels in Australia. With such exclusive resources, most companies that wished to work in Australia's irrigation network benefited from their services. Such an advantage also garnered technical support from universities and funding support from Australian

government. This kind of exclusivity is usually formed through two routes: special operation history – for example, company D starts up from a government-owned research institute; and, barriers to entry.

The collected data shows that the core-competence of a company should be in-line with a company's strategy. For successful SMEs, their competences were developed according to their strategy. On the other hand, SMEs also need to formulate correct strategy, according to their own core competence.

5.2.4 Focusing on Customers

Generally, scholars believe that focusing on customers is a key performance factor. It is strange that not every SME follows this advice. The reason might be that most small companies contact directly with customers. They do not think that communication with customers is an issue; however, the leading enterprises realized that customer growth is a key factor for long-term performance.

“Obviously we’re not going into business in the long term if we don’t have an underlying financial framework in place. But growth in terms of customer is obviously key factor for us in moving forward.”

– Executive D

“Customers’ reference is quite important in our industry. And that gained us reputations. I didn’t really have full-time sales until this year. That’s only because people have been directly contacting us upon our reputation.”

- Executive H

This study found that few SMEs collect customer feedback periodically (e.g. just 2 SMEs out of the 16 interviewed companies do), though most executives claim that they know their customers quite.

5.2.5 Managing Internal Process

Internal processes in the interviewed SMEs are not as important as that in the big organizations. Most executives believe that internal process is simple in SMEs. But the

leading SMEs have realized that internal processes are more important than the external business environment.

“Certainly internal process is really a key factor to make a long going business. And we’ve internally adapted the role and responsibilities of the company directors over the last 6 months, where my role is more formally on overseas market. We’re traditionally involved in Australian market. So I think the internal process really is a critical factor and in terms of us, it’s probably more important to external business environment. “

– Executive D

The issue of internal process in SMEs is usually around cost management.

“Well, internal process is generally very loose. They really evolved around the balance sheet. How much money have I got? How much more can I do? How much more that I can find at the moment? When we had a lot of people and a lot of money, we have a lot of issues. The more money we have, the more issues we could have. When revenues start running out, you have to start cutting projects. I guess in terms of internal processes, you need to have money.

- Executive C

Many executives understood that the internal processes are mainly on the project management, e.g. how to manage a team and manage a research project. Therefore, the internal process is simpler compared to that in large organizations.

“That depends on what process you mean. There is a technical process, and there’s a financial process. It’s obvious the financial reporting has to be good. As far as management type, if a small company like us is concerned, it’s just common sense. So the most management is the management of the technological building. Because it is research, everything should be properly documented. It has to be done in a very scientific manner; so managing that process to make sure the thing is done well. Not managing reporting. It’s just being very material and working together to make sure it’s done right, tested. So it’s a team process. We have a guy working on that. But we don’t call him CTO, we call him the team leader. He leads the team and helps the discussions and helps management. So the concept of management role is again different than in a company which sells products.

- Executive A

5.2.6 Optimizing Resources

Resources are something to which an organization has access; they can be either ‘tangible’ or ‘intangible’. The former are relatively obvious, including buildings, plant, equipment, exclusive licenses, patents, stocks, land, debtors, and employees. Intangible resources include skills, reputation, experience and knowledge of employees, advisers, suppliers, distributors, etc. (Mills, Platts et al. 2002). For SMEs, resources usually include: funding, technology specialist, managerial expert, strategic analysis, marketing, and sales channels.

Resources management is a key issue for small companies. Many interviewed executives agreed on this. In SMEs, many executives’ main job is looking for adequate resource to realize the company’s plans. Some executives believe that resource is the key to excellent performance.

“We’ve largely been a company that’s grown organically. So we’ve found developments in our products as we’ve won jobs for sell, but we’ve implemental built for capabilities of our products. So certainly that’s probably the most challenging area for a small company – to have resources available to build their dream. Obviously, a market strategy and then developing a product to match the strategy can be your reasonably academic consider. But if you haven’t got the balance sheet to found it, ever though you have the best idea in the world, and you have the best to implement it, but you can’t pay their wages, you are not going to set there. So resource is obviously a key issue for small company.”

– Executive D

On the other hand, other executives believe that, although resource management is a key factor, there is no direct correlation between company performance and resources. There are other things necessary to assure success.

“Capital is important. It is a main resource. However, there are other necessary conditions to achieve excellent performance. Resource is very important, but it is not the key factors (determinant). It is mainly a marketing strategy.”

– Executive F

In SMEs, resource utilization is a very important topic. Resource is usually very limited in SMEs. Wasting resource often leads to business failure because the SMEs have no second chance to correct their mistake.

“Well, you can’t waste your resources. You must use it efficiently. Because if you are inefficient, the other companies become more efficient than you are. You then lost your strategic advantages. So you need to make sure you are efficient in using your resource.”

– Executive C

In this study, the critical resources that influence an ICT SME’s performance have been recognized (see Table 5-1).

Table 5-1 Summary of the Key Tangible and Intangible Resource that Impact SME Performance

Tangible resources	Intangible resources
Technology	Employees
Capital	Team
Geography location	Culture/informal system
Physical resources	Knowledge/Expertise
	Network on resources
	Reputation

Source: Interview data collected for this study

5.2.7 Organizational Agility

This study found that environmental factors strongly influence SME performance.

“Because you are doing R&D, you’re doing the future, which is affected by the economy, by technological changes, by people, by all sorts of things. So for a company it’s important to be aware of the changes and to catch the important information in it and change with it. If you’re going in a wrong way, you have to stop. To catch all these kinds of change, we have a board director to talk about it. And we talk to other people, and we read. In our company, this is not one person’s job, everybody does it. And this way is much better. It’s not that ‘your job is research officer, you do what you should.’ –That’s what they do in big companies.”

– Executive A

“The environmental factors influence the performance. For example, the Internet users increased year by year, the broad-band widely spread; the competitive environment in

employment brings many opportunities to our company. We often consider how we can do if the economy slows down and enterprises do not recruit new employee.”

– Executive F

“Business environment is very important. Because here is Australia, we’ve been quite removed from overseas market. So it’s very difficult to get market penetration. A lot of the environment factors are actually external. Probably the most influential I guess is the competition environment (from other companies, overseas, what ever).

– Executive C

“The business environment is external for small companies. Most threats come from external sources. A lot of threats come from very big circle strategic companies as well. They may decide they are going to go into your strategic wish and just do those themselves.”

– Executive C

“The environment has huge impact on us as well. Right now, for example, we are very hard to find people ...”

– Executive H

From above statements, it can identify that the environmental influences come from:

1. Economic factors: economic influence depends on the products and services that SMEs provide. Results from the 17 interviewed executives were conflicting; 10 of them claimed that these factors are important, while others downplayed their importance.

“Well, we’re probably being a bit insulated in terms of what other things are going on in economy. That probably doesn’t matter too much to us, whether the industry is booming or whether it is not.”

– Executive D

2. Technological changes: most interviewed companies specialize in R&D. Technological changes obviously influence their performance. The important issue is whether these companies can track the technology changes.
3. Industry changes: the influence of a SME on the industry is very weak; on the other

hand, SMEs should follow changes in their relative industry.

4. Strategic partners: For SMEs, one big challenge is in finding the penetrative point; companies should enter at this point and collaborate with strategic partners. Of the interviewed companies, only one claims to have constricted their business because their strategic partner cancelled their established collaborative relationship.
5. Competitors: the competitive environment significantly influences SME performance. The competitive factors include: number of competitors in the market segment, barriers to entry, etc.
6. Geographic factors: every interviewed SME (in Australia) believed that Australia is not close to the global market, geographically. They believe Australia to be a small market, which influences their performance significantly. Therefore, every SME in Australia is eager to enter global market.

Compared to large companies, SMEs have a smaller influence on the market. They cannot lead technology and occupy market as some large organizations do. Therefore, for SMEs, the big challenge is to be adaptable to the external environment. According to the interviewed executives, the key to external competition is related to a company's strategic adaptability.

“The smaller you are the smaller influence you have in the market, and the less ambitious you are going to be. So you need to set small step changes to give you high profit-return. So you have to pick small influential products in your market position, but usually that are low-profit.”

– Executive C

“For a company, it's important to be aware of the changes and to catch the important information in it and change with it.”

– Executive A

“In our company, everyone is focusing on adapting and changing, and modifying the company to adapt to the market needs. We're continuously changing the way we structure”

– Executive H

5.2.8 Competitive Advantages and Innovation

There are two forms of competitive advantage: cost advantage and differentiation advantage (Porter 2004). Cost advantage exists when a company provides the same products or services as competitors, but at a lower cost. A differentiation advantage exists when a company can provide greater value at the same cost or lower than competitors. Competitive advantages can be created through innovation, efficiency, quality, and customer responsiveness. In ICT SMEs, the competitive advantage often comes from differentiation advantage. As discussed in the literature review, technological innovation is a typical feature in ICT SMEs. The interview data shows that a company's path to achieving technological advantage is difficult but critical. This is in accordance with the claims in Smith (2005), that innovation is “essential to the long-term survival of an enterprise and to the maintenance of its market share and competitive advantage.” The executives interviewed in this study also emphasize that innovation is an important dimension in ICT SMEs.

“So the key factor is to really have a unique product and to address unique challenges.”

– Executive D

“First, that product of your design should be the original and basically without much competition.”

– Executive A

“To measure a company's performance from technological aspect, it mainly checks whether the technology has competitive advantages or not”

– Executive A2.

The interview data shows the innovation in ICT SMEs presented following characteristics:

First, most studied SMEs (especially those started with venture capital) focused less on incremental innovation and more on radical innovation. This is due to an important feature of the ICT industry – technology is rapidly developed and products are easily substituted. In fact, four of the sixteen interviewed companies own patent; of these,

most focus on original technology development rather than incremental innovation. This finding is different from what Oke, Burke et. al. (2007) discovered. They identified the innovation types in UK SMEs, finding that they preferred incremental innovation.

Secondly, SMEs in the ICT industry focused more on product innovation than service innovation and process innovation¹. An interesting phenomenon is that, when an SME is conducting original technological research, the executives do not regard the process of innovation as priority. However, SMEs that make incremental innovation always take process innovation as priority.

“We think that product and service innovation is very important to the performance. In SMEs in the ICT industries, we think that there is not much work can be done on the managerial innovation”

– Executive P

Thirdly, SMEs tend to collaborate with other big companies and universities. Among the nine companies interviewed from Melbourne, Australia, five have collaborated with Melbourne University or Ericsson (one of the biggest telecommunication manufacturers in the world). It seems that SMEs focusing on product innovation are good at building their external research network. The results is not consistent with the conclusion by Mohannak (2007), who claimed that ICT SMEs rely primarily on their own internal capabilities in original product or service development, rarely exchanging ideas with other firms. The differentiation may be caused by the chosen sample. This study found that the radical innovation SMEs rely on external technology resources more than the companies that focus on incremental innovation.

Fourthly, it is hard for ICT SMEs to get the information about the progress of new products development. Even though interviewed executives state that products and technology should be unique and competitively advantageous, it is very hard to identify the competitive advantage of one technology. Compared to big companies, SMEs can only apply some simple methods in tracking these trends.

¹ Oke, Burke et al. (2007) divided innovation in SMEs into three types: product innovation, service innovation and process innovation. Product innovation includes new product offerings or improvement in existing products. Service innovation is described as new developments in those activities that are undertaken to deliver the core product and make it more attractive to consumers. Process innovation involves creating or improving methods of production, service or administrative operations.

“Yes, it is always a risk. We get the information from the Internet, search the products release information. It often happened that other companies released their products when we get the product concept. There is a benefit that we can test our idea through other companies’ product release.”

– Executive A2

Finally, it is challenging for SMEs to track customer needs and requirements. Hence, many SMEs’ product development does not strictly follow the NPD process². Their new product development is often derived from own original technology.

“First, communicate with customers. Second, from market, you should always focus on the competitors; you should know the progress of competitors. For our small companies, there is no special person to do this job.”

– Executive A2

The interview data provided the following critical factors for SME performance:

1. People and teams are the most important factors in ICT SMEs. Every SME executive agrees that high-quality people are critical to help them achieve excellence. At the same time, building a highly competent and competitive team is important. The interviewees mentioned that team atmosphere, structure, culture, communication, management style, and leadership significantly influence performance.
2. Strategy influences an SME’s performance and often dictates its survival; particularly those factors related to risk management.
3. Core competence is important and can come from new product development (NPD), the capability of manufacturer, or the capability on sales and distribution.
4. Customer satisfaction influences both short- and long-term performance. Customer satisfaction depends on product/service quality, delivery time, and communication with customers. For most SMEs, customer communication is not an issue because they usually contact customers directly. On the other hand, this

² A NPD process usual include: project generation; requirements capture; concept design; and implementation (Moultrie, Clarkson, et al. 2006).

study suggests that companies with high customer satisfaction develop superior communication. For example, Company O finds that their handle on customer satisfaction has more to do with communication between engineers and customers rather than products and services.

5. Internal processes influence SMEs efficiency and effectiveness. The common issues that SMEs need to deal with are cost control and collaboration between departments.
6. Resources determine SME development and performance. Resources are an important factor to help SMEs to build their exclusive competitive advantages. Correct utilization and development of resources help SMEs achieve excellent performance. For SMEs in the ICT industry, the most important resource management is in finance plans and IP protection.
7. Environmental factors influence SME performance, especially strategic partnerships, competitive factors, and technological development.
8. Innovation is the survival base of SMEs. Innovation in SMEs includes products (or service and technological) innovation, business model innovation, and sales-channel innovation.

5.3 COMPARISON OF PERFORMANCE DETERMINANTS IN CASE COMPANIES

According to the conceptual PM framework in Chapter 3 (see Figure 3-4), an SME's performance is determined by internal and external critical factors. They can be further categorized as: strategies, resources, capabilities, internal processes, innovation and learning, and environmental factors. Furthermore, the 16 interview cases were categorized into four groups (see Table 4-5 in Chapter 4): excellent companies, mediocre companies, frustrated companies, and new ventures. Table 5-2 to Table 5-6 illustrates these attributes in a comparative matrix, based on the aforementioned conceptual framework.

Chapter 5 – Determinants of Performance in SMEs in the ICT Industries

Table 5-2 Comparison of Performance Determinants in the SMEs that Achieved Good Performance

Case code Determinants	EP01	EP02	EP03	EP04	EP05
Strategy	Well-positioned products to be key part of the overall market. And cooperate with university (the technology provider).	Become the leader in the market segment, funding from VCs	Strategy was initially set on performance for a year or two later	Achieve the top level partnership with Giant in the industry	Focus on customer service (high quality, fast deliver time on budget), compete against low price
Resource	Have: technological experts; exclusive history data resource Need: People on sales; overseas network; intelligent information and the sales finance plan.	Need: correct and appropriate business model	Have: Reputations Need: Human capital, People	Top quality people, flexible working environment.	Have: Top class engineers, flat management; good customer relationship.
Capability	Directors have strong professional knowledge and skill at their area. And strong competence at board level	Catch the opportunities in the market swiftly.	Ability to deliver on customer expectation; Faster than competitors	High competence in IT.	Good performance measurement, comprehensive products rang, project management,
Environment	Niche market; a bit insulated of external environment; taking some benefits of having engineers recruit from IT blooms.	The rapid developing on internet technology and the amount of users provide many opportunities; the business also influenced by prosperous of job market.	Environment influence company to find people and client. The environmental factors influence the company's growth. The overall competitive doesn't impact that much.	The partnership influences the company's performance.	Good cooperate with customers, Niche business
Internal process	Internal process really is a critical factor, change the internal process to make a long going business need.	Internal process significantly influence cost control and efficiency, and enterprise culture	Internal process sometimes influences the level of quality, sometime influence; the capability building.	Flexible working, good facility	Multi-task management, regular meeting
Innovation and learning	Extend business to global market.	Business model innovation is important. Learning organization.	Continuing focusing on improvement; everyone is focusing on tracking technology changing and modifying the company to adapt to the market needs	Training once a month	Regular visit and communicate with customer.

Source: Interview data collected for this study

Note: EP: Excellent performance companies studied

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Table 5-3 Comparison of Performance Determinants in the SMEs that Achieved Mediocre Performance

Case code Determinants	MP01	MP02	MP03	MP04
Strategy	No definite strategy. Chang from trade company to manufacturer	Following the leaders of the market.	Succeed in products development in time and on budget.	Focus on a niche market and specific customers group.
Resource	Need: organizational structure; smooth working flow; management experts.	Have: Low cost human resource;	High capability employee; strong support from the strategic partner who is main player in the market segment.	Need: network of relationship with customers and governments;
Capability	Competence on marketing and sales, rapid and flexible responsibility to customers' needs.	The capability on R&D and the integrated capability on manufacture.	High competence on products development. The project management capability needs to be improved.	Extend to consulting service from providing software products.
Environment	Fast changeable environment requires to fast response and flexible internal process.	The fast development of IT provides much opportunity.	Fierce competitive environment require the company grasp the correct technology trend and response to the changes of market rapidly.	The niche market influence the company's development
Internal process	Need re-organize internal process. The communication and cooperation between departments is not good.	The collaboration between front departments and support departments should be improved.	The project management and training process need to be improved.	Good internal process, each job has specific working flow, responsibility, monitor mechanism and measurement
Innovation and learning	Need reform the R&D process and need to increase the efficiency of R&D	Need develop own products and brand	The efficiency of training is not good enough. Training is for training, not from the need of project.	Systemic training and measurement; clear promotion system

Source: Interview data collected for this study

Note: MP: mediocre performance companies studied

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Table 5-4 Comparison of Performance Determinants among the SMEs that Experienced Temporary Frustration

Case code Determinants	TF01	TF02	TF03
Strategy	Cooperate with strategic partners; Strategy has to be right-match the market expectation in 18 months or two years. Even into 5 years; For small & medium IT companies, the best thing to do is in align with the big companies	Strategy was not based on own core competence and resource. Chasing develop speed and failing in risk control and cost control.	Ambitious new product develop plan without regarding resource and capability
Resource	Resource was not used efficiently, lost strategic advantages. Need engineering staff; strategic analyser to perceive what the market needs to be; money-very important.	Have: a national wide sale plate and customer network; Comprehensive sales channel Need: fund; good relationship and cooperation with strategic partners. Risk management expert.	Have: high quality employees; VC fund, technological experts. Need: Good project management; human resource management; enterprise structure and collaboration culture.
Capability	Good key staff, well trained and strategically well placed, which provide the company with its core-competence; The competence is in aligning with the strategy.	Customer relationship management; overseas resource; engineering technology.	Strong technological capability; Most people come from technology background; shortness of management knowledge and skill
Environment	Environment influences attaining market penetration. The most influential is the competition environment (from other companies, overseas, what ever). Most threat comes from external sources. A lot of threats come from very big circle strategic companies as well.	The low threshold attracts many more young entrepreneurs enter this industry. The fierce competition leads company share a small market after the company enlarge scale in amount.	The environment of IT bubble breaking influences sustainability of fund. The prosperous of IT increase the company's human cost.
Internal process	Internal process is generally very loose. When revenues start running out, the company has to start cutting projects.	The internal process is very important even though the company is small The important internal process are customer relationship management and finance planning	Management skill is quite poor. Poor management leads to poor performance. The significant weakness was presented on human resource management. The management in NPD is poor.
Innovation and learning	Technological innovation and overseas marketing development	Business model innovation and service range extension	Technological innovation

Source: Interview data collected for this study

Note: TF: frustrated companies studied

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Table 5-5 Comparing Performance Determinants - the New Ventures

Case code Criteria	NV01	NV02	NV03	NV04
Strategy	Focusing on developing products for global market and grows quickly.	Short-term strategy is dictated by ' <i>what happens at the end of this funding round</i> ', identify and focus on those areas of market that give us reward in a short term phase. In a longer term, look for second fund round.	Active service mind; Research and develop own products; Technology innovation.	Choose a balance between speed and risk. " <i>If you find that you can get fund support, you can emphasize on speed</i> "
Resource	Have: excellent people on the board, top technological people Need: money; excellent, greater than average people; advice from expert;	Need: more people, money; physical space.	Have: high quality and skilled employee. Need: funding, sales network	Need: fund support,
Capability	Internet security core technology	Especially good at building software at a particular area; develop capability in a number of different area in balanced.	R& D capability	The capability to grasp market opportunities
Environment	The fast development of internet provide much opportunities for the company	Environment includes the network that you are part. The geographical position influence performance.	Government policy support	There are some factors that the company can not control. It is important to choose a appropriate market segment and deeply understand the technological development
Internal process	Simplify management; creative and clever people manage themselves	<i>"I would say-not just process-working culture is crucial for the performance. You have to have environment where people are happy working, where they cooperate well, interact well"</i> .	Culture building	Definite work flow to warranty responsibility of each position is clear
Innovation and learning	New product innovation	Product innovation	Engineer training	Emphasizing on technology innovation,

Source: Interview data collected for this study

Note: NV: new ventures studied

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Table 5-6 Comparison of Performance Determinants between the Excellent Companies and Other Companies

Case code Criteria	Excellent companies (EP01-05)	Mediocre companies (MP01-04)	Experienced frustration companies (TP01-03)	New ventures (NV01-04)
Strategy	Definite strategy and well positioned company in the industry is. Focus on long-term performance to build resource and capability.	No definite strategy, focus on day-to-day operations.	Ambitious strategies regardless of own resource and capability. Note: not all SMEs interviewed failed in strategy. Some SMEs failed for other reasons	Strategy focuses on products innovation. Long-term strategies are not clear.
Resource	Have specific and exclusive resource. Good reputation helps companies to achieve good performance. The resources needed are to develop business, not for survival.	Resources needed are mainly for internal management: forms of internal flows and process. Many resources are needed rather than one area	Short of resource on survival aspects.	Fund support is the significant resource need. The next resource need is people.
Capability	Formed clear competitive competences on their own area. Improve their competence continuously through innovation and learning	Have some competitive competences in the area, but these capabilities are not exclusive and unique. Busy with daily work and less works on employee improvement.	Often failed in the capability of risk control and IP protection.	Focus on specific area of technology capabilities and so on. Management and industry analysis capability are required.
Environment	Some insulate of external environment because of own competitive advantages. Have built good customer relationship and partnership in the industry.	Influenced by external factors, for example, the industry development speed, the government policies, and the size of the market.	Sensitive to external factors. The environment influence these companies' market penetration, human cost, and fund recruit.	Influenced by environmental factor absolutely, including technology development, barriers of market, geography position and so on
Internal process	Emphases on improving internal process continuously; have formal and clear definitions of position responsibilities, work flows and process. At the same time, as a SME, multi-task management and training are important.	Most companies believe their internal process need to be reorganized. The significant weakness is that the collaboration between departments should be enforced.	Did not realize the important of internal process except those companies have failed in internal management.	Most new ventures believe that the internal process and management should be loose and not strict. Too formal management would influence the working atmosphere. But they also realize that the position responsibility should be clear.
Innovation & learning	Continue focusing on innovation and learning; regular training and customer communication	Focusing on specific innovation, such as in product and technologic area. A few training given to employees.	Innovation on products or technologies	Innovation on products or technologies. A few of them realize the importance of learning.

Source: Interview data collected for this study

Differences in poor- and excellent-performance companies are outlined in Table 5-6.

The following summary is provided in a cross-case analysis:

1. An SME's performance is decided by both internal and external factors. Performance is an integrated result. Each factor interacts with others. The companies who achieved excellent performance always address the critical factors in balance. The companies that experienced frustration failed by one or two factors.
2. Strategy influenced SME performance. Excellent companies usually have definite and executable strategies. They choose a right position in the industry. Compared to successful companies, ordinary companies focus on their daily work; furthermore, their strategies are not clearly defined. The companies that experienced frustration often failed in their more ambitious strategies, regardless of their resources and capabilities.
3. SME resources can be divided into two groups: developing resource and survival resource. Excellent companies have specific and exclusive resources and focus on continually developing these. The resources required by excellent companies are often used for development; for example, excellent marketing and salespeople. In contrast, resources in mediocre companies are usually treated as survival resources; for example, those that improve organizational structures and those that reform and improve internal flows and processes. Mediocre companies also require more external resources because they do not develop as many on their own.
4. Excellent companies have formed competitive competences through innovation and learning. On the other hand, a mediocre company's competitive competences are not significant, even if they have some competitive advantages. This is partly because the competitive advantages are not exclusive and unique in mediocre companies. In contrast, excellent companies emphasize training and customer communication in order to improve their capabilities.
5. External factors impact SME performance. However, the impact on excellent companies is less significant than on that of mediocre companies. It may be because excellent companies have already built their own core business. The weaker a company is in terms of competence, the more sensitive it is to its environment.

6. Internal processes: excellent companies emphasize continuous internal-process improvements. They have formal and clear internal process. Hence, excellent companies have fewer internal-process issues. On the contrary, mediocre companies often face problems with internal processes. The significant problems are: (1) collaborations between departments; (2) the responsibility of position is not clear; (3) the gaps in the working flow and process; (4) lack of cost control and risk control. The excellent companies emphasize multi-task management and training. This is determined by the features of SMEs in the ICT industries: frequent change of employees and the small size of most companies.
7. Innovation and learning: innovation is the most widely used survival technique for ICT SMEs. Every SME needs innovation in regard to products, technology, and management; however, innovation and learning is different in excellent companies than that of mediocre companies. The former continually focuses on innovation and learning, emphasizing regular training and customer communication. Excellent companies also acknowledge that organizational innovation, business model innovation, and management innovation are very important. In contrast, mediocre companies often only focus on technology innovation and product innovation.

5.4 THE IMPORTANCE OF THE DETERMINANS

In pilot tests,

- 1) The impact of 12 Capability-related factors on PM in ICT SMEs are tested. The results show that 5 aspects focusing on customer service, resource management, adaptive to technology trend and market development should be further explored.
- 2) 17 Resource-related factors are enquired for PM in ICT SMEs. In the questionnaire, the importance of 12 factors around financial, technical and marketing resources are investigated.
- 3) 15 Environment-related factors are tested. In the questionnaire, 12 factors are enquired on. The emphasis is on the influence of industry concentration/fragmentation, market entry barriers, government regulation in the ICT industry and competitors development.
- 4) 13 Strategy-related factors are tested. In the questionnaire, totally 9 factors are included; more focuses are put on how strategies are developed and deployed.

- 5) 8 Process-related factors are tested. The importance of them to PM in ICT SMEs is assessed almost the same level from the pilot tests. All of them are kept in the questionnaire. Note items 5,6,7 are combined as one item (about key support process).

A detailed list of the SME related factors is presented in Table 5-7.

Table 5-7 the Factors that Impact SMEs' Performance

Capability	Response customers' needs swiftly
	Effectively manage people and resources
	Appropriate managerial system with improvement capability adaptive to resource and environment changes
	Deeply understand the technology trend and catch the changes
	Flexibility to adapt to new industry and market trends
Resource	Availability of capital
	The executives' managerial experience
	Access to overall low cost factors of production
	Technical resource (patents, exclusive technologies)
	Comprehensive and efficient organizational system, structure and planning.
	Expertise in product/service development
	Expertise in marketing
	Expertise in customer service
	Expertise in management
	Access to low cost distribution channels
	Enterprise culture
	Reputation
Environment	The company's customer groups and market segments are clearly defined and selected
	Understanding and learning about customers, anticipating customer needs, and developing business opportunities.
	A systematic and organized process exists for collecting, conveying customers complain, feedback about products and other information from customers. This information will be analysed to improve the business process.
	Measuring customer satisfaction periodically and the results are used to drive improvement
	Understanding the changes in technology
	The company knows the main competitors, and is aware of its own competitive position in the market.
	The company gathers competitors information continuously

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	Degree of industry concentration /fragmentation
	Entry barriers
	Size and growth rate of industry
	Stage of industry evolution
	Government regulation in the industry
Strategy	The company has a definite strategy
	Strategies are based on target customers, markets, environment
	The leader team considers employees' idea when planning the company's future.
	The strategy are developed, reviewed and updated periodically based on the information from customers, environment, and performance measurement.
	The strategy is deployed through a framework of key processes
	Provide new products to exist market
	Provide established product to exist market (differentiation on price, quality and other values comparing to competitors)
	Provide established products to new market
	Provide new product to new market
Process	The process ensures that any changes in customers and market requirements and technology can be incorporated into product design and/or service design
	Production/delivery processes meet customer, quality, and operational performance requirement
	Design, production and delivery processes are coordinated to ensure trouble-free and timely introduction and delivery of products/service
	The processes are evaluated and improved continuously to achieve better performance
	The key support processes (for example, finance and accounting, IT support, personal, legal, risk management and so on) are well defined. They support the key product and service process to achieve the company's performance outcome and objectives
	The supplier and partner processes are well managed to ensures that the collaboration goal can be achieved to enhance the company's position in industry
Measure and analysis	The key financial and non-financial information and data are selected, managed and used to support overall business goals achievement
	The competitive information (competitors, environment and technology and so on) are collected and are analysed to improve the performance of overall activities
	Employee and departments know how to measure their performance. The results can be use to guide their activities to improve the performance
	Appropriate performance measurement tools and skills are employed.
	Each goal and objective will be measured according to designed measure criteria

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	Strategy is clearly translated into company, department, unit and individual goals and objectives.
	Employees knows his/her job goal and objective and the contribution to the whole strategy and objectives
Innovation and learning	The investment in new products development (NPD)
	The efficiency of NPD process (the input of NPD/output of NPD)
	The company emphasize on the employees training.
	The organizational structure and system is renewed based on the environmental changes.
	Flexibility to adapt to new industry and market trends

A questionnaire survey was designed to rank the level of importance of the factors (see Appendix B). Question 14 is designed to test performance determinant factors. The reliability of the items (analysed through SPSS) is estimated by Cronbach's alpha. As shown in the table 5-8, the alpha values are above 0.8, which imply high reliability of the question items.

Table 5-8 Internal Consistence Results

Factor	Critical success factors	No. of items	Alpha value	Items for deletion	Alpha if item deleted
F1	Capability	5	0.8518	None	0.8518
F2	Resource	12	0.9400	None	0.9400
F3	Environment	12	0.9387	None	0.9387
F4	Strategy	9	0.9186	None	0.9186
F5	Process	6	0.8772	None	0.8772
F6	Measure and analysis	7	0.8967	None	0.8967
F7	Innovation and learning	5	0.8534	None	0.8534
Total:	56				

Note some questions in the questionnaire seems similar, we have the following considerations:

- In Question 10, the selection “not enough performance measurement knowledge” is expected to be chosen by the SMEs at the start-up stage. They don't have people who knows why PM and how PM works. While the selection “not know how to tailor complex PM tools to suit our company” is expected to be ticked by the ones who already have noticed the importance of PM and want to put some energy in this area. They have people know something about PM, but not experts that are confident enough to modify existing PM tools to be implemented in the companies.

- In Question 10, four most mentioned problems in implementing PMS in the interviewed 16 SMEs are listed for the sample companies to select. While Question 11 is an open question, it allows the sample companies to list the three most serious problems of implementing PMS, the answer of which could be out of the four selections given in Question 10. The intention is to make sure all the significant problems in PM implementation in SMEs are to be discovered.
- The reason for both Question 7 and Question 12 are put in the questionnaire is similar as the reason given above for including both Question 10 and Question 11: to make sure no important performance indicators are missed from analysis.

The survey result is summarized in Table 5-9.

Table 5-9 the Perceived Importance of Critical Success Factors by Respondents

	Minimum	Maximum	Mean	Std. Deviation
Responding to customers' needs swiftly	3.00	5.00	4.30	.80
The company's customer groups and market segments are clearly defined and selected	3.00	5.00	4.21	.63
The reputation of the company	3.00	5.00	4.11	.56
Adapting to new industry and market trends	2.00	5.00	4.05	.88
Enterprise culture	3.00	5.00	4.00	.67
Strategies are based on target customers, markets, environment	3.00	5.00	4.00	.69
The company has a definite strategy	3.00	5.00	4.00	.69
Understanding the changes in technology	2.00	5.00	4.00	.91
Production/delivery processes meet customer, quality, and operational performance requirement	2.00	5.00	4.00	.94
Understanding and learning about customers, anticipating customer needs, and developing business opportunities.	2.00	5.00	4.00	.97
Being aware of its own competitive position in the market.	2.00	5.00	3.94	1.1
Expertise in marketing	3.00	5.00	3.90	.64
The efficiency of NPD process (the input of NPD/output of NPD)	3.00	5.00	3.88	.70
Ensuring that any changes in customers, market and technology can be incorporated into product and service design	2.00	5.00	3.88	.93

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The company's emphasis on the employees training.	3.00	5.00	3.87	.72
Understanding the technological trend and catch the changes	3.00	5.00	3.85	.67
Managing people and resources effectively	2.00	5.00	3.85	.99
Size and growth rate of the industry	3.00	5.00	3.83	.71
Stage of the industry evolution	3.00	5.00	3.83	.79
The key financial and non-financial information and data are selected, managed and used to support achieving the overall business goals	3.00	5.00	3.82	.73
Evaluating and improving operation process continuously to achieve better performance	3.00	5.00	3.82	.73
Strategies are developed, reviewed and updated periodically based on the information from customers, environment, and performance measurement.	2.00	5.00	3.82	.88
Providing new product to new market	3.00	5.00	3.81	.75
Providing established products to new market	2.00	5.00	3.81	.83
A systematic and organized process exists for collecting, conveying customers' feedback.	3.00	5.00	3.78	.81
The key support processes (e.g. finance and accounting, IT support, personal, legal, risk management, etc.) are well defined.	2.00	5.00	3.76	.83
The investment in new products development (NPD)	2.00	5.00	3.75	.77
Innovation in management	3.00	5.00	3.75	.86
Comprehensive and efficient organizational system, structure and planning.	1.00	5.00	3.75	.91
Measuring customer satisfaction periodically and the results are used to drive improvement	2.00	5.00	3.72	.89
The degree of industry concentration /fragmentation	2.00	5.00	3.72	.96
Gathering competitors' information continuously	2.00	5.00	3.72	1.02
Every employee knows his/her job goal and objective and the contribution to the whole strategy and objectives	3.00	5.00	3.71	.77
Design, producing and delivery processes are coordinated to ensure timely introducing and delivering of products/service	2.00	5.00	3.71	.92
Expertise in customer service	1.00	5.00	3.70	.86
Availability of capital	1.00	5.00	3.70	1.17

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Providing established product to exist market (differentiation on price, quality and other values comparing to competitors)	3.00	5.00	3.69	.60
Providing new products to exist market	2.00	5.00	3.69	.70
The supplier and partnering processes are well managed	2.00	5.00	3.65	.86
The competitive information (competitors, environment and technology, etc.) are collected and analyzed to improve the performance of overall activities	2.00	5.00	3.65	.93
Strategy is clearly translated into company, department, unit and individual goals and objectives.	2.00	5.00	3.63	.89
The executives' managerial experience	1.00	5.00	3.60	.99
Expertise in product/service development	1.00	5.00	3.60	1.19
Appropriate performance measurement tools and skills are employed.	2.00	5.00	3.59	.87
Employees and departments know how to measure their performance. The results are used to improve the overall performance	1.00	5.00	3.56	1.15
The entry barriers of the industry	2.00	5.00	3.50	1.04
Expertise in management	1.00	5.00	3.50	1.15
The leader team considers employees' idea when planning the company's future.	2.00	5.00	3.47	.94
The organizational structure and system is renewed based on environmental changes.	1.00	5.00	3.47	.94
Strategies are deployed through a framework of key processes	1.00	5.00	3.41	1.06
Each goal is measured according to designed measure criteria	2.00	4.00	3.35	.61
Technical resource (patents and so on)	1.00	5.00	3.35	.99
Appropriate managerial system with improvement capability adaptive to resource and environment changes	1.00	5.00	3.25	1.25
Government regulation in the industry	2.00	5.00	3.22	.88
Access to low cost distribution channels	1.00	5.00	3.15	.99
Access to overall low cost factors of production	1.00	5.00	3.15	1.04

The survey shows that the top ten most important factors for success, as rated by the interviewed executives, are:

1. Responding to customers' needs swiftly
2. Clearly defined and selected customer groups and market segments
3. The reputation of the company
4. Adapting to new industry and market trends
5. Strategies based on target customers and relative market/environmental factors
6. A definite strategy
7. Understanding changes in technology
8. Production/delivery processes that meet customer, quality, and operational performance requirements
9. Understanding and learning about customers, anticipating customer needs, and developing business opportunities
10. Enterprise culture

The survey result shows that a swift response to customers is believed to be the most important factor; in fact, customer service is believed to be most SMEs' key competitive advantage, as compared to large organizations. Swift responses to customer needs require an understanding and anticipation. At the same time, the company's internal processes should meet the customers' requirements on product quality, delivery times, and so on. The importance of the last two factors was also confirmed by this survey. They were ranked in place 8 and 9 among the first ten most important factors. Besides the capability to response to customers' needs, clearly defined and selected customer groups and market segments is the second important factor that impact SMEs' performance. This is regarding to the strategy in SMEs in the ICT industries.

Among the first ten important factors, three items are relevant to strategy, suggesting that they play a super-important role.

5.5 DISCUSSION: DETERMINANTS OF SUCCESS AND PERFORMANCE MEASUREMENT

Critical success factors determine SME performance. This section discusses how a company can measure its performance against others in the field.

5.5.1 Measuring Team Performance

Teams in SMEs in the ICT industries have common features. The innovative R&D team is most typical. Heinz, Baga et al. (2006) studied cooperation and leadership as the success factors in innovative R&D projects on electronic platform. Their research shows that leadership and cooperation, including goal clarity, decision autonomy, team trust, team spirit, open vertical communication, frequency of personal contacts, are positive to team performance. The interview and case studies conducted in this study show that most SMEs in the ICT industries can be treated as an innovative team.

Measuring a team's performance is a challenging issue. Meyer (1994) mentioned that teams usually spend too much designing measures that employ incorrect measurement metrics. For example, many teams often use measures like profits, market share, and cost, which are often used to score the performance of business. Meyer (1994) pointed out those inappropriate measures usually do not help team to improve their performance. Instead, scholars have shifted to other measures. For example, Hacker and Lang (2000) introduce three performance objectives for high-technology virtual teams: 1) performance against schedule (indicator: work load/work hours balance, action required to complete work on-time); 2) internal customer satisfaction (indicators: customer survey: on-time delivery, quality, communication overall satisfaction); and, 3) overall team health. In their research, the critical indicators for team performance include: team member survey, meeting attendance, individual performance reviews, division of labour, commitment letters. Indeed, Hacker and Lang's (2000) PM components suit the attributes of an innovative team.

Further, Mendibil and MacBryde (2005) employ four performance dimensions to measure a team's performance: team effectiveness (process outcome), team efficiency (internal team process), team learning and growth, team member satisfaction. Interview data collected for this study show that most SMEs in the ICT industries often employ intangible measures to evaluate the health of a team, for example: the atmosphere in working environment, the morale of the team, etc. The collected data seems to conform to Mendibil and MacBryde (2005)'s four dimensions (see section 5.3).

5.5.2 Strategy and Performance Measurement

An appropriate strategy depends on many factors, the most important of which is innovation. The main innovation includes technology innovation, products innovation, and management innovation. In Porter's Five Forces model (Porter 2004), threats of substitutes are an important motivator for innovation. In addition, it offers potential benefits to competitive market entrants. To achieve this kind of substitute, an enterprise needs improve their management to achieve the targets of decreasing cost, speed up delivery, and improve quality – referred to as a 'managerial substitute'. Second, the substitute utilizes new products, technology, and materials to compete with the current products in the existing market. This kind of substitute needs technological innovation in the products, called a technological substitute. In general, ICT SME strategies for innovation are broken into the following four categories (as shown in Figure 5-1).

Figure 5-1 Strategy Types in SMEs in the ICT Industries

Innovation	Product	Technology Substitute Provide new product to exist market	Explorer Provide new product to new market
	Management	Managerial Substitute Provide established product to exist market	New entrant Provide established product to new market
		Exist	Emerging
		Market	

Note: product include products, service and technology

- 1) Explorer: the strategy is to offer new products (or service; technologies) in order to construct a new market. This strategy has the toughest requirements on resource and capability among the four categories.
- 2) Technological substitute: providing new products (or service; technologies) to the existing market.
- 3) Managerial substitute: providing same products to the existing market with a lower price, higher quality, faster delivery, and better service, as compared with competitors.

- 4) New entrants: SMEs also can play the role of new entrants. This strategy allows them to provide products to a new market.

Most SMEs employ the Technological Substitute strategy, due to their limited resources. The Explorer strategy is associated with a higher risk; as such, few SMEs attempt it.

Kaplan & Norton (1996) mentioned that the Balanced Scorecard is often used by managers to accomplish the related management processes, which include clarifying and translating the vision and strategy; communicating and linking strategic objectives and measures; planning and target setting to align with strategic initiatives; enhancing strategic feedback and learning. It is obvious that all of these functions are derived from strategy. Indeed, the function of performance measurement can be evolved to the function of a strategic management tool. As described by Neely, Adams et. al. (2001) , the role of performance measure is four-fold: first, to help managers to track whether or not the strategies they have chosen are actually being implemented; second, to help managers to communicate these strategies within the organization; third, to encourage implementation of strategy; and fourth, to analyse whether the strategies are working as planned.

PM helps enterprises conduct planed strategies throughout the entire organization. For large organizations, strategy implementation is a critical issue. In large organizations, PM is a powerful tool. However, PM in SMEs should be used not only to conduct planed strategy, but also help in its choice. Among the companies interviewed in this study, there are three cases that suggest problems with strategy formulation and implementation in SMEs in the ICT industries.

Case 1: Company N is an online marketing service company. Its main business is to help Internet search companies set-up and manage advertisement-based selling networks. During the first three years, the company developed quite well; the employee base grew from 7 to more than 100 and it became an exclusive agent of many large web-searching companies in China. The success at the initial stage encouraged this company to implement a more ambitious strategy. It signed contracts with many web-searching companies that allowed them to overtake additional advertising channels.

However, the company shrank after investing too much in the business. The CEO admitted that they focused on business growth at the expense of profit. Before failure, the company did not take profit measures as an important indicator. Their strategy was to enlarge the sales volume to achieve scale effectiveness. However, it did not work.

Case 2: Company M is an electronic parts manufacturer with nearly 100 employees. The managing director is now worrying about the company's strategic objective. In contrast, the owner claims that the company will IPO in 2010. However, there is no implementation plan for realizing this strategic objective.

Case 3: Company F is a company specializing in telecommunication product research and development. Venture capital companies have funded the company and, since, it has undertaken a very ambitious strategy – it conducts research in three different product categories, simultaneously. The company is now falling into trouble because of insufficient resources.

The above cases show that strategy planning and management in SMEs are quite different from that of a large organization. Moreover, PM in SMEs should play a different role in the strategy management. This is based on the following observations:

1. SMEs do not take every aspect of external and internal factors into account when they make strategies. Their strategy plan does not follow a comprehensive and in-depth analysis. They seldom employ a formal and feasible strategy management framework to plan strategy. Strategic objectives in many SMEs were often decided by their owners' personal idea.
2. After choosing the strategy objectives, many SMEs have no executable plan. Few of them implement PM framework to help conduct the strategy management.
3. Many SMEs are not aware of the risk involved in each strategy..
4. Some SMEs do not choose correct performance objectives when they implement a specific strategy. Misleading performance indicators often confuse SMEs.

Generally, the role of PM not only helps SMEs conduct strategy, but also helps them weed through relevant data.

Broadbent (1999) introduced Miles & Snow (1978) matrix and the appropriate performance measure for different organizational types. The Miles & Snow (1978) matrix categorizes organizations into three classifications; each type of organization emphasizes different performance measures (Table 5-10).

Table 5-10 Organizational Classifications and Performance Measures

Organizational type	Characteristics of strategic decision-making			Performance measures
	Dominant objectives	Preferred strategies	Planning and control	
Defender	Desire for a secure and stable niche in market	Specialization; cost-efficient production; Marketing emphasizes price and service to defend current business; tendency to vertical integration	Centralized, detailed control; emphasis on cost-efficiency; extensive use of formal planning	Cost-efficiency of production methods, Customer satisfaction, Maintenance of relative market share
Prospector	Location and exploitation of new product and market opportunities	Growth through product and market development; constant monitoring of environmental change; multiple technologies	Emphasis on flexibility, decentralized control, use of ad hoc measurements	New product from research; Speed to market for new products; Development of new technologies; Flexibility of manufacturing capability; Growth of market share
Analyzer	Desire to match new ventures to present shape of business	Steady growth through market penetration; exploitation of applied research; followers in the market	Very complicated; co-coordinating roles between functions (e.g. product managers); intensive planning	Development of market share; Product development

Source: Broadbent (1999)

Above table shows the performance measures vary according to the organizational types. Most SMEs in the ICT industries can be categorised into the type of prospector, for which, the chosen performance measure emphasizes on innovation.

Based on above discussion, the relationship between strategy and PM can be summarised as follows:

1. Compared with that in large organizations, the emphasis of managing strategy in SMEs is focused on strategy selection. SMEs need to analyse internal and external factors, and deeply understand the industry and value chain. Proper PMs are important in providing reliable information.
2. PM helps planed strategy to be implemented. First, PM helps SMEs transform the strategic objectives into executable sub-objectives. Second, correct performance measures helps SMEs check whether the strategies were well executed in the organization. Third, measurable performance objectives

stimulate employees to participate in work, in order to achieve strategic objectives and motivate communication among employees and departments.

3. For SMEs, flexibility of strategy planning is critical. PM helps SMEs to be aware of internal and external changes and to provide instant response to the changes.

To meet the requirement for strategy management in SMEs, adjusting PM to fit ICT SMEs' nature is necessary. First, PM should provide enough support for SMEs to choose correct strategies. This means PM must provide both correct internal and external information for SMEs. Second, PM should fit to the nature of strategic flexibility in SMEs. Therefore, the PM structure is also a flexible and changeable system. Institutional and traditional PM structure does not suit SMEs. Third, selecting appropriate performance measures to evaluate strategy implementation is very important. Incorrect performance measures will mislead SMEs' strategic direction. According to SMEs' organizational nature, intangible measures are more important than tangible indicators, growth indicators are more important than absolute scales, and long-term measures are more important than short-term measures.

Based on above discussion, measuring an SME's strategy can be conducted by assessing the following aspects:

- 1) Does the SME have a well-definite strategy?
- 2) Does the SME's strategy fit to its resource and capability?
- 3) Does the SME take environmental factors into account when they select strategy?
- 4) Does the lead team consider employees' idea when planning the company's future?
- 5) Is the strategy developed, reviewed, and updated periodically, based on the customers, environment, and performance measurement?
- 6) Is the strategy clearly translated into company, department, unit, and individual goals and objectives?

5.5.3 Capability and Performance Measurement

There are many definitions about the capability in enterprises.

Chandler and Hanks (1994) studied the relationship between resource-based capabilities and firm performance. They defined ‘capability’ as the capacity for a coordinated set of resources to perform a specified task or activity. Even though Chandler and Hanks identified the difference between ‘resource’ and ‘capability’, they admitted the difficulty inherent in divorcing the concepts. In fact, their research dictates that measures of resource-based capability can not be differentiated from its resource. For example, they use six items describing resource supportive of innovation: (1) innovative marketing people, (2) marketing expertise, (3) employees that are very good at marketing the product, (4) expertise in product development, (5) technical expertise, and (6) innovative employees. They use five items to represent resource-based capabilities related to quality: (1) employees trained to provide superior customer service, (2) expertise in customer service, (3) top quality customer service training, (4) managerial expertise, and (5) expertise in process technology. Finally, they use seven items to represent resource-based capabilities related to cost-leadership: (1) low-cost raw materials, (2) low-cost distribution channels, (3) low-cost labour, (4) low-cost factors of production, (5) availability of capital, (6) highly productive employees, and (7) leading edge plant, equipment, and production facilities.

Grant (1991) believed that resources are the source of a firm’s capabilities, while capabilities are the main source of its competitive advantage. He defines the linkage by saying that the *“capabilities of a firm are what it can do as a result of teams of resources working together. A firm’s capabilities can be identified and appraised using a standard functional classification of the firm’s activities (p. 120).”*

Neely, Adams et. al. (2002) defined capability as the “combination of an organization’s people, practices, technology and infrastructure that collectively represents that organization’s ability to create value for its stakeholders through a distinct part of its operations”. From this definition, capability is the ability of combining resources and utilizing them to create value. Neely, Adams et. al. (2002) Believe measuring the capability should be from six different components:

- A customer order handling capability
- A planning and scheduling capability

- A procurement capability
- A manufacturing capability
- A distribution capability
- A credit management capability

Based on the theoretical framework of resource-based view (RBV), Eisenhardt and Martin (2000) introduce the concept of dynamic capability as “the firm’s processes that use resources-specifically the processes to integrate, reconfigure, gain and release resources-to match and even create market change. Dynamic capabilities thus are the organizational and strategic routines by which firms achieve new resource configurations as market emerge, collide, split, evolve, and die.” They claim that dynamic capabilities are a set of specific and identifiable processes such as product development, strategic decision-making, and alignment. Their value for competitive advantage lies in the ability to alter the resource base: create, integrate, recombine, and release resources. Further, the “dynamic capabilities” approach stresses an exploitation of existing internal and external firm-specific competences, in order to address the changing environment (Teece, Pisano et al. 1997).

This study employs the concept of dynamic capabilities (Eisenhardt and Martin), i.e. a firm’s capability is based on its ability to integrate, reconfigure, gain, and release resources. The firm’s capability can be viewed as ‘organizational capability’, as compared to its use and development of personal capabilities. Note that ‘personal capabilities’ can be categorized as a firms-resource. For example, an executive’s managerial experience is considered to be firm-related resources, over and above its use as in organizational capability. The most significant difference between ‘capability’ and ‘resource’ is that the latter can be ‘rented’ from an external source; in contrast, capabilities can only be developed by a company itself. Resources can be ‘purchased’ using tangible costs. However, capabilities can only be formed through a long operation-period. Capabilities can be improved through managerial innovation, learning, and training; they also have inherent potential competitive advantages. Besides the ability to organizing resources, organizational capability is also responsible for perceiving, controlling, and reacting to external environmental factors. For SMEs in the ICT industry, following components of organizational capability are very important: 1) the products / service development capability, 2) the market developing capability, 3)

operation /process designing and innovation capability, 4) risk controlling capability, 5) distribution capability, 6) decision making capability, and 7) resource-gaining capability. Hence, the following items should be measured in order to correctly assess a firm's organizational capability:

- 1) Has the company identified its core components?
- 2) Does the company have a comprehensive view on its capability (strengths and weakness, gaps, and so on)?
- 3) Does the company have a systemic development plan (training system, learning compensation system, knowledge management system)

5.5.4 Customer Satisfaction and Performance Measures

Customer satisfaction refers to how well the products or services meet customer expectations. Measuring SMEs' customer satisfaction in the ICT industries present following features:

First, since many ICT SMEs' customers are enterprises that are not end-users, perceived value³ is a key measure in assessing customer satisfaction. Second, measuring customer service employees need to be technically savvy – a general customer satisfaction survey cannot obtain sufficient and specific information. Third, because technology is upgraded quickly and the products are easy to be substituted in the ICT industries, customer loyalty measurements are sometimes invalid. Instead, by focusing on specific products in order to conduct customer satisfaction measurements, one can gather more valuable feedback. Fourth, as mentioned in literature review chapter, ICT industries are highly collaborative; hence, internal customer satisfaction is as important as external.

Based on these features of customer satisfaction measurements in ICT SMEs, the following industries should be considered when designing customer satisfaction measurement metrics:

Firstly, companies should setup formal customer information collecting, recording and analyzing mechanism; they should collect and process customer feedback daily.

³ Perceived value refers to the overall price divided by quality or the overall quality divided by price. Perceived value is measured in many ways including overall evaluation of value, expectations of price that would be paid (Smith 2007).

Secondly, it is better for an ICT SME to design customer satisfaction measurement on its own; it should not be outsourced. Third, companies should measure both internal and external customer satisfaction.

5.5.5 Managing Internal Process and Performance Measurement

Neely, Adams et. al. (2002) has identified one key reason for strategic failure – the organization's processes are not aligned with its strategies. The business processes can be generally put into four separate categories: develop products and services, generate demand, fulfil demand, and plan and manage the enterprise. Each category includes many sub-processes. These processes can be measured using five kinds of measures in terms of quality, quantity, time, and ease of use and money. These measures can be categorized into two groups: effectiveness and efficiency (Mills, Platts et al. 2002).

Cooperation between enterprises is very popular in ICT industries. Most SMEs in the ICT industry, particularly SMEs specialised in R&D play a small role in the whole value-chain process. Process management often needs extend to their strategy partners. This practice increases the difficulties on process performance measurement in SMEs in the ICT industries. Porter's value-chain and process structure are appropriate analysis tools in analysing SMEs' process in the ICT industries.

When measuring processes, one should ask:

- 1) Do a company's processes collaborate with its strategic partners? The company's core process should be aligned with its business partners' process. The extended process can be formed into an integrated process for terminal customers.
- 2) Do the processes ensure that any change in customer requirements and technologies can be incorporated into products and service design and development?
- 3) Do the company's key support processes provide strong support for its core process?

5.5.6 Optimizing Resources and Performance Measurement

Resources constrain an ICT SME's performance. Measuring resource can help to predict and perfect its future performance. ICT SMEs at different development stages

have different resource requirements. The resource requirements also vary with companies. In section 5.2.6, the common resources needed for SMEs in the ICT industries have been investigated. The most important issue is that ICT SMEs need develop their resource continually, in order to achieve a sustained competitive advantage. This is consistent with the opinion from the resource-based view (RBV)⁴ on which resource-development performance is based.

Measuring SMEs' resources can be made from following aspects:

- 1) Has the company identified those resources they need to conduct the definite strategy?
- 2) Has the company identified the resource gap between need and acquisition?
- 3) Does the company have clear plan on resource requirement and development?

5.5.7 Organizational Agility to Environment and Performance Measurement

Many scholars have studied the relationship between performance and environment. Different variables have been used to describe environment. For example, Nicholas O' Regan (2005) used nine attributes to describe the operating environment of firms. They are: stable environment, dynamic environment, subject to technological change in processes, threat of substitutes, threat of new firms, threat from oversea firms, technological change in products, decreasing product cycle, and a changing regulatory environment. They apply Miles and Snow's (1978) typology to construct their categorical types. For instance, prospector-type firms prefer perceiving their environment as dynamic one, whereas defender-type firms perceive their environment as stable one. From another aspect, their research implies that the SMEs in different external environment would choose different strategies. Also based on Miles & Snow's (1978) typology framework, Hambrick (1983) examined multiple environments and found that defenders are better than prospectors on profitability and cash flow. Their research indicates that situational environmental variables had main effects on performance.

⁴ The **resource-based view (RBV)** is a theory used to determine the strategic resources available to a firm. The fundamental principle of the RBV is that the basis for a competitive advantage of a firm lies primarily in the application of the bundle of valuable resources at the firm's disposal (Wernerfelt 1984).

Matthews (1990) found that environmental uncertainty plays a role in influencing a firm's decision making and ultimately influence the firm's performance. Furthermore, the perception of environment is an important dimension in organizational decision-making.

Following Tsai, MacMillan et al.(1991) and Zahra (1993), Wang and Ang (2004) selected two particular dimensions of munificence and hostility to describe the environment. Their research shows that there is a positive relationship between environmental munificence and venture performance. Environmental munificence reflects the richness of opportunities for firms. It is a multidimensional concept including dynamism, the abundance of technological opportunities, industry growth, and the demand for new products in the environment. Environmental hostility shows the adversity of environmental force for firms. Hostility was reflected by unfavorable change and the intensity of competitive rivalry in the industry.

In this study, the concept of Task Environment is adopted; it has previously been used by Govindarajan (1984), in reference to the portion of the total environment that is relevant for organizational goal-setting and goal-attainment. In their research, the task environment is composed of four major sectors: (1) customers, (2) suppliers of material, labour and capital, (3) competitors for both markets and resource, and (4) regulatory groups such as governmental agencies, unions, etc (Neely, Adams et al. 2002). SMEs in the ICT industry often compete in the environment composed of the above four major sectors. Therefore, based on literature review, this study supposes that ICT SMEs should understand their customers' needs; capture the needs (and their changes, if applicable), and react to these needs efficiently, through internal processes. This will help to achieve a competitive advantage. At the same time, SMEs need keep a good relationship with suppliers and regulatory groups, maintain a good supply and regulation network, and gauge competitor products and abilities.

Therefore, an SME's performance in response to the environment should include: customer satisfaction performance; competitive performance; supplier cooperation; and the relationship with its community. Among these four aspects, customer satisfaction measurements are discussed in Section 5.5.4; however, this section only discusses the last three aspects.

Competitive performance measurements

In the ICT industries, competition in technology is popular. For SMEs in the ICT industry, competitive performance measurements are necessary. Competitive performance can be measured from following dimensions:

1. Competitive advantages in product and service: those which regard product or service value, as compared to competitors.
2. Marketing performance: a company's marketing performance and position comparing to competitors, e.g. the relative market share, the customer satisfaction comparing to competitors.
3. Managerial measures: managerial indicators that are relative to the competitors; for example, the employee turnover relative to competitors, the profit relative to competitors, or the NPD speed relative to competitors.

Suppliers' Cooperation Performance Measurement

An excellent supply network is a competitive advantage. ICT industries are rapidly changing, so good supplier cooperation helps SMEs get good competitive position and keep with the development of the industry.

Measuring supplier cooperation can be taken from following aspects:

- 1) Has the company join in the main player's network in the industry?
- 2) Does the company keep good relationship with its suppliers?
- 3) Has the company formed a strategic partner relationship with its suppliers?

Regulatory Groups' Relationship and Performance Measures

Regulatory groups' policy significantly influences SME performance. There are both positive and negative impacts on SMEs from the policies. For example, governmental industry policy often influences SMEs' funding, market, and technologic choosing. Therefore, maintain a good relationship with regulatory group and establish a supporting network is critical for SMEs. The measures include:

- 1) Does the company's strategy follow the governmental industry policy?

2) Does the company have good relationship with regulatory groups?

5.5.8 Innovation and Measurement

There are two comprehensive frameworks that refer to innovation performance measurement: EFQM Innovation (Legenvre 2008) and Oslo manual (OECD 2005). However, they measure innovation performance from a comprehensive perspective which is designed for large organizations. In literature, measuring innovation are mostly based on result-based measures, such as sales, profit, stock price, or market valuation. They do not suit SMEs' innovation practices. Indeed, the interview data shows that most ICT SMEs operate with very tight resource availability. Failure on product innovation always threatens an SMEs' survival. Hence, innovation measurement in ICT SMEs should be timely. Measuring innovation performance in ICT SMEs should consider the innovation-features analysed in Section 5.2.8.

Table 5-11 Innovation Performance Dimensions and Measures in ICT SMEs

Dimensions	Measures	Indicators
Innovation outcomes	Product quality	Satisfaction (from the opinion of the company); Satisfaction (from the opinion of the customers); The usability; The service cost; the competitive advantages...
	Profitability	Margin; profit
Innovation capability	Innovation training	
	Innovation culture building	
Innovation resource	People	
	Network building	
	Competitive information	
Innovation strategy	Integrate the innovation strategy with overall strategy	
	Strategy based on resource, capability and environment	
Internal innovation process	Project management process	Quality, delivery time, budget..
	Idea collection process	
	Customer feedback innovation info. Collection process	
Innovation measure and management	Innovation policies	
	Innovation process and procedure design	

Source: author

Note: the indicators for each dimension only provide some examples. An SME needs to quantify indicators based on its own situation.

The typical project measure is the “cash curve” (Mankin 2007). According to Moultrie, Clarkson et. al. (2006), both NPD processes and new product outcomes should be

measured in order to assess NPD performance. Based on the conceptual framework for present study, SME innovation performance also depends on the SME innovation resources, capabilities, environmental factors, innovation strategies, and innovation processes. The Table 5-11 helps break-down these recommendations:

5.6 SUMMARY

This chapter summarizes and discusses results from the interview data and questionnaire survey regarding performance determinants. The empirical study results show that SME performance is not only determined by its own internal factors, but also by many external factors. Analysis of the data collected in this study confirms that an SME's performance depends on its capability, resources, environment, strategies, internal processes, and innovation. Therefore, both internal and external performance factors should be taken into account when measuring an SMEs performance.

The interview data shows that high-performance SMEs have a common feature: they all focus on long-term performance improvement rather than short-term performance. High-performance SMEs emphasize improving their own capabilities while efficiently utilizing available resources. At the same time, they can formulate right strategies and improve internal process continually. Furthermore, high-performance SMEs quickly respond to changes from the external environment. They focus on their customers' requirements and continually improve their core competitive competence.

The capability of response to customers' needs is the most important factor for an SME's performance. In addition, the capabilities of catching technology changes and adapting to environment are critical success factors for SMEs in the ICT industries.

Internal process plays an important role in ICT SMEs' performance even though SME internal processes are usually quite simple. The internal processes usually include three divisions: quality management, cost control, and delivery management.

The interview and survey data also show that there are specific features in performance determinants in SMEs in the ICT industries. These features should be considered when measuring the performance determinants. To further reveal the PM features in the ICT

SMEs, the next chapter will investigate the implementation of PM in ICT SMEs based on the questionnaire survey data.

CHAPTER 6 IMPLEMENTATION OF PM IN SMES IN THE ICT INDUSTRIES (I)

6.1 INTRODUCTION

This chapter discussed the findings from interviews and questionnaire surveys about implementation of performance measurement in SMEs in the ICT industries from following aspects:

- The performance-measurement areas targeted by the SMEs;
- The performance frameworks and tools employed;
- The reasons to implement PM;
- The barriers to implementation of PM;
- The significant problems in implementing PM;
- The key performance indicators employed ; and
- Suggestions on PM from interviewed executives.

The research objective in this chapter is to investigate the current implementation of PM in SMEs to reveal the requirements and features of PM implementation.

6.2 THE AREAS OF IMPLEMENTATION OF PM IN SMES

According to the concept of balanced control, an innovated enterprise not only measures traditional financial measures, which tells the story of past events, but also needs to measure future value in customers, suppliers, employees, processes, technology, and innovation (Kaplan and Norton 1996). To investigate how SMEs in ICT industries keep a balance in the four aspects of finance, customers, internal process, and learning/innovation, four typical implementation areas were surveyed: (1) financial performance measurement; (2) personal performances appraise; (3) customer satisfaction measurement; and, (4) quality management measurement (i.e. ISO9000, TQM and so on). The results (see Table 6-1) show that in the surveyed SMEs, financial performance and customer satisfaction measures are the most frequently utilized. Of the

45 companies that participated in the survey, 34 (75.6 percent) stated that their company measures financial performance; 66.7 percent measure customer satisfaction, 62 percent measure personal performance, and 22.2 percent reported that they conduct quality measurements. Finally, 3 companies among the 45 indicated that they do not measure anything.

Table 6-1 the Performance Areas that ICT SMEs Measure

The areas measured		Percentage	Responses
Financial performance measurement		75.6	34
Personal performance appraise		62.2	28
Customer satisfaction measurement		66.7	30
Quality management measurement (i.e. ISO9000, TQM and so on)		22.2	10
Other		11.1	5

Source: data from questionnaire survey (n =45)

Among 16 interviewed companies, 15 companies measure their financial performance except one company that is still at starting up stage, 4 companies employs an ISO9001-2000 quality certification system, and 1 company measures customer satisfaction, and 1 company formally appraises employee performance.

The above survey and interview data show that few SMEs in the ICT industries employ balanced performance measurements, even though many have accepted the balanced control mindset. Financial indicators are still the first choice for most SMEs. However, SMEs have started to measure customer satisfaction and personal performance. This result is contrary to McAdam (2000), which argued that “it seems that measuring customers satisfaction is an ‘adding bureaucracy’” because that SMEs contact with customers straight.

6.3 PM MODELS AND TOOLS EMPLOYED BY ICT SMES STUDIED

Table 6-2 presents the performance models and tools employed by interviewed companies. It shows that the KPIs system, or ISO9001-2000, is widely used in SMEs that have achieved success. In contrast, mediocre companies usually have no systemic PM implementation.

Table 6-2 the Performance Models and Tools Employed by the Companies Interviewed

Case code	PM employed
EP01 Company D	ISO9001-2000
EP02 Company E	KPIs
EP03 Company J	KPIs
EP04 Company I	KPIs
EP05 Company H	KPIs. ISO9001-2000
MP01 Company K	ISO9001-2000
MP02 Company M	ISO9001-2000
MP03 Company N	PM based on project management. i.e. from the aspects of time, quality and budget.
MP04 Company O	Personal business commit; Performance measurement through business plan, monitor, record and evaluation
TF01 Company C	No PM system. Doesn't have plan; just do it running from one thing to next.
TF02 Company L	Measure profit. At the first stage, chasing the sales growth lead to business failure
TF03 Company F	The performance measurement process is based on project management.
NV01 Company A	No any financial indicators and non-financial indicators to measure performance; make budget and schedule every month; have very good financial planning
NV02 Company B	Don't use any of formal PM framework
NV03 Company G	Project management
NV04 Company P	Besides measure financial indicator, measure the stability of team, the competitive advantages. The development of competitor is being watched.

Source: interview data collected for this study

From the survey data result (see Table 6-3), key performance indicator systems and bench markings are used by 46.7 percent and 33.3 percent, respectively. The Balanced Score Card is used by 13.3 percent, while 8 respondents (17.7 percent) have obtained an ISO9000-2000 Certificate. Only a few SMEs employ the Business Excellence Model, e.g. the Australia Business Excellence Model (2 companies), EFQM (1 company), or the Baldrige approach (1 company). Among the companies responding the questionnaire, 35.6 percent claim that they did not use any performance measurement and management models or tools. Three SMEs report that they use their own designed performance system or other industry quality frameworks, such as the Australian Quality Training Framework (AQTF).

Table 6-3 the Performance Models and Tools Employed by the SMEs Surveyed

Models and Tools employed		Percentage	Responses
ISO9000-2000 certification		17.7	8
Total quality management (TQM)		8.9	4
Australian Business Excellence (ABE) framework		4.4	2
EFQM Excellence model (European business excellence model)		2.2	1
Baldrige Business Excellence criteria		2.2	1
Balanced Scorecard		13.3	6
Key performance indicators (KPI) system		46.7	21
Benchmarking system		33.3	15
None of above		35.6	16
Other		6.7	3

Source: data from questionnaire survey (n=45)

The collected data shows that few SMEs have employed comprehensive performance measurement and management system. This result is consistent with the research result of Barnes, Coulton et. al.(1998).

Most of the SMEs surveyed employed KPIs and bench-marking systems as their main performance measurement tools; this was noted as the most flexible approach, hence its use. This result was expected, as the literature review has made apparent: SMEs need a tailored, resource-saving performance measurement that aligns to their changing environment (Hudson, Smart et al. 2001).

The Bench Marking System and BSC are not commonly used by ICT SMEs. This result is in contrast with the study result by Kald and Nilsson (2000), which shows that the Balanced Scorecard (BSC) is widely adopted (61 percent of Nordic companies may be using a scorecard in the next two years). However, this survey result confirmed with the analysis of McAdam (2000), which claims that BSC's mechanization and inflexibility does not fit the flexible environment of SMEs and their inherent informal organizational structure.

Business excellence models include EFQM, the Baldrige Business Excellence criteria, and Australian excellence business framework. As mentioned in the literature review,

business excellence models are non-prescriptive and adaptable self-assessment models. The typical characteristic of an excellent PM framework is that all performance criteria are categorized into two groups: performance enablers and results. Stephens (Stephens 2000) evaluated the implementation of the Baldrige criteria in SMEs; his research shows that operational items are highly ranked, which might come from the fact that managers in small firms spend more time on day-to-day operations. Among the forty-five returns in this study, four SMEs employed business excellence frameworks (two employed the Australian business excellence framework while one employed the EFQM and Baldrige, respectively). In the end, the study has shown that business excellence models are not utilized by many SMEs although many relative modules are claimed to be developed for SMEs. In fact, performance criteria in business excellence models need further modification to suit the features of SMEs.

In the manufacturers among the SMEs, the ISO9000 quality management system was often used. This is most likely due to the diversity of the system. In the companies that achieved good performance, the criteria of ISO9000 are strictly conformed to and the system is regarded as a managerial tool. In contrast, some companies regard it as a certification rather than a quality management system because the implementation of ISO9000 is imperfect. For example, in company K, MD mentioned that “after getting the certificate of ISO9001-2000, the manager who is in charge of the job left, the quality control system in the company become very loose, many processes are short of monitoring.”

A company with a business that produces a wide-range of ICT products (e.g. design-production-sale) needs a comprehensive performance system. In contrast, SMEs that play a small role on the value chain need specific performance measurement systems. For example, SMEs whose business is product design, a project management tool is preferred, i.e. measuring performance from product quality, delivery time, and budgets. For manufacturers, quality management systems, such as ISO9001:2000, is popular. For ICT service companies, high customer satisfaction and good service quality are the main objectives. For ICT distributors, cost control, risk management, and inventory turnover are more connected to performance targets. Indeed, analysing PM implementation in the ICT SMEs should consider the SME's position in the value chain.

6.4 THE REASONS TO IMPLEMENT PM IN SMES

Table 6-4 summarizes the reasons why the ICT SMEs studied implement performance measurement and management system. Of the forty-five respondents, twenty companies (44.4 percent) indicated that they measure performance to ensure customer requirements. Twenty companies (44.4 percent) reported that measuring performance is for decision support – support the decision making at top-management level and the operating level. Nineteen companies surveyed in this study (42.2 percent) claimed that they implement performance measurement to highlight quality problems. The next three important reasons are: to provide feedback for people to monitor their own performance, to identify possible needs for changes in strategy, and to provide feedbacks for driving the improvement effort. Other reasons include: To determine the bonus to management and/or staff (24.4 percent), and to justify the use of resources (20 percent).

Table 6-4 Summary of the Primary Reasons for ICT SMEs surveyed to Implement PM

Reasons to implement PM		Percentage	Responses
To ensure customer requirements		44.4	20
To provide feedbacks for people to monitor their own performance levels		40.0	18
To highlight quality problems and to determine which areas need attention most		42.2	19
To identify possible needs for changes in strategy		37.8	17
To justify the use of resources		20.0	9
To provide feedbacks for driving the improvement effort		35.6	16
To help decision making at the top-management level		44.4	20
To help decision making at the operating level		44.4	20
To determine the bonus to management and/or staff		24.4	11
Others		8.9	4

Source: data from questionnaire survey (n=45)

The findings show that SMEs surveyed in this study realized that customer satisfaction and decision making are fundamental factors that impact a company's performance. This confirmed the conclusion from interviews, that choosing appropriate strategies are more important than strategy execution. The above survey result also confirmed the

research result of Kald and Nilsson (2000), which shows that helping decision support at the top-management level and the operating level are the first two reasons for performance measurement in Nordic SMEs.

6.5 THE BARRIERS TO IMPLEMENTATION OF PM IN SMES

Garengo, Biazzo et al. (2005) discussed the factors influencing performance measurement in SMEs, which include: (1) Lack of human resources; (2) Managerial capacity; (3) Limited capital resources; (4) The lack of explicit strategies and methodologies to support the control process; (5) The lack of a managerial system and formalized management of the processes; and, (6) Misconception of performance measurement. Not surprisingly, similar barriers to PM implementation (see Table 6-5) were cited by the interviewed executives.

Table 6-5 Summary of the Barriers of Implementation of PM in SMEs

Barriers to implementing PM		Percentage	Responses
No enough performance measurement knowledge		28.9	13
No time and no resource to do them		73.3	33
lack of a managerial system and formalized management of the Processes		31.1	14
Performance measurements do not suit our firm		13.3	6
The performance measurement tools or model are complex, we did not know how to tailor them to suit our company		6.7	3
Other		4.4	2

Source: data from questionnaire survey (n=45)

The biggest barrier in implementing PM in SMEs is related to “no time and no resource” (73.3 percent companies claimed it). The second barrier is “lack of a managerial system and formalized management of the processes”. The third barrier is “not enough PM knowledge and no idea how to tailor the PM framework for their company”. This survey shows that 13.3 percent of surveyed SMEs think that PM does not suit their company. Above survey results support the opinion that the PM framework for SMEs should be resource-saving (Hudson, Smart et al. 2001). The survey also shows that SMEs seek to implement PM systems in their company, but they lack sufficient knowledge (partly because most of the PM system is designed for large organizations).

6.6 THE PROBLEMS IN IMPLEMENTING PM IN SMES

Table 6-6 listed the PM issues that the interviewed company faced. In summary, the key problems include:

- a. SMEs have no PM knowledge.
- b. The formal existing PM system is not suitable to SMEs.
- c. It is hard to measure intangible and subjective area.
- d. It is hard to decide the KPIs in ICT SMEs because of the fast changeability of ICT industries.
- e. SMEs are busy with their daily work. They have no time to conduct PM system.

Table 6-6 Summary of Key Problems with Implementation of PM in SMEs Interviewed

Case code	The problem on PM implementation
EP01	Focus on day-to-day job. Not much time to think PM
EP02	Hard to setting PM objectives
EP03	The inability to measure subjective areas
EP04	Not much knowledge on PM;
EP05	Not enough communication with customers and not involving customers in PM system.
MP01	Hard to decide the KPIs because of the flexibility of market. No PM information to help reform internal process; no PM information for reward system; no definite strategies
MP02	Difficulty with performance objectives setting
MP03	Shortness of PM information to support project management
MP04	Rewards are not based on performance measurement information
TF01	Short of measuring resource deployment and resource utilization
TF02	The trend of ICT industries is changeable, how to utilize PM to help company keep in right direction.
TF03	Even though the VCs measure each stage's performance, the criteria of performance have not been executed appropriately.
NV01	Small creative teams are quite different from the teams in big companies; They have much more freedom, creativity. Too much managerial control not good for them
NV02	Problem with performance monitoring and coaching
NV03	The performance measurement system is not very formal
NV04	It is hard to measure intangible performance

Source: interview data

Table 6-7 illustrates the survey results about the problems commonly faced in PM implementation. The question that measured this was open-ended and, based on the responses, it is clear that problems exist in every stage of PM implementation.

It is not surprising that most respondents report that the biggest issue is related to resources such as time, cost, and PM-implementation learning curves.

The second highly ranked problem is cultural acceptance. Responses reflect that it is hard for employees in SMEs to accept a formal and comprehensive performance measurement and management system. Even when an SME has employed a performance measurement and management system, the required procedure of implementing PM is not strictly followed. At the same time, four respondents report training and education as a big issue in implementation.

Setting reasonable measures and performance objectives is another problem in implementing PM in SMEs. Actually setting appropriate metrics is a challenging issue for SMEs, as the choice is often confusing (due to the fact that most PM systems are tailored to large organizations). Additionally, small organizations have problems collecting and analysing data. For example, many respondents report that “clients are reluctant to give feedback” and there are issues in analysing the response. Finally, many SMEs doubt the efficiency and effectiveness of implementation PM in SMEs. Some believe that “there is no significant benefits if a SME employ PM,” and that “time and effect is an issue”; furthermore, “Continuous Improvement systems are complicated and the paperwork is heavy”.

The survey result is consistent with Sousa, Aspinwall et. Al. (2006), which revealed that PM-adoption obstacles are most often: training of employees, difficulty in defining new measures, information system currently used, cost, leadership, and flexibility of present quality system.

Table 6-7 Summary of the Problems in Implementation of PM and the Frequency of Respondents of in SMEs surveyed

Problems	Frequency
Resource/time/money; Lack of dedicated role; Too busy with day-to-day business, cost	18
Short of support from employees; Some employees don't take it seriously, do it just as a matter of formality; Cultural Acceptance; Mentality that business related job is more important than process	4
Inadequate time/effort from the management to train/educate people on PM	4
The required procedures of the measurement systems are not strictly followed	2
The reasonable of measures setting; Identifying appropriate metrics	2
Setting reasonable objectives	1
The efficiency of PM communication	1
The modification of PM	1
How to get real performance and identify weakness	1
Connect performance to reward	1
Benchmarking outcomes	1
Time vs Productivity	1
PM can not be implemented absolutely	1
Limited experience with setting up processes from scratch	1
Effectiveness	1
clients reluctant to give feedback	1
analysing the responses	1
Lack of experience	1
No knowledge	1
Not enough obvious benefit	1
Time to stop and measure	1
Lack of reporting	1
Subjectivity	1
Quantification	1
Over complete data entry requirements	1
Employee Knowledge and commitment to process	1
Continuos Improvement system is complicated and paperwork is heavy	1

Source: data from questionnaire survey (n=45)

6.7 KPI EMPLOYED BY THE STUDIED ICT SMES

The data collected from the interviews show that performance indicators vary from company to company. However, the following financial and non-financial indicators are often implemented (see Table 6-8).

Key financial indicators:

- 1) *Profit*: Profit is one of the most important indicators for SMEs. As a result, many measure it. But for some new ventures that are specialized in R&D (where few if any products are sold in the market) this indicator is not appropriate.
- 2) *Growth indicators*: Growth indicators are most important to SMEs. The key growth indicators include: sales growth, income growth, profits growth, cost growth, and margin growth.
- 3) *Cash –flow*: a few SMEs have noted the importance of the measurement of cash flow. For the companies that have operated for several years, especially for the companies that have temporary financial frustrations, cash-flow is taken as a very important indicator.

Key non-financial indicators:

1) *Customer orientation*:

Customer service calls; sales volume from single customer, the growth of customer number, the distribution of customer; the important customers recruit; customer satisfaction.

2) *Market orientation*:

Market penetration (such as increasing the market share), new market entry, etc.

3) *Product orientation*:

The products are original with competitive advantage; the quality of a product.

4) *Project management orientation*:

Match R&D outcome to product development; the goals in the project plan; delivery time; quality; and, budget.

5) *Staff orientation*:

Number of staff; the specialists on the board; culture-building; the atmosphere of the working team; the employee's morale; the stability of team; etc.

6) *Intellectual property orientation*:

The number of IP a company has; the protection of IP; the value of the IP.

Table 6-8 the KPIs Employed in the Companies Interviewed

Case code	Key performance indicator (KPIs)
EP01	• Revenue • Profit • Margin • Customer calls
EP02	• Sales volume • Profit
EP03	Financial: • Revenue • profit ability Non-financial: • consult ability • employee education • Indicators on HR
EP04	• Revenue • Sales • Staff satisfaction
EP05	• Delivery time (schedule) • Budget • Quality • customer satisfaction
MP01	• Cost (budget) • ROI • Profit
MP02	• Growth on sales • New product development
MP03	• Quality • Delivery time • Budget
MP04	• Sale and • Money return; • Cost control • Building of the team; • Customer satisfaction • Cooperate between departments
TF01	• Growth of margin, • Cost • Intellectual property
TF02	• Profit • The atmosphere of working
TF03	• Work plan • Delivery time • Budget
NV01	• Budget • Schedule
NV02	• Total sales • Customer quality
NV03	• Sales volume • Profit
NV04	• Cash flow

Source: interview data collected for this study

The questionnaire survey results (see Table 6-9) show that SMEs take customer satisfaction seriously; this is because that most ICT SMEs are service firms. They contact with customers directly. In terms of financial indicators – profits, revenue, and cash flow are the first most popular indicators for SMEs. Besides market shares and growth, product quality and staff satisfaction are also key performance indicators.

Table 6-9 the Key Performance Indicators that Employed by SMEs Surveyed

Measures	KPI	Frequency
Financial indicators	Profit after tax; Profit; overall profit; Profitability	6
	Sales volume	3
	Revenue	3
	Turnover	2
	Cash flow	2
	Return Business	2
	Revenue growth	1
	The growth on annual sales volume	1
	Revenue per customer	1
	Increase profit	1
	Increase sales	1
	Forward Revenue Projections	1
	ROI	1
	The effective and efficiency on liquidity	1
Customer orientation	customer satisfaction, Client satisfaction, Customer satisfaction with our offering	7
	the rate of customer second visiting	1
	The rate on visiting customers	1
	Acceptance by peers & clients	1
	Revenue per customer	1
	Customer Referrals	1
Market orientation	The shares on market, market share	2
	the order number	1
	Market share growth	1
Product orientation	Efficiency/Productivity	1
	Efficient processes/systems	1
	Client retention	1
	Quality of product and service	1
	Quality of product service	1
	Reduced issue with the use of the product	1
	Reduction in time to support customers	1
	Adoption by new markets	1
Project management orientation	project completion	1
	Complete projects on time and within budget	1
	Turnaround time for project	1
Staff orientation	Employee Training	1
	Employee Turnover	1
	Working Capital	1

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	Billable hours	2
	Business expertise & experience	1
	Work satisfaction	1
	training completed	1
	Staff satisfaction	1
	Variety of jobs that take advantage of our skills	1
	individuals performance	1
	Attracting partners to take on larger jobs	1
Intellectual property orientation	Innovation	1
	Product development	1
	Social/Cultural Fit	1
	Time spent on bringing in/completing jobs versus administration	1

Source: the data from questionnaire survey (n=45)

Note 91.1 percent of the surveyed SMEs are ICT service companies (see Table 4-4). Fitzgerald, Johnston et. al.(1991) studied the performance measurement in the service industries. A results and determinants framework (see Table 3-4 in Chapter 3) has been introduced by them. Corresponding to Fitzgerald, Johnston et. al.(1991)’s opinion, this study suggests that financial performance measures are always important metrics for ICT SMEs. Profitability is the most important. Sale volume and cash flow are next. For SMEs, the former is important because of hard-to-get financial support. In addition, , Fitzgerald, Johnston et. al. (1991) argued that “relative market share and position,” “sales growth,” and “measures of the customer base” should be measured.

The strategy that a company employs influences its performance measures and indicators. Broadbent (1999) and Miles & Snow (1978) have categorized organizations into three classifications and each type of organization emphasizes on different performance measures (see Table 5-10 in Chapter 5). In Chapter 5, this study discussed these strategies, as four types: explorer, technological substitute, managerial substitute, and new entrants (see Figure 5-1). It is found that these four types usually emphasize different measures (see Table 6-10). An explorer employs measures that relate to the advance of the technology and the exclusiveness of the products or services. The technological substitute emphasizes on the new product development; the speed to market for new products. Managerial substitutes emphasize the measures on cost control, delivery time, the customer satisfaction, and the quality of products and

services. New entrants employ the measures on market share growth and new market entry.

Table 6-10 Organizational Types and Performance Measures in SMEs in the ICT Industries

Organizational type	Performance measures
Explorer	Advance of the technology; The exclusiveness of the products or services
Technological substitute	New product development; Speed to market for new products; Growth of market share
Managerial substitute	Cost control; The delivery time; The customer satisfaction; The quality of products and services
New entrants	Growth of market share; New market entry

Source: interview data and questionnaire survey data collected for this study

Based on the empirical study results and above discussions, the results-based performance measures in an ICT SME are summarized in Table 6-11.

Table 6-11 Results-based Performance Measures in SMEs in the ICT Industries

Financial indicators	Profitability, Sale volume, Sale growth, Cash flow
Non-financial indicators	Customer satisfaction, Employee satisfaction, Market shares
Specific indicators	Depend on the strategies the SME employed and the position of the SMEs in the value chain.

Source: finding from interview data and questionnaire survey data collected for this study

6.8 SUGGESTIONS ON PM FROM EXECUTIVES

The senior executives from the interviewed companies provided many suggestions on PM for SMEs (see Table 6-12). The executives interviewed believe that structured and formal performance measurement is not suitable for small companies.

“I think it needs not use a framework to manage. We need minimize management when the company is small”

– Executive P

“We did not use any performance model; we were doing things like customer relationship management. I guess when you are small, you don’t have plan. You just do it-running from one thing to next.”

– Executive C

“No, we don’t have any (performance measurement model) at the moment. It’s interesting the balance-score card was all over the world a few years ago, is it still

popular ... We are starting to focus more on (performance measurement). But right now we don't have any in."

– Executive D

It is claimed that micro and small companies' performance measurement is quite different from that in large organizations. The typical difference is that measuring micro and small companies' performance is measuring future. For example, executive A mentioned: *"The main difference is if you buying an existing industry company, you measure by sales and pockets; but if you invest a company like us, you are investing the future. We are making answers to the future problems. So the measurement is 'do we win when the future comes?'"*

For micro and small companies, measuring project or business plans is good practice, especially for product-design or R&D companies. However, plans should be flexible; some executives addressed this, such as B and C:

"One way of measuring our performance is whether we are consistent with our business plan. Since our business plan has maps out various things we will do on the product development side, on the marketing, sales, building a team-so I suppose the first general committee is our performance is measured against our plan we made out in the business map."

– Executive B

"I guess you should have project plan, and the project plan has goals for each stage, and to check whether we realize all those goals."

– Executive C

Some SMEs have realized that it is good for a company to setup a performance measurement mechanism. One critical issue is that a PM mechanism should work well with monitoring and training systems.

"It needs to clarify the responsibility of each position, and set up formal working flow. Set up performance measurement mechanism. Performance measurement system should connect to reward system, motivation and training system. Training is important."

– Executive O

In regard to performance measures, every executive interviewed agrees that non-financial indicators should be taken into consideration. The non-financial indicators include: staff satisfaction, professional staff growth, customer satisfaction, quality, and delivery.

“If considering performance measure, one should be staff-integration: how many staff you have, and how much is the growth income. The other thing I look at is profit growth.”

– Executive C

“Well, apart from the financial measures, delivering project on time and on budget, meeting the customers expectation are obviously critical factors in terms of how we measure our overall performance. Occupation of health and safety is an important aspect, professional staff growth, culture, the working environment, team work at all.....In a small company, the key factor of performance- we want to create the environment here where people are happy to work, and they gain satisfaction from their work. Well, there is of course underlying commercial remuneration that everyone counts.”

– Executive D

“People should be involved (in a performance measurement), the team of management. The quality and delivery, and budget of course, financial performance, really budget from our customers. For example, our customer may say we need this to be delivered with this budget. We need go to check what the bottom line is; if we agree, then we will say it okay, we will delivery it with this budget. If something happened, it will hit our bottom line. Measure the schedule ..., the budget .., quality of course.”

– Executive H

“We have revenue growth, short-term versus long term discussion. We must suspend levels of growth what’ve been over-stressing the quality. Quality is a very subjective problem. One is that we haven’t got objective... but that’s something we’ll be able to focus on one day. Give general survey in responsible people. So now that’s not something we’ve been able to focus on.”

– Executive J

Generally, the measures should be measurable and objectives. *“The key indicators in KPI that we use are that it must be measurable. The fact is that it must be something that is objective.”*

– Executive J

Table 6-12 Summary of Suggestions about PM Implementation Made by Executives in SMEs Interviewed

Case code	Suggestion and comments on PM
EP01	A SME needs measure following performance measures: • Financial measures • Delivering time • Budget • Meeting the customers expectation • A happy working environment • Employee satisfaction
EP02	PM objectives should be SMART (specific, measurable, attainable, relevant, time-based)
EP03	The key things: • The ability to engage with different clients. • Good team to be able to do that. So I guess teams underpin it. • Culture capability • Flexibility. • KPIs are measurable, objective; reliable. • PM based on role and job responsibility.
EP04	Employee learning and training are important for performance
EP05	• People should be involved in decision making • The team of management. • The quality of product • Delivery time • Budget • And financial performance
MP01	Setup a comprehensive PM system to break barriers between departments, to help managers to set up the knowledge of PM
MP02	The PM system should take current situation and future needs of the company into consideration
MP03	PM should guarantee good project management.
MP04	• It needs to clarify the responsibility of each position, and set up formal working flow. • Set up performance measurement mechanism. • Performance measurement system should connect to reward system. • Training is important
TF01	Measure goals based on project plan
TF02	From my experience, the BSC, KPI mostly were used in business unit level. For example, the sales indicators, the cash back and so on, they are all from the business unit.
TF03	Clarify job responsibility and PM criteria
NV01	The measures is quite different from exist industry companies. The measures in the companies like us are measuring that in the future.
NV02	Measuring our performance is on whether we are consistent with our business plan. Since our business plan has maps out various things we will do on the product development side, on the marketing, sales, building a team and so on
NV03	Simple PM system is better for small company
NV04	I think it need not use a framework to manage. We need minimize management when the company is small.

Source: interview data collected for this study

6.9 FURTHER DISCUSSION

This section presents further discussion on the features of, and requirements for, SME PM in the ICT industry.

This study shows that in the past ten years, a fair number of SMEs have employed a PM system in their operations. For example, 35.6 percent of SMEs surveyed have implemented KPIs, benchmarking, ISO9000 certificate, or BSC. The business excellence models (ABE, EFQM and Baldrige) are also employed by some SMEs. Based on the examined literature, small-firm leaders tend to spend more time dealing with day-to-day operational concerns and, SMEs as a whole tend to emphasize operational and financial performance measures rather than a balanced measurement (Stephens 2000). The most common reasons PM is employed (as found via survey) are: (1) to ensure customer requirements, (2) to help decision making at the top-management level, and (3) to help decision making at the operating level.

On the other hand, this study confirms some PM-related drawbacks noted in the literature. For example, many SMEs admit that PM systems are not well utilized in their operations and most of them implement PM to deal with discrete events. This supports the observation by Barnes, Conulton et al. (1998) – that performance indicators tend to be discretely employed, rather than continuously. This is not surprising, considering the PM resource requirements that typically hinder SMEs. Indeed, it is identified in the literature review chapter that SMEs, because of their shortage of resource and informal planing and managerial process, do not need institutional PM systems. The same reason explains the survey results in this study that KPIs is a popular measurement system for SMEs. Flexibility is the advantage of KPIs over other PM systems.

As seen in Chapter 3, the requirements for PM in SMEs present the following key elements:

- Manage uncertainty (measure both internal and external environmental factors)
- Help innovation on product and service
- Sustain evolution and change process
- Measure competitiveness
- Develop strategy

Align with ordinary processes on planning and reporting

Balanced

Tailored

Flexible adaptability

Dynamic adaptability

This study makes the following suggestions about the requirements for PM in SMEs, which are more executable than the above requirements (derived from the literature review).

- o SMEs employ PM to ensure customer requirements. A PM framework should help SMEs manage and measure customer satisfaction and help SMEs identify customer needs.
- o SMEs employ PM to support decision making at top-level and at operating level. PM helps SMEs to develop strategies and provide appropriate and adequate information for decision making.
- o SMEs employ PM to highlight quality problems and to determine which areas need more attention. PM helps SMEs to continue improving its processes.
- o The PM for SMEs should be a resource-saving system. To save resources, the PM framework should be tailored to SMEs, i.e. the PM framework should provide appropriate and adequate information. It can not over-measure and mismeasure the inappropriate information. At the same time, the PM process should align with an SME's daily processes, such as ordinary planning, budgeting, and reporting.
- o To overcome implementation barriers, the PM framework for SMEs should be flexible and dynamic.

6.10 CONCLUSION

This chapter analyzed and discussed the implementation of PM in SMEs, based on the results of interviews and questionnaires. The aspects of PM implementation were presented in: the performance measurement and management framework and tools that are being used in SMEs; the KPIs that SMEs employed; the issues SMEs met when they implement PM; and, the related suggestions from these companies.

It is shown that many SMEs employ performance measurement. Among the SMEs surveyed in this study, 75.6 percent companies have employed financial performance measurements, 66.7 percent companies have conducted customer satisfaction measurements, and 62.2 percent companies have implemented personal performance measurements. In terms of performance measurement, the Key Performance Indicator system is the most popular (46.7 percent surveyed companies have selected it, while 35.6 percent companies did not use any performance framework).

This empirical study shows there are no universal performance indicators that can be employed by all SMEs in the ICT industry. The indicators an SME should employ are based on the SME's type and the strategies the SME chosen (because of most SMEs in the ICT industries are ICT service companies, customer satisfaction is often employed).

The implementation of PM in SMEs has been further investigated based on four multi-case studies in next chapter.

CHAPTER 7 IMPLEMENTATION OF PM IN SMES IN THE ICT INDUSTRIES (II): MULTI-CASE STUDY

7.1 INTRODUCTION

In Chapter 6, the implementation of PM in SMEs was investigated based on the analysis of survey data and interview data. The result revealed that existing PM frameworks and tools do not fit ICT SMEs. This chapter presents four case studies for further investigation. The multi-case study in this chapter allows for an in-depth and comprehensive examination surrounding the implementation of PM in SMEs. The following issues were explored:

- Critical success factors that impact the case companies' performance
- The PM system the case companies chose
- Problems that SMEs met when they implemented PM
- Lessons and experience in implementing PM in the case companies

The analysis is presented and structured in the following way:

- a. The background of the case company;
- b. The PM issues the case company faced;
- c. The performance management systems that case companies implemented (the performance metrics, the performance framework & tools); and,
- d. Lessons and experiences regarding PM implementation in case companies

7.2 CASE STUDY 1

Background of the Company

Company A is a high-technology enterprise whose main business is designing and manufacturing electronic parts in the field of over-voltage protection. The company was established in March, 1999 in south China. The company has its own patents and has setup both a manufacturing centre and new product development (NPD) centre in two big Chinese cities. Its products are widely used in communication, power supplies, CATV, switching, household appliances, and so on. The company has 95 employees, half of which work in manufacturing and nearly 20 that work in product design. This company obtained the certification of ISO9001-2000 on design and manufacturing in 2004.

After nearly 10-years in business, the company is facing many issues and problems with its management. This case study will focus on the performance issues that the company experienced and how the company implemented PM to address them.

The Performance Issues Identified

After a period of quick development, the company confronted many issues in terms of its performance. The most significant issues are: (1) high cost in manufacture, (2) low response speed to customers' needs, (3) inefficiencies, and (4) low morale/high employee turnover.

The managing director (MD) claimed that these issues are the results of:

(1) No clear strategy or implementation plan

The company has a very aggressive vision. It declares that within five years, it should become one of the five biggest manufactures in its field; additionally, it will IPO before 2010. However, the company has not made any specific or feasible executive plan for these strategic objectives. In fact, there is no formal annual development plan. Senior managers are often confused by the owner's new ideas because they do not take any long-term performance objectives into account.

(2) The Serious Problems with the Internal Processes

Even though the company has obtained the certification of ISO9001-2000, it has many issues with its internal process. “There are two processes which have significant problems. One is the process of NPD; the other is the internal supply chain. Both are short of a systematic way to monitor and measure the routine operations.” (Source: MD in the case company, translated from Chinese)

The problems in NPD include:

- 1) Market information is not collected and conveyed to NPD team (resulting in little time to plan for its future products);
- 2) No performance measurement – because of the rushed operations, product metrics are not collected. Many quality problems have not been checked and troubleshoot, resulting in an unacceptably high number of defective units. More seriously, some products have not even been tested before being shipped.
- 3) Record keeping is virtually non-existent during the NPD processes. Employees often work in a status short of associated paper work, which sometimes makes it very hard to identify the problems in manufacture. Actually, there exists a process gap between NPD and manufacture.

The issue exists in the internal supply chain is that there is no plan on products’ material supply. “The supply department often stores too much of one material which is not demanded very urgently; on the other hand, the shortage of some other material is very serious, such that the amount cannot meet the normal producing requirement. This bad arrangement of internal supply chain raises the total cost of manufacturing” (Interview data, MD, translated from Chinese).

(3) Lack of Systematic Performance Management

There is no performance measurement system in the company. Hence there are no clear criteria set for people promotion and rewarding, which results in “many good employees have no chance for profession training and being promoted” (Interview data, MD). Without PM employees are less likely to take-on risk and responsibility because their actions are not properly identified and rewarded. At the same time, each position’s responsibility, capability requirement, and performance measurement criteria are not

clear. “It leads to a situation where people cannot be appointed to right position” (from the interview data, MD).

The Implementation of PM in the Company

To improve performance and solve the above problems, the company decides to implement KPIs and BSC to help conduct performance measurement.

First, PM is designed to cover middle- and top-level managers, each of whom are trained in the basic knowledge of performance measurement.

Second, the company’s KPIs and each business unit’s KPIs are setup. “The KPIs mainly derive from strategy, and then we use BSC to deploy the KPIs in each business unit. After that, we build up the KPIs for each position” (from the interview data, MD). The implemented KPIs are listed in the following table.

Table 7-1 the KPIs in the Company A (from the interview data, managing director)

The KPIs at company’s level	More than 10 indicators: ROI, budget...	
The KPIs at business unit level	Manufacture department	The delivery time, the products’ passing rate according to professional standard
	Supply department	The quality of raw material, the complaints from customers
	Finance department	Cost control
	Marketing and sales department	The finishing rate of sales task, the money return, the profit of sales
	HR department	Employee satisfaction, lost rate of people, and the achievement rate of training
	NPD department	Changeable, Up till now, there is no appropriate indicator measuring NPD’s performance

Third, manager performance is appraised according to adherence to the above indicators.

The Results of Implementing Performance Measurement in the Company

The company tries implementing PM in the whole company. The result does not live up to the expectation. There is no evidence showing that the company’s products quality improves or that customer complaints decrease significantly.

Through the implementation of performance measurement, the company realizes that performance measurement is not a separated managerial tool. It needs a holistic solution. Performance measurement is a system that spans the whole planing and

controlling process in an organization. Therefore, the company plans to work with consulting companies to design a new performance measurement system.

The Lessons and Experience in PM in the Company

This case provides an example on what performance issues face an SME as it attempts to solve issues through performance measurement and management. Data from interview and surveys show that the managerial issues that the company faced are quite typical for ICT SMEs. These issues include: (1) after a fast development period, how does an SME build its formal and standard management system from a bare-bones management process? (2) How does an SME setup its PM system to improve performance?

The practice of performance measurement is a systemic project that needs a holistic approach. The implementation should be based on the company's own practices. The company employed KPIs and the Balanced Card. This performance measurement failed to work, most likely because the approach was not tailored to the company. In fact, any existing large-organization system cannot correctly be implemented in an SME. In addition, the SME tried to measure too many things (more than 10). Because of resource constraints in SMEs, it is impossible to improve performance in all the perspectives in a very short time. Therefore, focusing on one or two main performance objectives is a more sensible solution.

To set performance objectives and indicators, a performance factor analysis should be carried out, which helps map out the performance cause-and-effect linkage. The leader's decision is quite important for successfully implementing PM; however, the involvement of all of the employees in the implementation of PM is even more important. In this case, only managers were involved in the action. This might be the main reason for the failure of the implementation of PM. Because of the enterprise politics, managers respond to managerial innovation more actively than common employees. However, every employee needs to understand his/her responsibility in order to contribute.

From this case, following lessons can be learned:

In Section 2.5, the PM implementation process is reviewed. The normal implementation processes include: (1) Strategy formulation; (2) Mapping of interrelated, strategic objectives with cause-and-effect linkages; (3) Choosing performance metrics and indicators, which reflect on performance objectives; (4) Analysis of performance information; and, (5) Action based on performance measurement results.

Generally, the performance objectives are derived from a company's vision and strategy. In this company, strategy was not clarified. No detailed analysis was given toward the company's external environment, its competence, or resources. In addition, the company did not systematically analyse its performance factors. The interrelated strategic objectives were not clearly defined. For example, the managing director identified serious problems in the internal process; however, the company did not take any measures to reconfigure the organizational structure and reform the internal processes. Finally, the company did not involve all employees in the PM action that was actually taken.

7.3 CASE STUDY 2

This SME copied a big company's PM system.

The Background of the Company

Company B specializes in providing professional telecommunication solutions in the VoIP area. It is a young SME with 6 years of history. It employs around 90 people, more than 40 percent of which work in new-product development (NPD) department.

The Implementation of PM in the Company

As a new, young SME, Company B has established its own culture and formal managerial system. Its core team members come from a big telecommunication company. In fact, the company copied the big company's culture and managerial processes.

1. Performance Management Infrastructure

In the literature, the Procurement Executives Association suggests that two fundamental components should be in place to manage performance effectively: 1) the right organizational structure, and 2) the ability to use performance measurement results to

actually bring-about change. To implement performance measurement and management effectively, the company has built fundamental performance management infrastructure which includes: (1) tailored and flat organizational structure, (2) flexible and effective internal process, (3) performance measurement and management mechanism, (4) performance reward and promotion mechanism.

- (1) Tailored and flat organizational structure – For an innovative ICT SME, simple and flat organizational structures are necessary. The main business units in the company are NPD and Marketing& Sales. Employees can move from NPD to Marketing & Sales flexibly. The company executes a three-level managerial structure (CEO-Managers-Employee). All managerial activities are carried-out through an IT-based system. For instance, employees can send emails directly to the CEO.
- (2) Flexible and effective internal processes – The Company’s internal processes are streamlined and accountable. The key processes go through the NPD to customer service, based on “the back-processes are the customers of Pro-processes.” The collaboration between departments is improved, based on performance information. Customer feedback is reported to the NPD and other responsible units although it is usually first processed in the customer service department.
- (3) PM mechanism – The company had setup a comprehensive performance measurement mechanism that includes:
 - a. Businesses plan (i.e. performance commission): each business unit and each employee are required to make out quarterly performance plan, which should first be discussed with and approved by its supervisor. The business plan (performance commission) is a performance measurement objective that is assessed at the end of the quarter;
 - b. Clarify performance measurement structure and relationship: The supervisors are responsible for measuring and appraising subordinate units and employee performance. On the other hand, the related performance assessment not only depends on a supervisor’s record and observation, but also on the feedback from subordinate processes and the collaboration with other business units;

c. Clarify performance measurement measures and measurement criteria –

Each department and position has clear job responsibility and performance measurement criteria. All performance measures and appraising criteria come from each employee, collaboration with the other business units, and the committed business plan (performance commission).

(4) Performance reward and promotion mechanism – In the company, the performance appraising results are used to determine the annual reward and promotions.

2. The PM Tools and Framework Employed

The case company employs a KPI system to setup its key performance indicators and implement the BSC indicators. With its R&D background, company B's key performance indicators relate to four aspects:

- (1) The growth in the sales.
- (2) The profit per person
- (3) The decrease of the cost on sales
- (4) Customer satisfaction

In fact, the company performance indicators are selected by analyzing the company's survival-status. To do so, the following questions were asked:

- 1) What is our strategy?
- 2) What are the critical success factors for our company?
- 3) What are the key indicators for our company development?

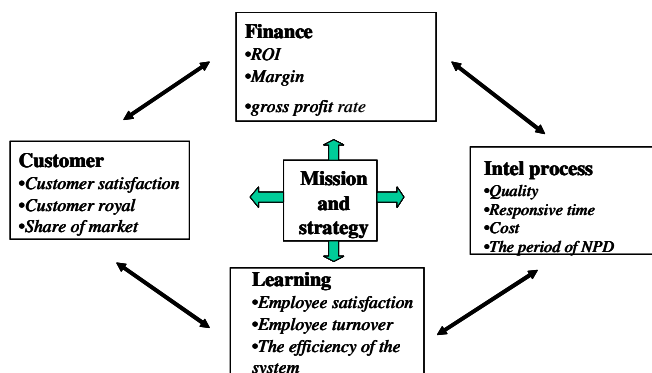
After answering the above, it was determined that the following business aspects should be kept in balance:

- 1) The relationship between development and risk control
- 2) The growth in volume and the growth in potential competence

- 3) Balanced development and putting great emphasis on some special developing strategy
- 4) The quantitative measurement and qualitative measurement

The translation of the strategy objectives into the performance measures in Company B was done through the BSC, as shown in Figure 7-1. The translation was based on the BSC's four dimensions. The company chose three or four performance indicators in each dimension, where: financial performance is measured by ROI (Return of Investment), Margin, and Gross profit rate. Customer service is measured by customer satisfaction, customer loyalty, and the company's share of the market. The learning dimension refers to employee satisfaction and training, which measures employee satisfaction, turnover, and HR efficiency. For internal Processes, the measured attributes are product quality, response time to customer requirements, operation cost, and NPD efficiency.

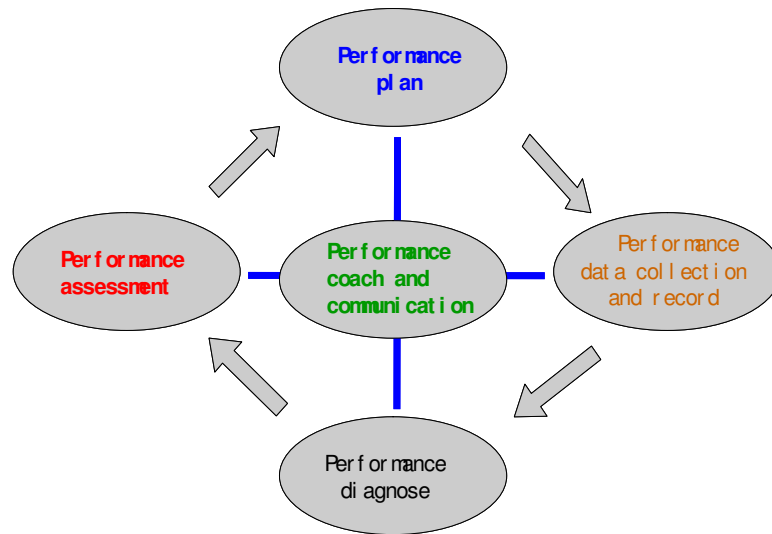
Figure 7-1 Performance Measures in Case Company B Based on BSC



3. The PM Procedure

Four steps are employed in the performance measurement in Company B: 1) performance plan, 2) performance information collects and observe, 3) performance diagnose, and 4) performance assessment. The performance measurement process follows the cycle of 1) to 2) to 3) to 4) to 1), as shown in Figure 7-2.

Figure 7-2 Performance Procedure in Case Company B



Step 1: Business (performance) plan

In this phase, the main task is to decide performance objectives. Company B decided that performance objectives should mirror the SMART approach (Specific, Measurable, Attainable, Relevant, and Time-based). When a business unit is making the performance plan, a SWOT (Superiority, Weakness, Opportunity, and Threat) analysis is also required.

The company's KPI come from the company's performance objectives based on analysis of key performance factors. Then, the company's KPIs were deployed into business units' KPIs and personal performance objectives. The performance objectives were deployed through two paths:

- (1) Deploy the objectives according to the organizational structure – This deployment follows the rule of “objectives- means”, i.e. the functional process necessary to achieve specific objectives.
- (2) Deploy the performance objective according to the internal process – This deployment follows the rule of “the next process is your customer.”

As previously mentioned, Company B makes performance plan and assesses performance results once a quarter. At the end of each quarter, every business unit and

every staff-member needs to assess its performance results and make the next term's performance objectives via a discussion with supervisors. The performance plan includes three columns: the KPIs, the means that need to be taken to achieve the KPIs, and the required cooperation with other colleagues and business units. Usually, the KPIs are those performance objectives that can be assessed by quantitative measures. The means are specific measures that a business unit/staff plans to take to achieve the performance objectives. Because of the features of the ICT industry, most jobs are executed by teamwork. The cooperation measures are taken as a very important indicator. It helps to break the barriers between businesses units, which make sure the company's strategy can be executed throughout the whole organization. At the same time, these measures also help employees to understand their own responsibility in the process.

Step 2. Collecting performance information – All performance information is collected daily. This information includes the supervisor's observation, fault record, the financial indicators from financial unit, the feedback from subordinate processes and the collaboration with other business units.

Step 3. Performance diagnoses – Supervisors and managerial units monitor performance in real-time. If there are big issues with the performance plan, supervisors and managerial units will analyze the issue and identify the reasons that lead to low performance. For this, the performance diagnose box (as shown in Figure 7-3) is often implemented.

Figure 7-3 the Performance Diagnose Box in Company B

Knowledge	Skill
Attitude	The external barriers

Step 4. Performance measurement (assessment) – The Company measures performance once per quarter. The measurement criteria are the performance plans which are set by employees and business units. The measurement not only compares the performance

results with the KPIs (which are set at the beginning of the quarter), but also takes into account the cause-and-effect analysis (which analyzes the issues met, identifies the causes that influence the performance, and suggests the measure that should be taken to overcome such issues).

The Results of Implementing Performance Measurement in the Company

Company B has a formal PM system in place. It is derived from those used in large organizations. Both positive and negative influences of PM were realized.

The positive aspect is best visible through the changes in the managerial structure. The PM system plays an important role in formulating an employee's behavior. Each understands the company's objective and personal performance objectives and has a clear responsibility and relative working-flow. The performance measurement provides the reference for employee promotion and reward.

The negative influences present in following perspectives: first, a large amount of paperwork was introduced (which wasted a great deal of employee and manager time). Second, the mechanistic system hindered the company, as it was unable to catch quick changes in the market. For example, when customers required the company provide a primary solution to some special operating situation, the company often was overtaken by competitors because of the slower flow of information. Third, most employees were unable to fully participate in the PM system because they did not have enough time. Fourth, performance objectives were not consistently set. Fifth, employees often focused on the specified performance measurement criteria while ignoring others.

Lessons and experiences from PM implementation in Company B

Although it is claimed that the performance measurement and management system was tailored from PMS in large organizations, it is not properly tailored to suit the SME's specific status. The PM in Company B is more like a smaller version of that in big companies. Almost all the procedure is kept without an in-depth analysis of the difference between the requirements on PM in large companies and that in SMEs. For example, in large companies, the tedious paperwork involved in PM does not influence the working efficiency because special PM team takes care of the work. While in SMEs, being short of human resource, all these work have to be undertaken by staffs who

should also focus on other professional work. Hence delays some urgent task. In fact, such kind of PM issues can be solved by adjust the procedure of PM implementation, e.g. setting different PM monitoring levels, the way to deal with PM problems can vary with the working status of different project teams.

Indeed, this case highlights the importance of flexibility of PM in SMEs.

7.4 CASE STUDY 3

This case provides an example on how a software designing SME implements PM to improve its service quality, customer satisfaction, and marketing performance.

The Profile of the Case Company

Company C is a software design and consulting company established in 1999 in south China. The company's core business is designing software to help enterprises to declare import/export at customs. The company's customers are import/export companies. To survive, Company C must obtain authentication to its import/export declaration software from the customs. Such an authentication is granted based on feedbacks from import/export companies.

The company has four branches and employs 80 people. Among them, 20 work in new product development departments, while 40 work as service staff. The company's R&D capability is strong enough to meet the design requirements regarding the related software.

There are big competitors in this field. Company C was originally a unit of government customs and, hence, it has a good relationship with the now separate customs unit. Despite this competitive advantage, it has not gained significant market share. In order to remedy this, Company C is eager to improve their performance regarding product quality, service quality, and customer satisfaction.

The Performance Problems Facing Company C

The performance problems are the following:

Firstly, customer satisfaction – By comparing with its main competitors, the company found that the biggest factors that influence customer satisfaction are not related to product quality, but rather quality of service.

Second, strategy – The company’s strategy is to provide high quality software to import/export enterprises. The small market dynamics and sudden changes in customs policy are potential threats to the company’s development. The company is currently making decisions regarding future strategy: to expand the business in a related field, e.g the implementation of ERP software, or providing more expert consulting service on custom import/export report software.

Third, service performance – It is hard to monitor the service process directly because of most employees work in customers’ place individually.

The PM system the company implemented and modified

The company implemented a KPI system to measure performance. The company-wide KPIs came from the company’s strategic objective; then, the KPIs were given to each business unit’s KPIs, in order to support the achievement of strategic objective. The KPIs in Company C are shown in Figure 7-4.

Figure 7-4 KPIs in Company C

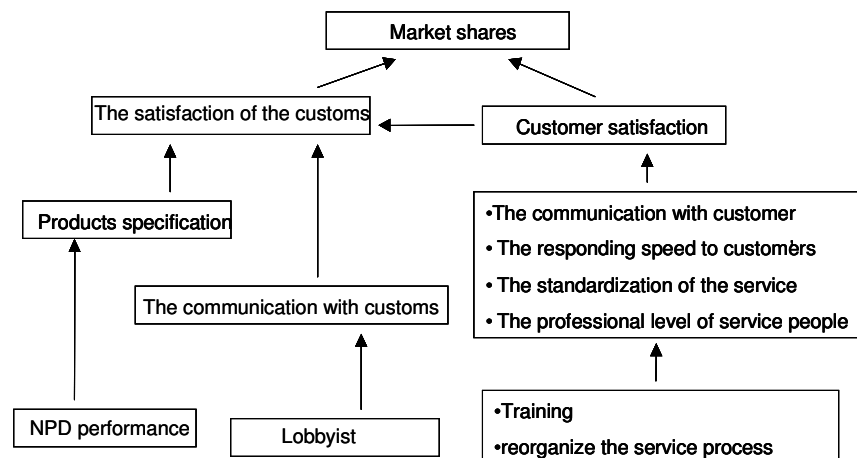
Company KPIs:	• Market shares • Customer satisfaction • Turnover per person
Business unit KPIs:	NPD: •Customer service call (times) •The new products deliver time •The cost controlling
	Service : •Customer satisfaction •Customer complain
	HR : •Human cost control •Satisfaction of training
Personal KPIs:	

To face the challenges of new competitive situation, the company undertook the following PM modifications:

First, it reviewed its strategic position. Expanding the business to enterprise management software like the ERP solution, the market of which has already been occupied by large software companies, might be over-ambitious for the company. Instead, deeply ‘cultivate’ its small market is more reasonable. In the end, the company focused on: the current market, developing current products, improving expert consulting services; in order to increase market share.

Second was a critical-factor analysis. To achieve the strategic objective, the company mapped the critical factors based on expected cause-and-effect (see Figure 7-5).

Figure 7-5 the Map of Performance Cause-and-Effect in Company C



Customer satisfaction is the key factor to gain market share expansion. Their customer satisfaction survey shows that following factors influence service quality significantly:

- 1) Communication with customers
- 2) Responding speed to customers’ issues
- 3) Standardization of the service
- 4) Customer service professionalism

Gaining authentication from national custom is the key to enlarge market share. Satisfying the national share depends on the product quality and the communication with customs. Based above analysis, the following measures should be taken: (1) Train and reorganize the service process to increase customer satisfaction, (2) Recruit a

lobbyist(s) to improve the relationship with customs; and, (3) improve NPD performance.

The actual changes included:

1. Employee training. This training not only emphasizes technological knowledge, but also includes customer-communication improvement. This company found that most of its poor customer-satisfaction examples were not due to product and service quality, but rather the communication skills of service engineers.
2. Reorganize the service process. This included a rebuild of the customer relationship management (CRM) IT platform, e.g. a specific module is designed in the company. This IT platform ensures: A) customer issues are recorded in the IT system; and, B) that customers can monitor and feedback the service quality through the IT service.
3. To collect and record performance data, everyday. Engineers are required to load everyday work and customer feedback in the CRM system. It ensures that every service job and step can be reviewed and monitored. At the same time, customer issues and associated feedback is collected and recorded.
4. All performance information is used to appraise employee, business team, and company performance.

After implementing above performance improvement activities, the company's customer satisfaction increased significantly and the market share has been increased by 5.4 percent.

Lessons and Experiences from PM Implementation in Company C

This case provides an example on how a software company implements the performance measurement to help its market share.

Recall in the literature review chapter, it is claimed that three elements should be involved in PM system: (1) a set of metrics and indicators; (2) a reporting process and delivery tool; and (3) a diagnostic and analysis tool (see Figure 2-8). The implementation of PM in Company C provides the evidence that confirms the above three elements. First, Company C has a set of KPIs employed throughout the entire organization. Second, the company collects PM information everyday, which was

reported, delivered, analysed, and utilized through an IT platform. Furthermore, the information collection was expanded to the customers. Third, diagnostic and analysis tools (e.g. the cause-and-effect map, SWOT, and critical factor analysis) were used by the company. This PM strategy was objective-oriented and, as a result, successful (see Chapter 3 for details on why this is a positive attribute).

The performance plan was based on an in-depth analysis of the internal and external performance factors, supporting the notion that a holistic, detailed plan is more likely to be successful.

7.5 CASE STUDY 4

A Brief Introduction of the Case Company

Company D was established in late 1990's in Melbourne, Australia. It. The company designs a range of products, from hardware and software to telecommunication equipment. Its core business is providing ASIC (Application Specific Integrated Circuits) and FPGA's (Field Programmable Gate Arrays) to a leading telecommunications company. Among the six founders, three had been full-time employees with the company's largest customer – the aforementioned telecommunications company. The other three founders are specialists in hardware and software design, and project management. Most of its competitors are from Indian and Chinese design companies, which typically have an advantage in low-cost human capital. Company D has maintained its position in the industry. The growth history is impressive: \$250,000 in 2003-04 to \$10.76 million in 2005-2006. In the same period, the staff grew from 5 people to 47 (as of 2006).

There are many factors contributing to the growth and success of the company. Based upon the present study, this research considers the PM system that, in this case, is focused on testing: 1) product quality control; 2) product deliver speed, and 3) budget management. Good communication between the company and its customers/ team leaders and team members assists the implementation of the PM.

The Critical Success Factors for Company D

The CEO is very proud of his company's performance. Talking about the key factors, he mentioned: "High competence people; very low overhead and flat management system – people in small company multi-task ... The other thing is, we have very good customer service – we always deliver our design on time. Excellent quality, high competence, and good communication and management strategy are our key factors."

Indeed, the company relied on:

- 1) People. They are always a very important factor that influences SME performance. In fact, teams conduct most tasks in an IT project. Therefore, a high-quality team composed of highly competent people is important.
- 2) In ICT SMEs, flat management is quite important. The ICT industry changes very quickly, so the senior manager should understand the technology and should communicate directly with the design engineer.
- 3) Good customer service is always a key performance factor.
- 4) Good communication with customers and team members is another key performance factor.

Implementation of PM in the Company

A tailored and effective performance measurement system is implemented in Company D. The PM system includes: (1) involving customers in performance plans and monitoring; (2) customer satisfaction survey; (3) internal performance communication; and, (4) an extended quality manager.

Customer-involved performance plan and performance monitoring

The company plans every project for which it contracted. Regarding performance, the plans include: the delivery day (schedule), budget, and what the company should do to try to complete the project as early as possible on budget. All plans are confirmed with its customers and are measured by the customers. The company's core business is hardware and software design, which is almost impossible to ensure with a zero-fault guarantee. To deal with such an issue, a strict test-process is established within the company. Since such designs will be used in the customers' final products, the company meets the customers regularly to discuss the issues on scheduling, budget, and test

results. In-depth communication with customers helps the company understand the customers' needs and achieves a high customer satisfaction.

Customer satisfaction survey

Besides regular customer meetings, the company executes customer satisfaction survey periodically. In fact, the company designs the questionnaire and sends it to customers by itself.

Internal performance communication

Several communication modes are employed in the company. Regular meetings are most often preferred. The company also uses a suggestion box, from which employee-ideas are considered. As a team-building exercise, the company has a pizza-day on one Friday of each month. On that day, all the employees have lunch together, trading ideas about the company.

An extended quality manager

The company employs a part-time quality manager. The manager looks after all of the quality issues, which is based on his expertise. He reviews what the company is doing regarding quality recording and quality management. The quality manager not only checks the quality issues in technological designs, but also reviews the quality of business operations, financials, employee satisfaction, IT systems, etc. Based on the reviews, the quality manager will communicate varying recommendations to project managers.

The Lessons and Experience in PM in Company D

This case shows that communication is quite important. On the one hand, communication with customers can help a company understand its customers' needs and improve the customer satisfaction. At the same time, customer feedback helps the company identify the performance issues quickly. Finally, internal communication is also very important and perhaps improves employee satisfaction.

To improve performance, a company should look after both internal and external performance factors. The customers involved in a performance plan have a beneficial effect on the company's performance improvement. A regular meeting with customers

is usually a positive way to communicate with customers. A personable quality manager can also help SMEs check the performance issues. Furthermore, a flat management system and highly-skilled team is also a great help.

This case provides the evidence that customer satisfaction is vital for product-driven SMEs. In this case, involving customers in and team-members in the production process has helped Company D improve its performance.

7.6 CROSS-CASE DISCUSSIONS

Among the four case companies: Company A is an electronic manufacturer with its own patent; Company B is a research and development enterprise in the telecommunication industry; Company C is a software designer and consulting company; and Company D specializes in telecommunications hardware and software design. Importantly, they each have features common to ICT SMEs, i.e. technological innovation is its critical success factor. In another respect, the four case companies represent the three typical ICT company-types discussed in the literature review: manufacturer with its own technology; research & development in new products; and, services and consulting.

Each company was unique in their PM implementation: Company A has setup its basic PM system, according to a theoretical framework; Company B has setup its PM system to mirror that of a large organization; Company C formed its PM system after analysing its own features; Company D setup an efficiency PM system with its own features. From the results of implementation, Company C and Company D's PM systems are more efficient than Company A and Company B's. It indicates that the implementation of PM in SMEs should be aligned with the company's features.

According to Tesoro and Tootson (2000), a practical performance measurement system is composed of three simple elements: (1) a set of metrics and indicators; (2) a reporting process and delivery tool; and (3) a diagnostic and analysis tool (as shown in Figure 2-8 in Chapter 2). Following paragraphs compare the PM implementation in the four case companies from these three elements.

Performance metrics and indicators. Company A employed the ones that are applied in large organizations, which are not derived from its own strategies and objectives. This is the main reason why the company's PM system is not successful. Company B employed BSC, with performance indicators categorized into four aspects: customers, finance, process, and learning; however, BSC's inflexibility does not fit well with SMEs (McAdam 2000). Company B's failure is perhaps linked to their PM choice. Company C's performance measures and indicators; however, are based on its own business features and strategies. The measures and indicators included: company and employee capability-development; resource utilization; environment adaptation; internal processes; innovation; and, performance results. Company D's performance measures and indicators were also derived from its business features and the company's strategies. The company tailored most performance indicators, such as budget, delivery time, and quality. At the same time, the company improved the communication with customers to achieve better customer satisfaction.

In Chapter 3, a conceptual framework for this study has been introduced. The framework argued that measuring SME performance should measure both internal and external performance determinants in addition to performance results. However, in the case of Company A, most measures and indicators employed are still financial indicators. Although Company B emphasizes the process measurement in addition to measuring the performance results, conducting the institutionalized PM costs a lot of resources. In contrast, Company C and Company D measures both leading and lagging performance indicators. The intangible leading indicators help SMEs identify the necessary actions in order to achieve proposed objectives. The lagging indicators are employed to exam whether the company is on the right track.

Reporting processes and delivery tools. Company A did not setup clear PM reporting processes. The performance information was not systemically collected, delivered, and utilized. Company B collected performance data and information following formal processes; they did have clear performance measurement structures. Company C designed its own IT platform to record and deliver performance information; the performance information was transmitted in real-time. Company D follows formal templates to collect, record, and communicate performance information. The performance information was delivered quickly within company and to customers.

Diagnostic and analysis tools. Company A employed BSC to deploy performance objectives. The company also imported an ISO 9000-2000 quality certification system. The reasons for the undesirable implementation result are that the diagnostic and analysis tools were not tailored to the company's needs. Actually, the quality certification system was not strictly followed (i.e. it was not a managerial tool in the company). In company B, the BSC and KPIs system were employed. Appropriate diagnostic tools were used. However, the diagnostic and analysis tools were used more as performance appraisal tools rather than managerial tools. In company C, performance information was collected and analysed instantly by the IT platform. In company D, the formal performance templates ensured that performance data was recorded and analysed effectively

7.7 CONCLUSION

This chapter presented and analysed the implementation of PM in four ICT SMEs. The multi-cases study shows implementation of PM in SMEs is a systematic project that needs holistic analysis. The implementation should be based on the company's own practice and situation.

This multi-case study confirmed that four components should be considered when an SME implements PM: build the performance management infrastructure; formulate strategy; analyse critical success factors that impact SME performance; and, choose the right measures and indicators.

The multi-case study shows that communication is important in implementing PM successfully. Communication with customers can help a company understand its customers' needs and improve the customer satisfaction. At the same time, customer feedback helps the company identify the performance issues quickly. On the other hand, internal communication is very important for a good teamwork. Indeed, employee satisfaction is based on the effective internal communication.

PART IV DEVELOPING PM FRAMEWORK AND IMPLEMENTATION PROCESS

INTRODUCTION

In the previous parts performance determinants, performance measures, and PM implementation were investigated. In this Part, a new PM framework for SMEs will be developed; implementation process of the PM framework will be discussed.

The following is divided into two chapters: Chapter 8, which develops a new PM framework, and Chapter 9, which provides a procedure that outlines the suggested PM implementation.

CHAPTER 8 DEVELOPING AN EFFECTIVE PM FRAMEWORK FOR ICT SMES

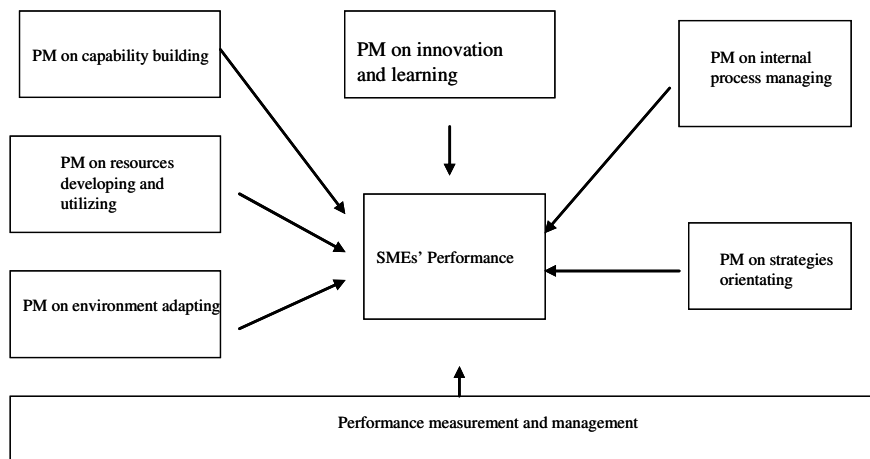
8.1 INTRODUCTION

An effective PM framework for ICT SMEs is developed from two components: performance determinants and performance results (based on the empirical studies in the chapters 5, 6 and 7. In Section 8.2, the performance determinants and the relative measures are derived, as well as the performance results and the relative measures. In Section 8.3, a PM framework is presented; the resulting PM model is based largely on the structure of the Business Excellence Models (i.e. the structure of ‘enablers’ and ‘results’). Section 8.4 makes a brief conclusion.

8.2 PERFORMANCE DIMENSIONS IN ICT SMES

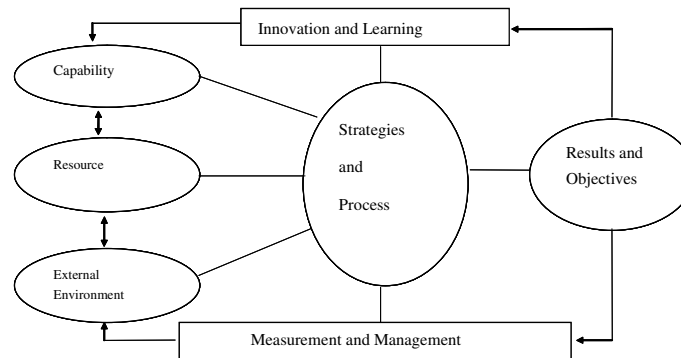
Based on analysis of PM features of SMEs and the theoretical approaches to designing PM for SMEs, the PM conceptual framework developed in Chapter 3 (see Figure 3-4) argued measuring an SME’s performance should measure both lagged and leading indicators. Three interrelated components: (1) internal performance determinant factors, (2) external performance determinant factors, and (3) performance results, should be involved in a PM framework for SMEs. The related PM dimensions are shown in Figure 8-1.

Figure 8-1 PM Dimensions in SMEs in the ICT Industries



The relationship of the above performance dimensions are described in Figure 8-2. The strategic objectives of an SME are based on the company's capabilities, resources, and external environment. After setting strategic objectives, the company should build and manage internal processes. Two main feedback chains are involved in the framework: one where capability is improved through innovation and learning, and one where strategies and process can be developed through PM measurement.

Figure 8-2 the Relationship between Performance Dimensions in ICT SMEs



Source: Wu (2006)

8.2.1 Performance Determinants and Measures

In Chapter 5, the critical success factors for ICT SMEs were investigated. The empirical study results reveal eight critical success factors that impact ICT-related SME performance. The factors are: competitive teams, best-practice strategies, core competency, competitive advantages, customer-focus, internal process management, resource utilization, and organizational agility. In the conceptual PM framework, it was assumed that SME performance depends on whether the company can adopt appropriate strategies and processes, in order to align the internal/external resources with the environment.

As discussed in Chapter 5, the important intangible resources for ICT SMEs are: employees, team, culture, knowledge/experience, network and reputation (see Table 5-1). Hence, team building can be categorized into the dimension of resource-building and utilization. In terms of tangible resources, capital acquisition is the most important.

Interview data confirmed that core competency and competitive advantages are important to SME performance. Among competency, the questionnaire survey data shows the capabilities regarding customer response, resource management, response to technological change, and flexibility with environment are the first four important factors that impact SME performance.

In terms of environmental factors, the empirical study shows that clearly defining customer-groups, anticipating customer needs, building a partnership, and competitor awareness are the most important.

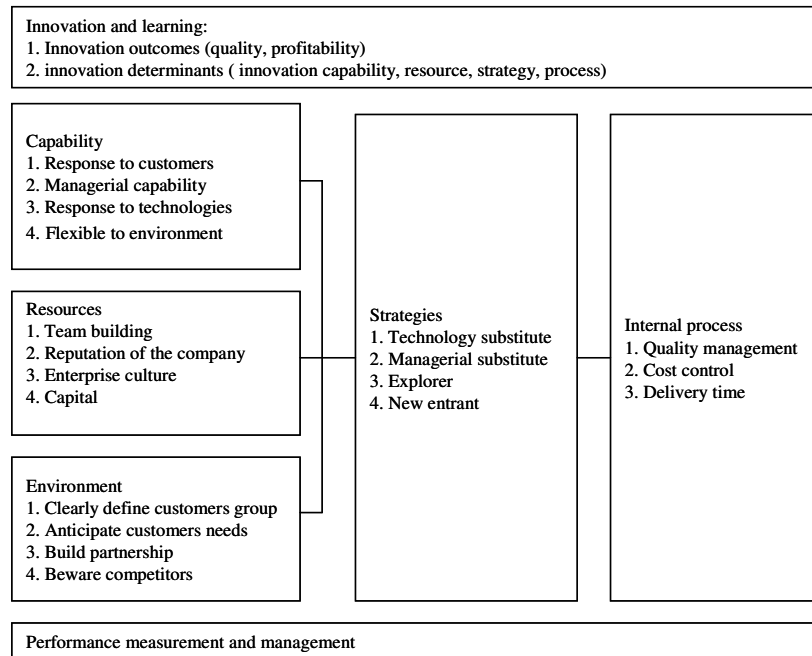
Through analysis of features of SMEs in the ICT industries, the study identified SMEs' strategies can be categorized into four types: technological substitutes; managerial substitutes; explorer; and, new entrant (see Chapter 5). The empirical study data confirmed that a definite strategy, based on target customers, markets, and environment, has a greater likelihood of being successful.

Internal processes are also important, especially for measuring metrics such as budgeting, quality, cost, and delivery time. To be most effective, the PM system should be combined with the SMEs' daily planning and reporting processes. In SMEs, the performance regarding internal processes can be represented by three dimensions: quality, cost and delivery time.

Additionally, innovation is fundamental and should be encouraged. Discussions in Chapter 5 show SMEs' innovation performance can be measured from six dimensions: innovation outcomes; innovation strategy; innovation processes; innovation capability building; innovation resource development; innovation measurement and management.

In summary, the performance determinants in SMEs in the ICT industries are listed in Figure 8-3.

Figure 8-3 Performance Determinants in SMEs in the ICT Industries



8.2.2 Performance Results and Measures

In Chapter 6, the performance indicators employed by existing SMEs were investigated. The data shows no universal performance metrics that suit all ICT SMEs. However, some common performance indicators were suggested by executives. Table 8-1 lists these performance results, measures, and indicators, the structure of which is based on Neely, Marr et. al.'s concept of third-generation PM. In short, Neely, Marr et al.(2003) believed that first-generation PM is balanced, like BSC and the Performance Prism. The second-generation measurement framework include strategy maps developed by Kaplan and Norton (2001). The third-generation PM framework “requires organizations to seek greater clarity about the linkages between the non-financial and intangible dimensions of organizational performance and the cash flow consequences of these” (Neely, Marr et al. 2003).

In Table 8-1, financial performance is measured. Growth indicators, profitability, and liquidity were thought to be important. In terms of non-financial performance dimension, this performance framework suggests measuring performance-related aspects like customers, employees, competitiveness, and strategic partners. For example,

in R&D enterprises, NPD efficiency and project-management performance are measured; for service enterprises, customer communication and inter-company collaboration should be measured.

Table 8-1 Performance Dimension; Measures and Indicators for SMEs in the ICT industries

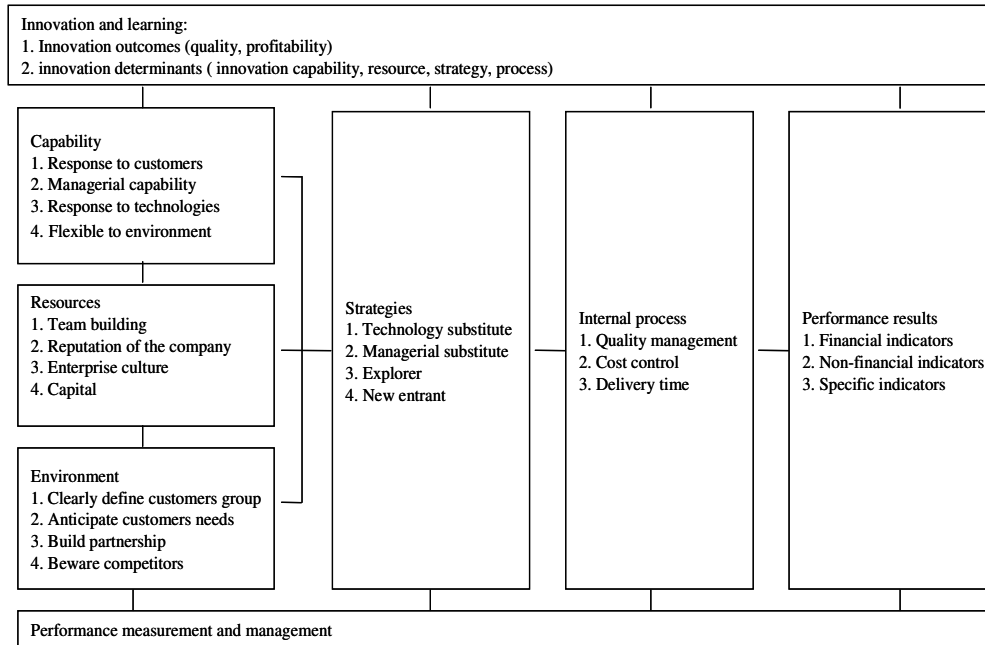
Dimension	Measures	Indicators
Financial performance	Growth	Growth on profit, Growth on revenue, Growth on annual sales volume
	Profitability	Return on sales; Profit after tax, Overall profit
	Liquidity	Net cash flow, Cash flow relative to competitors, Case flow to sales
	Efficiency	Return on assets; Return on investment; Return on equity; Average return on sales
	Revenue	Gross revenues; Revenue per customer;
Non-financial and intangible performance	Customers orientation	Customer satisfaction; Customer service calls; sales volume from single customer; The growth of customer number; The distribution of customer.
	Employees orientation	Employee satiation; Employee turnover/royalty; Remuneration and benefits benchmark; The culture building; Employee training; the collaboration between employee and business units.
	Competitiveness orientation	Growth on market share; Position in the market; New market entry; Customer satisfaction relative to competitors; Other indicators relative competitors...
	Strategic partners orientation	The indicators on cooperation with supplier; The process align with other organizational processes on the value chain; The network building on NPD or sales;
Specific performance	Depends on the SMEs' business types and strategies employed	

8.3 DEVELOPING AN EFFECTIVE FRAMEWORK FOR SMES IN THE ICT INDUSTRIES

In Chapter 6, the PM frameworks and tools employed by SMEs in the ICT industries were investigated. The results show that most SMEs use key performance indicators. Parmenter (2007) clarified the concept of Key Performance Indicators (KPIs), arguing that “KPIs represent a set of measures focusing on those aspects of organizational performance that are the most critical for the current and future success of the organization”. KPIs measure not only performance results, but also critical factors that influence an organization’s current and future success. This indicates that the ‘Enablers’ and ‘Results’ structure is valid and reliable for SMEs, helping to measure both performance results and determinants.

Drawing on the discussion on the performance determinants and performance results, the PM framework for SMEs is summarized in Figure 8-4.

Figure 8-4 the PM Framework for SMEs in the ICT Industries



It is observed that the eight components in the framework can be categorized into two groups: “enablers” and “result”. Each component involves a couple of performance dimensions which are derived from the critical success factors for SMEs’ performance. The strategies depend on organizational capability; resource and environmental factors. After formulating strategies, build and manage effective internal process in order to achieve the proposed performance objectives. According to the performance results achieved, improve capability and resources; adjust strategies and optimize the internal processes continually through innovation and learning.

The proposed framework provides the following benefits, helping to guide a company through its development:

First, it provides a comprehensive and holistic framework to look into SME performance. To achieve excellent performance, an SME should keep sight of all of the performance determinant factors, i.e. improving organizational capability; obtaining resources and utilizing resources effectively; adapting to environment; choosing the correct strategy based on internal and external factors; and modify internal process continually.

Second, this framework suggests that SMEs not only measure performance results, but also evaluate the internal and external performance factors. It fits the features of SMEs in the ICT industries. The ICT industries are a rapidly developing industry in which lagging performance results do not accurately reflect real-time market competitiveness. To overcome this shortcoming, continual assessment is necessary.

Third, this framework can also be used as a performance planning and diagnostic tool. For example, before setting a performance objective, an SME needs first to think about its own capability and resources in order to identify changes in the environment; then, it can formulate the correct strategies. After that, it should assess whether the internal process provide enough reliability to achieve the objective and strategies. Finally, based on measuring the achieved objectives, it should take action to improve its processes. The above procedure can be accomplished by running the proposed framework, step by step. In fact, it can also be used in a business unit, work-team, and on every employee (it is visually represented in Figure 8-5).

Figure 8-5 Performance Planning and Diagnosing Box

Resource Do we have the resource to achieve the performance objectives? If not, how to obtain it?	Capability Do we have the capability to achieve the performance objectives? If not, how to improve it?
Strategies and process Have we had correct strategy and processes to achieve the objectives? If not, how to build them?	External barriers What are the external barriers to our achievement of the objectives ? How can we adapt to them ?

Finally, this framework provides flexibility. It allows SMEs focus on one or several performance aspects at a time instant to improve their performance whilst providing a comprehensive view on PM.

8.4 CONCLUSION

This chapter discusses the PM requirements in SMEs in the ICT industries through comparing the conceptual and empirical results. After discussing the measures and indicators on performance determinants and performance results, based on the empirical result in part III, a new PM framework was developed (using the structure of ‘enablers’ and ‘results’).

CHAPTER 9 PM IMPLEMENTATION PROCESS FOR ICT SMES

9.1 INTRODUCTION

In Chapter 8, a new PM framework for SMEs has been developed, based on the conceptual PM framework given in Chapter 3. This chapter discusses how SMEs can implement the proposed PM framework. Particularly, following issues are addressed:

- What are the specific requirements for performance process in SMEs?
- What are the practical steps to implement the proposed PM framework in SMEs?
- How does a SME set its performance objectives based on analysis of its capability, resources and external environment?
- What is the performance infrastructure that can be built for implementing PM?

Above questions relating to the PM implementation processes are investigated among the 16 SMEs interviewed. In fact, the above questions are related to the continuous improvement methodology (Zangwill & Kantor, 1998), which requires managers change their leadership styles for better team work and employees learn a model for successful team process. This is the basic for the PM implementation process suggested in Section 9.3.

9.2 THE REQUIREMENTS OF THE PM PROCESS IN SMES IN THE ICT INDUSTRIES

Hudson, Lean et al. (2001) summarized the requirements of PM process in SMEs; they are as follows:

- Encourage regular measurement and performance feedback
- Introduce changes incrementally
- Be structured and facilitate business planning
- Be fast and resource efficient
- Give short as well as long term benefits and be flexible enough to accommodate strategic changes

Neely, Marr et. al. (2003) argued for a third generation PM system – one of practicality and organisational alignment, wherein “the measurement system is fully integrated with other organisational processes such as planning, budgeting, reporting as well as performance reviews.”

Indeed, the above viewpoints are supported by the interview and survey data in this study, which highlight the following points:

First, PM process should help SMEs formulate strategy. In the survey results, forty-four percent of the companies reported that they measure performance in order to support the decision making at top-management level. For ICT SMEs, the most important issue related to PM is to formula right strategy based on an organization’s resource, capability and external environment.

Second, PM process for ICT SMEs should be flexible enough to suit the rapid changes in customer needs, market situation and technologies. In the survey conducted for this study, the first critical factor marked by executives that impact SMEs’ performance is ‘response to customer need swiftly’. In addition, ‘adapting to new industry and market trends’ and ‘strategies are based on target customers, markets and environment” were in position 3 and 4. The flexibility requirement on PM process also comes from the internal changes (e.g. the change in managerial mechanism and organizational structures). The survey results confirmed that KPIs system is most widely used (46.7 percent) because of its flexibility.

Third, PM process should be resource-saving. According to the survey, the first barrier in implementing PM is ‘no time and no resources’ (73.3 percent of respondents). In order to help SMEs save resource when implementing PM system: (1) design a simple PM template, which saves much paper work; or (2) combine PM process with daily plan work; this also helps SMEs formalize its plan work. In fact, PM process can not be separated from planning process. Further, PM process should extend to individual employee’s plan process.

9.3 PROCEDURE FOR IMPLEMENTING PM IN SMES

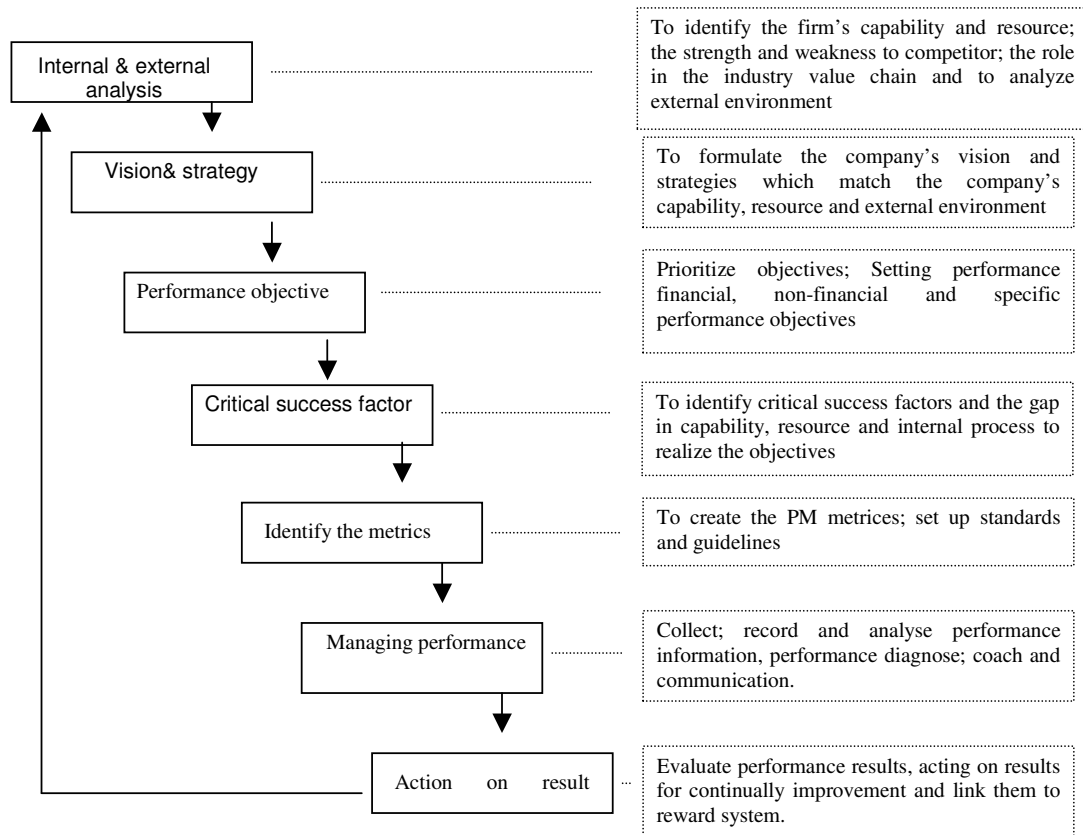
Following requisites for a good performance measurement system are mentioned in (Kanji, 2008):

- Provision of holistic and systems approach.
- Provision of multipurpose and interrelated activities.
- Links to the organizational values and strategy.
- Links to the organizational critical success factors and quality principles.
- Provision of valid, reliable and easy-to-use models and methodologies.
- Comparison of results and monitoring of progress.
- Improvement opportunities and improvement strategy.

Considering above requirements of PM process in SMEs in the ICT industries, we see that whole procedure of PM could be a big feed-back loop, in which the actions at one step is taken based on the results of other steps. Note the whole process should not work in a simple flow-plot form (which usually just give “Yes” or “No” information from measurement); instead, it should supply valuable assessment with suggestion on performance management. It should be noticed that the measurements in some steps could share performance related information to produce a fair assessment result. In brief, the PM procedure is suggested to works as follows:

- 1) Internal and external analysis is the basic of the whole procedure;
- 2) Vision and strategy checking should mainly based on the result from 1);
- 3) Performance objective setting is directed by the result from 2) and based on the filtered information of 1) passed on by 2);
- 4) The identification of critical success factors and creation of PM metrics gives detailed applicable indices from result from 1), 2) and 3);
- 5) Performance management is carried on following 4) and actions are taken based on the results. Then a new process starts, go back to 1).

Hence, we have the following procedure for implementing PM framework in (See Figure 9-1).

Figure 9-1 Proposed Performance Management Procedures for SMEs

1. Internal and External Analysis

According to the survey, the first two reasons behind implementing PM systems in SMEs are “to support decision making at top level” and “to identify possible needs for changes in strategy.” The interview data shows that one reason for PM failure is related to choice of inappropriate strategies. In order to avoid this pitfall, an in-depth internal and external analysis should identify: (1) the capability that the firm has; (2) the resource the firm needs and the resource the firm has; (3) the firm's strengths and weakness comparing to its competitors; (4) the firm's position in the industry value chain; and, (5) the environmental factors, including competitors, suppliers, customers, and regulations.

2. Choose Appropriate Vision and Strategies

A fundamental feature of SMEs in the ICT industries is innovation. Strategy can be chosen based on an SME's capability, resources, and its external environment. In

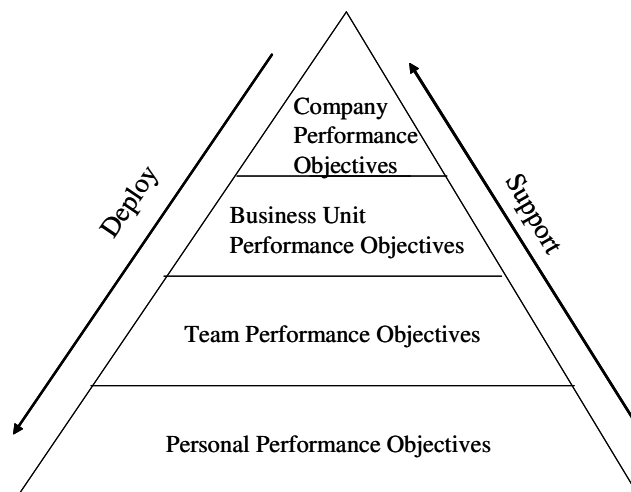
general, an ICT SME's strategy can be categorized into: Technologies substitute, Managerial substitute, Explorer, New entrant (see Chapter 5).

3. Setting Performance Objectives.

Objectives must be specific, measurable, achievable, relevant, and time-bound (SMART) (Booth 1998; Broadbent 1999). As discussed in Chapter 7, SME performance objectives include three components: internal performance drives, external performance determinants, and financial and non-financial performance results. The setting of performance objectives should support realization of strategies based on external and internal analysis.

In order to make performance requirements manageable, SME performance objectives can be divided into four levels (as shown in Figure 9-2): company performance objectives, business unit performance objectives, team performance objectives, and individual performance objectives. The lower-level performance objectives, on the one hand, are deployed from upper-level objectives. On the other hand, they support the accomplishment of the upper-level objectives.

Figure 9-2 Performance Objectives Structure



To save resource, it is advisable that a company choose at least 2 or 3 objectives to execute in terms of the performance improvement plan. Correspondingly, each business unit, work team, and employee makes 2 to 3 relevant performance improvement objectives.

4. Critical Success Factor Analysis

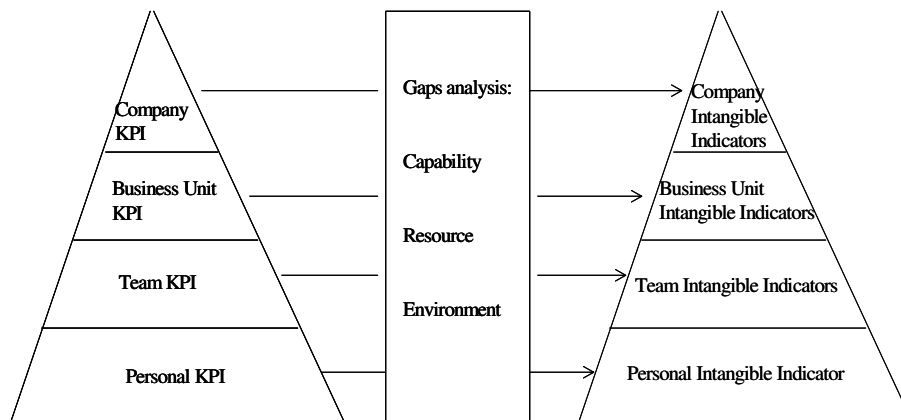
Critical success factors are those attributes which drive the success of the firm (Broadbent 1999). The critical success factors for ICT SMEs were investigated and identified in chapter 6. However, specific SME has its specific critical success factors. To identify the critical success factors, an effective tool is performance map base on case-and-effect chain.

5. Identify the Metrics

After setting performance objectives, appropriate performance indicators should be identified. An SME needs to ensure that metrics are relevant, valid, and feasible. The indicators not only reflect the performance results, but also measure the processes that achieve results and future performance. The performance indicators include the KPI and the associated activities (intangible indicators); their relationship is shown in Figure 9-3. KPIs are derived from the company's performance objectives. Each performance objective can be measured by several key performance indicators. Evans et. al., believed that KPIs can play an essential role in developing a widespread understanding of how the various parts of the business operate together. Furthermore, the KPIs delegate corporate goals into an accountability matrix of individual management responsibility.

Usually, there exists a gap between the proposed KPIs and the reality of an SME. To achieve proposed KPIs, an SME should improve its capability, resource, and processes. The intangible indicators measure the improvement of capability, resources, and process. They also guide the SME's improvement activities.

Figure 9-3 the Relationship between KPIs and Intangible Indicators



6. Managing Performance

Performance measurement and management is not an appraisal process that focuses on results. It is a managerial control process that spans across organizational activities. Managing performance includes making performance plan (i.e. work plan), performance data collection, delivery and analysis, performances diagnosis, and performance coach.

After setting-up performance objectives, performance management should be based on the following practical steps:

- 1) Build up the organizational structure for performance measurement and management. This includes clarifying the performance examination-structure among the whole company, formalizing the reporting process, and so on.
- 2) Choose the performance measurement tools; construct the performance information collecting and delivery network.
- 3) Develop the performance measures to support the achievement of the performance objectives. Setup the performance criteria for measures.
- 4) Review and revise the performance objectives based on the changes of the company's capabilities, resources, and external environment.
- 5) Collect the performance data according to the pre-defined performance information collecting and reporting process.
- 6) Analyse the performance data. At this step, the raw performance data should be transformed into performance information. Performance data analysis is not just a specialist's job. Employees and business units in the company are encouraged to participate in the related work. Different tools can be chosen to analyse the information data.
- 7) Communicate the performance information. Forward the analytical results to performance information user (executives, planner of marketing, finance and so on).

7. Action on Results

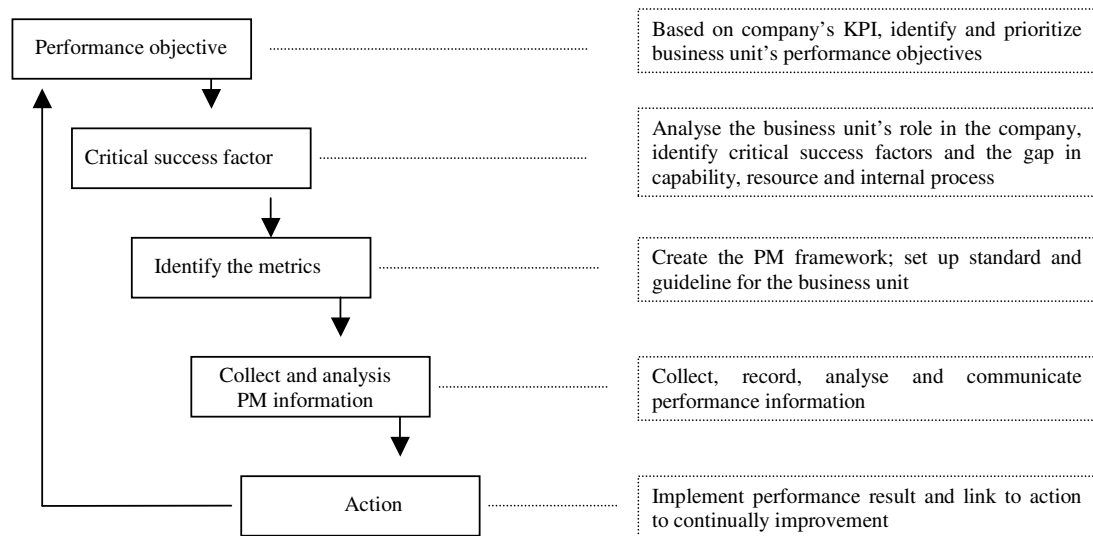
In a company, the performance results can be utilized in many areas. Godener & Soderquist (2004) summarized the implementation of performance results into four categories:

- Use of performance measurement results for personnel evaluation, promotion, and incentives (promotion prospects, salary, project participation, bonuses)
- Use of performance measurement results for resource allocation (project participation, forming/dissolving teams, assignment of new projects and of resources)
- Use of performance measurement results for control/correction (control, correction, reorganisation)
- Use of performance measurement results for learning/continuous improvement.

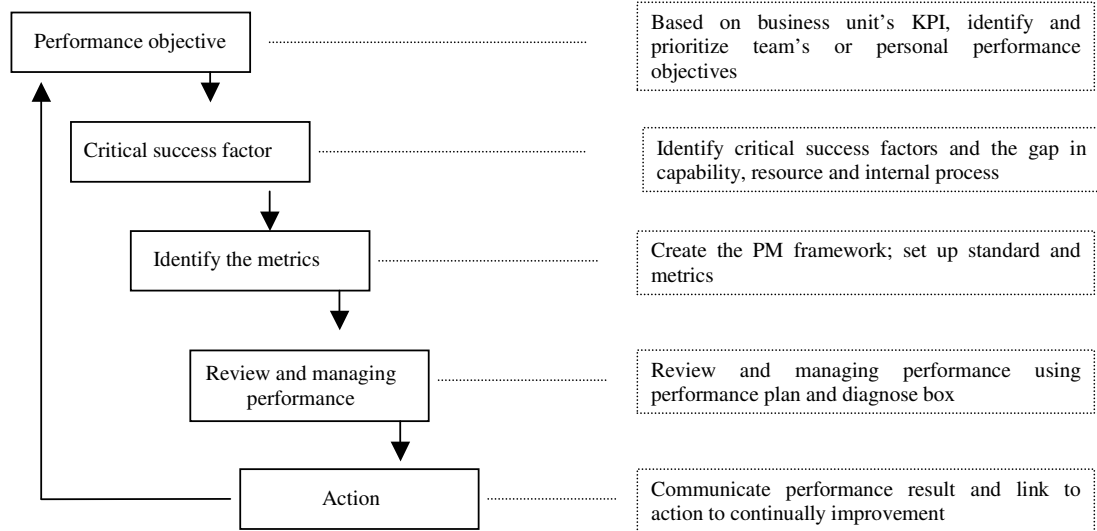
For SMEs, the most important PM issue is to link performance results to action, in order to continually improve performance.

The performance measurement and management procedure – at the business unit level – is proposed in Figure 9-4. Note the business unit's performance objectives follow the company's objectives.

Figure 9-4 Proposed Performance Management Procedures for Business Unit



In terms of teams' and personal performance management in SMEs, performance management procedure needs to be combined with daily work. The individual performance measurement process is shown in Figure 9-5.

Figure 9-5 Proposed Performance Management Procedures for Teams and Individuals

9.4 CONCLUSION

This chapter analyses the requirement of PM process and suggests the steps for implementing the PM framework proposed in chapter 8. It is important for SMEs to setup comprehensive strategic objectives and to execute an effective performance measurement throughout the whole company. To set up appropriate strategies, the priority is to consider capabilities, resources, and external environments. After setting performance objectives, analysing critical success factors helps to identify performance measures and metrics.

PART V SUMMARY AND CONCLUSIONS

CHAPTER 10 SUMMARY AND CONCLUSION

10.1 SUMMARY

The theories and practices of performance measurement have been advanced during the last two decades. However, there are no commonly accepted solutions to the issues of ‘what should be measured’, ‘what can be measured’ and ‘how to measure’. Because of the features of SMEs and the ICT industries, these issues are more challenging.

10.1.1 Key Contributions and Findings

A practical PM framework for ICT SMEs has been developed, which answers the following key questions – What is an effective PM system for SMEs in the ICT industries? What are the factors that should be considered in PM in SMEs in the ICT industries? What are the appropriate performance measures and indicators for SMEs in the ICT industries? How can SMEs implement PM system successfully?

The key findings from the present study include:

- The PM systems used in large organizations do not suit SMEs.
- Measuring and managing ICT SMEs’ performance should be done from a systematic perspective, i.e. both internal and external factors should be taken into account.
- Business Excellence Models and System Theory are more appropriate than other theoretical approaches to designing a PM framework for ICT SMEs.
- A combination of three components: internal performance determinants, external performance determinants, and performance results, should be included in a PM system for SMEs.
- An SME’s performance depends on whether the company can formulate appropriate strategies and process, to align its internal/external resources with its environment in order to achieve desirable results and objectives.
- Eight critical success factors have a profound impact on SME performance. They are: competitive team; right strategies; core competency; competitive

advantages; customers focusing; internal process management; resource utilization; and, organizational agility.

- Eight factors should be considered in a PM framework for ICT SMEs: capability development; resource utilization; environmental adaptability; strategy implementation; internal process management; innovation and learning; performance measurement and management; and, performance results. The eight factors can be categorized into three groups: internal performance determinants, external performance determinants, and performance results.
- The key financial indicators that are employed by SMEs in the ICT industries are: growth indicators, profitability, and cash flow.
- The key non-financial indicators that are employed by SMEs in the ICT industries are: customer satisfaction, employee satisfaction, and team building.
- Key Performance Indicators system (KPIs) is the solution for SMEs in managing performance because of its flexibility and resource saving.
- Leader-support and communication are critical in implementing PM.
- When an SME, a businesses unit, or a staff member analyses performance, the following four questions should be asked: 1) Do we have the capability, resource, correct strategies and processes to achieve the performance objectives? 2) If not, how to compensate for it? 3) What are the external barriers to achieving the performance objectives? 4) How to overcome these barriers?
- Four components should be clarified when implementing PM in SMEs: 1) building the performance measurement infrastructure; 2) formulating the strategy; 3) analysing the performance factors; and, 4) choosing the right measures.
- A seven-step performance management process should be followed: Internal & external analysis → Choose appropriate vision and strategies → Set performance objectives → Critical success factor analysis → Identify the metrics → Manage performance → Take action from results.

10.2 LIMITATIONS OF THE STUDY

There exist in the study some limitations (in terms of research methodology).

A multi-case study was used at the qualitative research stage. As the researcher mentioned before, it is an appropriate method to conduct the exploratory research on

SMEs. However, case studies have biases, due to sample selection and retrospective reporting (Ahire and Golhar 1996). Only 16 sample companies are studied at the qualitative stage, which might lead to biases. In addition, biases can arise from the retrospective reporting of data. Participants may not accurately recall events and important items. To minimize this problem, the results of 35 previous research works in this area were reviewed and studied, through meta analysis.

The primary limitation on data collection is the small sample size. Even though triangular data resources were employed, the findings in this study might be biased because of the small responses in the questionnaire survey.

At the quantitative research stage, bias would be arising from variables choice. Some important variables may not be taken into account. To minimize the effect of such problem, the variables were selected after careful study of the literature and qualitative results. Another potential source of bias is that the interviews schedules and the questionnaire design are based on a conceptual PM framework. This may cause some important factors being ignored in data collection. To minimize this risk, many open questions are used in interviews and questionnaire.

Moreover, the data was collected from a manager self-reporting survey. As Ahire & Golhar (1996) mentioned: there is a potential bias of self-reporting when one collects data from managers, particularly about managerial aspects with which they are closely associated. To address this issue, senior managers were invited as participants in this research as they have more comprehensive view about PM in their company.

It should be noted that, given the small sample size, this study does not intend to make a generalization from the results. Instead, the emphasis is on analysing the complexity of a range of PM issues in ICT SMEs, which has hardly been researched or reported in the literature.

10.3 CONCLUSIONS AND DIRECTIONS FOR FURTHER RESEARCH

An effective performance measurement system is tailored, considers the external environment, is integrated into a company's existing daily processes, is flexible, and gives special attention to customer needs (especially in a product-driven company).

The suggested PM framework created by this study addresses all of these issues and is based on the Business Excellence Models and the Theory of System Management. Furthermore, it incorporates the concept of Key Performance Indicators.

Specifically, the constructed PM framework measures both internal and external performance determinants and performance results. The performance determinants are represented in following dimensions: capability building; resource developing and utilizing; environment adapting; strategies formulating; internal process managing and PM on innovation and learning. The performance results' dimensions include the financial results to satisfy the investors, the customer satisfaction indicators to meet customers' needs, the competitive indicators to reflect the competitive advantages, and the collaborative measures to measure the partnership.

Based on the present study, the following future research is suggested:

Firstly, identify, and test the validity of performance determinants and performance measurement metrics using quantitative approaches. This study mainly employed the qualitative approaches.

Second, relationships between internal and external performance determinants and performance results warrant further investigation. By means of quantitative methods, the relationships and the nature of the relationships can be measured and quantified.

Finally, additional PM designs could be created; using the key methods presented here and those discovered in future research, the resulting model would serve to further aid SMEs in organizational development and planning.

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APPENDIX A

Interview outline

Interview outline I (for SMEs in the ICT Industries)

1. In your opinion, what are the key factors that impact your company's performance?
2. What are the key financial indicators and non-financial indicators to measure the performance in your company?
3. In your opinion, what are the most important performance indicators that measure a company's performance?
4. What are the crucial factors for a successful performance measurement?
5. How does the strategy influence a company's performance?
6. What are your company's core competences? In your opinion, how does one company's capability influence the company's performance?
7. What are the resources your company needs? Whether would shortness resource influence the company's performance? How to overcome this issue?
8. How does business environment influence a firm's performance?
9. How does the internal process influence a company's performance?
10. What is the main performance measurement model being used in your company?
11. What are the weaknesses and problems of the performance measurement model currently being used in your company?
12. In your opinion, what is an effective performance measurement model?
13. How to modify the performance measurement model in your company?

Interview outline II (for VCs which mainly invest in ICT industries)

1. From your company's opinion, what are the key factors that influence SMEs performance?
2. What are the features of SMEs in the ICT industries compared with the entrepreneurial firms in other industries?
3. What is the existing monitoring mechanism to SMEs (including monitoring frameworks, theories, methods, key indicators, and key factors)?
4. What are the problems of the existing monitoring system?
5. What are the key factors which should be involved in a successful monitoring system and the adding-value activities?
6. What are the key financial indicators and non-financial indicators which are used by VCs to measure the performance in SMEs?
7. Does the VCs characteristics and cooperation influence the performance of SMEs? How are they influence the performance of SMEs?
8. What are the most important indicators that VCs employed?

APPENDIX B

The questionnaire of the survey about performance measurement in SMEs

Section (I) Company Profile:

1. Please estimate the number of employees in the firm.
☐ 1-5 ☐ 6-20 ☐ 21-100 ☐ 101-250 ☐ 251 or more
2. What is the type of your firm?
☐ Manufacture ☐ Service ☐ R & D ☐ whole sales ☐ Retailer
3. How many years has your firm been in business?
☐ Less than 3 years ☐ 3 -7 years ☐ 7- 15 years ☐ More than 15 years
4. Your job position in the firm:
☐ Owner manager ☐ Manager ☐ Supervisor ☐ Employee
5. Is the company private or publicly held?
☐ Private ☐ Public
6. Has your company been invested by venture capital companies?
☐ Yes ☐ No

Section (II) Implementation of performance measurement

7. What kind(s) of performance measurement is(are) implemented in your company?
 - ☐ Financial performance measurement
 - ☐ Human resource performance measurement (personal performance appraise)
 - ☐ Customer satisfaction measurement
 - ☐ Process management measurement
 - ☐ Strategy measurement
 - ☐ Sustainability measurement (impact on society)

- ☒ Innovation measurement
- ☒ Other: _____

8. Which performance measurement model or tools are used in your company?

- ☒ ISO9000 certification
- ☒ Total quality management (TQM)
- ☒ Australian Business Excellence (ABE) framework
- ☒ EFQM Excellence model (European business excellence model)
- ☒ Baldrige Business Excellence criteria
- ☒ Balanced Scorecard
- ☒ Key performance indicators (KPI) system
- ☒ Benchmarking system
- ☒ Other: _____
- ☒ None of above

9. What are the initial reasons for your company to implement that performance measurement system?

- ☒ Ensuring customer requirements
- ☒ Providing feedback for people to monitor their own performance levels
- ☒ Highlighting quality problems and determining which areas most need attention
- ☒ To identify possible needs for changes in strategy
- ☒ Justifying the use of resources
- ☒ Providing feedback for driving the improvement effort
- ☒ For decision support at the top-management level
- ☒ For decision support at the operating level
- ☒ To determine the bonus to management and/or staff
- ☒ Other:

10. The barriers of implementation of performance measurement in your company include

- ☒ No enough performance measurement knowledge
- ☒ No time and no resource to do them
- ☒ Performance measurements are not useful
- ☒ The performance measurement tools or model are complex, we did not know how to tailor them to suit our company

11. List the first 3 significant problems of implementing performance measurement in your company

- (1) _____.
- (2) _____.
- (3) _____.

Section (III) The implementation of measures

12. Please write down 5 most important key performance indicators in your company

- (1) _____.
- (2) _____.
- (3) _____.
- (4) _____.
- (5) _____.

13. How do you estimate your company's performance

☐ **Excellent** ☐ **good** ☐ **not bad** ☐ **not good** ☐ **no idea**

14. How important do you believe the following factors or managerial practices for your company are to achieving excellent performance?

Not important Middle Very important

1 2 3 4 5

		How important-..				
		Low	Middle	High		
Capability	Response customers needs swiftly	1	2	3	4	5
	Effectively manage people and resources	1	2	3	4	5
	Appropriate managerial system with improvement capability adaptive to resource and environment changes	1	2	3	4	5
	Deeply understand the technological trend and catch the changes	1	2	3	4	5
	Flexibility to adapt to new industry and market trends	1	2	3	4	5
Resource	Availability of capital	1	2	3	4	5
	The executives' managerial experience	1	2	3	4	5
	Access to overall low cost factors of production	1	2	3	4	5
	Technical resource (patents, exclusive technologies...)	1	2	3	4	5
	Comprehensive and efficient organizational system, structure and planning.	1	2	3	4	5
	Expertise in product/service development	1	2	3	4	5
	Expertise in marketing	1	2	3	4	5
	Expertise in customer service	1	2	3	4	5
	Expertise in management	1	2	3	4	5
	Access to low cost distribution channels	1	2	3	4	5
	Enterprise culture	1	2	3	4	5
	Reputation	1	2	3	4	5
Environment	The company's customer groups and market segments are clearly defined and selected	1	2	3	4	5
	Understanding and learning about customers, anticipating customer needs, and developing business opportunities.	1	2	3	4	5

Appendix B

	A systematic and organized process exists for collecting, conveying customers complain, feedback about products and other information from customers. This information will be analysed to improve the business process.	1	2	3	4	5
	Measuring customer satisfaction periodically and the results are used to drive improvement	1	2	3	4	5
	Understanding the changes in technology	1	2	3	4	5
	The company knows the main competitors, and is aware of its own competitive position in the market.	1	2	3	4	5
	The company gathers competitors information continuously	1	2	3	4	5
	Degree of industry concentration /fragmentation	1	2	3	4	5
	Entry barriers	1	2	3	4	5
	Size and growth rate of industry	1	2	3	4	5
	Stage of industry evolution	1	2	3	4	5
	Government regulation in the industry	1	2	3	4	5
Strategy	The company has a definite strategy	1	2	3	4	5
	Strategies are based on target customers, markets, environment	1	2	3	4	5
	The leader team considers employees' idea when planning the company's future.	1	2	3	4	5
	The strategy are developed, reviewed and updated periodically based on the information from customers, environment, and performance measurement.	1	2	3	4	5
	The strategy is deployed through a framework of key processes	1	2	3	4	5
	Provide new products to exist market	1	2	3	4	5
	Provide established product to exist market (differentiation on price, quality and other values comparing competitors)	1	2	3	4	5
	Provide established products to new market	1	2	3	4	5
	Provide new product to new market	1	2	3	4	5
Process	The process ensures that any changes in customers and market requirements and technology can be incorporated into product and service design	1	2	3	4	5
	Production/delivery processes meet customer, quality, and operational performance requirement	1	2	3	4	5
	Design, production and delivery processes are coordinated to ensure trouble-free and timely introduce and deliver products/service	1	2	3	4	5
	The processes are evaluated and improved continuously to achieve better performance	1	2	3	4	5
	The key support processes (for example, finance and accounting, IT support, personal, legal, risk management and so on) are well defined. They support the key product and service process to achieve the company's performance outcome and objectives	1	2	3	4	5
	The supplier and partnering processes are well managed to ensures that the corroboration goal can be achieved and improved to meet the requirement of the company	1	2	3	4	5
Measure and analysis	The key financial and non-financial information and data are selected, managed and used to support overall business goals achievement	1	2	3	4	5
	The coemptive information (competitors, environment and technology and so on) are collected and are analysed to improve the performance of overall activities	1	2	3	4	5

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	Employee and departments know how to measure their performance. The results can be use to guide the activities and to improve the performance	1	2	3	4	5
	Appropriate performance measurement tools and skills are employed.	1	2	3	4	5
	Each goal and objective will be measured according to designed measure criteria	1	2	3	4	5
	Strategy is clearly translated into company, department, unit and individual goals and objectives.	1	2	3	4	5
	Employee knows his/her job goal and objective and the contribution to the whole strategy and objectives	1	2	3	4	5
Innovation and learning	The investment in new products development (NPD)	1	2	3	4	5
	The efficiency of NPD process (the input of NPD/output of NPD)	1	2	3	4	5
	The company emphasis on the employees training.	1	2	3	4	5
	The organizational structure and system is renewed based on the environmental changes.	1	2	3	4	5
	Flexibility to adapt to new industry and market trends	1	2	3	4	5

15. To what extent does your company emphasise on the following measures?

Not important

Middle

Very important

1

2

3

4

5

	How important ...				
Performance Result & Outcome	Low	Middle	High		
Growth in sales	1	2	3	4	5
Growth in market share (GMS)	1	2	3	4	5
Growth in profit	1	2	3	4	5
Return on assets (ROA)	1	2	3	4	5
Return on equity (ROE)	1	2	3	4	5
Average return on sales	1	2	3	4	5
Gross revenues	1	2	3	4	5
Total asset turnover	1	2	3	4	5
Net cash flow (NCF)	1	2	3	4	5
Cash flow to sales □CFS□	1	2	3	4	5
Inventory turnover	1	2	3	4	5
Debt to equity	1	2	3	4	5
Times interest earned	1	2	3	4	5
Market share	1	2	3	4	5
Customer satisfaction	1	2	3	4	5
Customer satisfaction relative to competitors	1	2	3	4	5
Employee satisfaction	1	2	3	4	5
Employee turnover/royalty	1	2	3	4	5
Remuneration and benefits benchmark	1	2	3	4	5
The company helps employees help their community	1	2	3	4	5

Thank you very much for your participating!